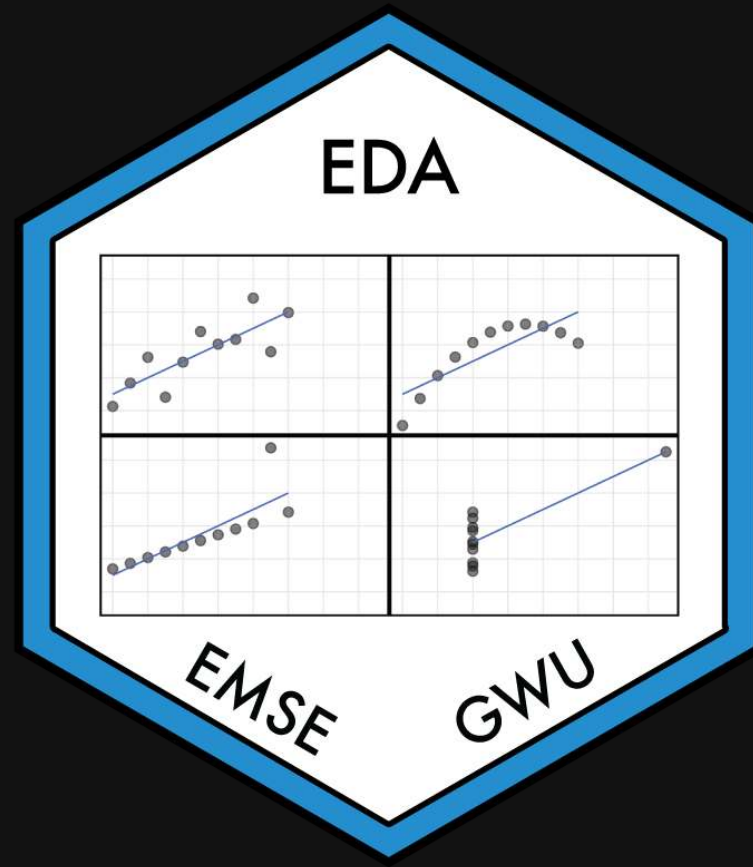




Visualizing Information

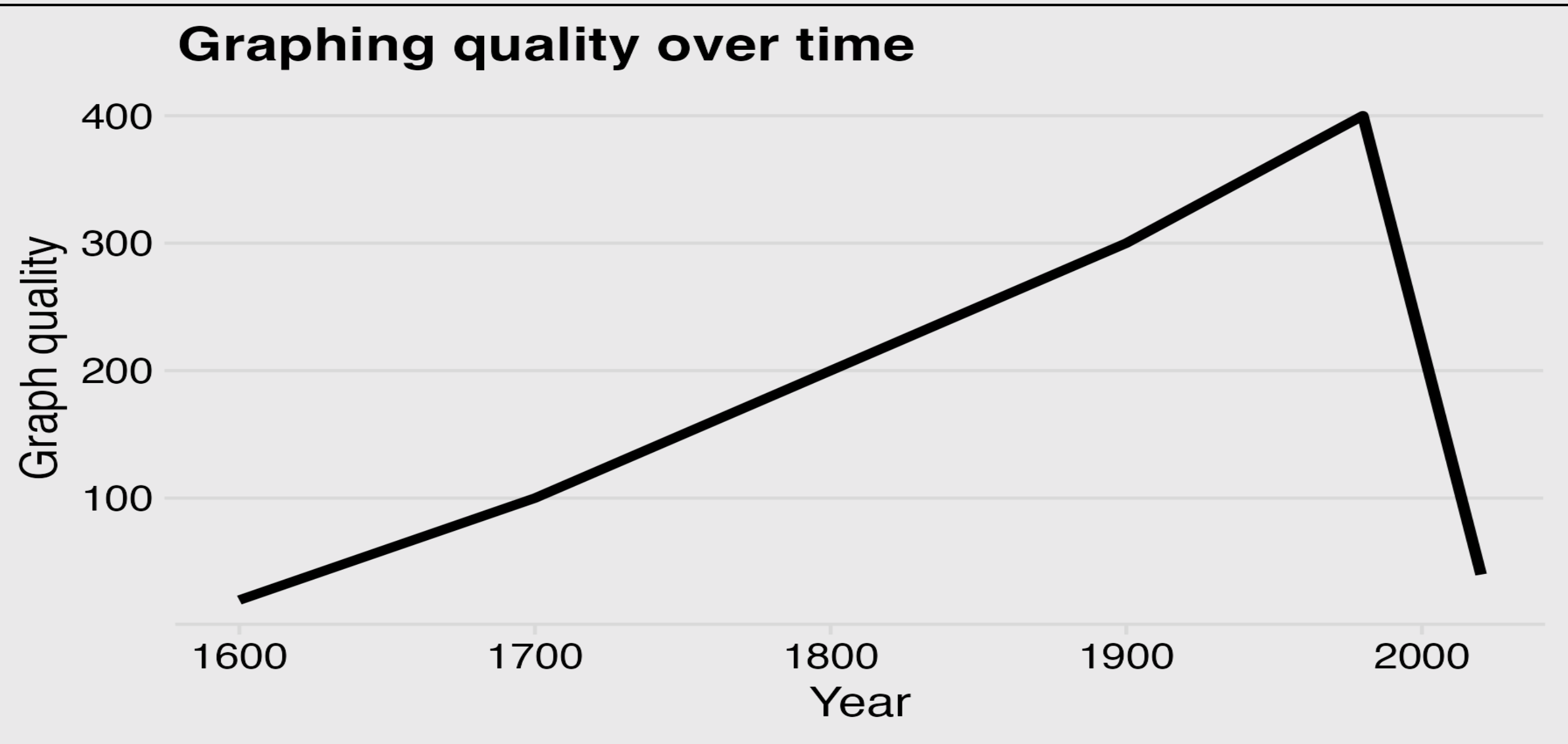
🏛️ EMSE 4572/6572: Exploratory Data Analysis

👤 John Paul Helveston

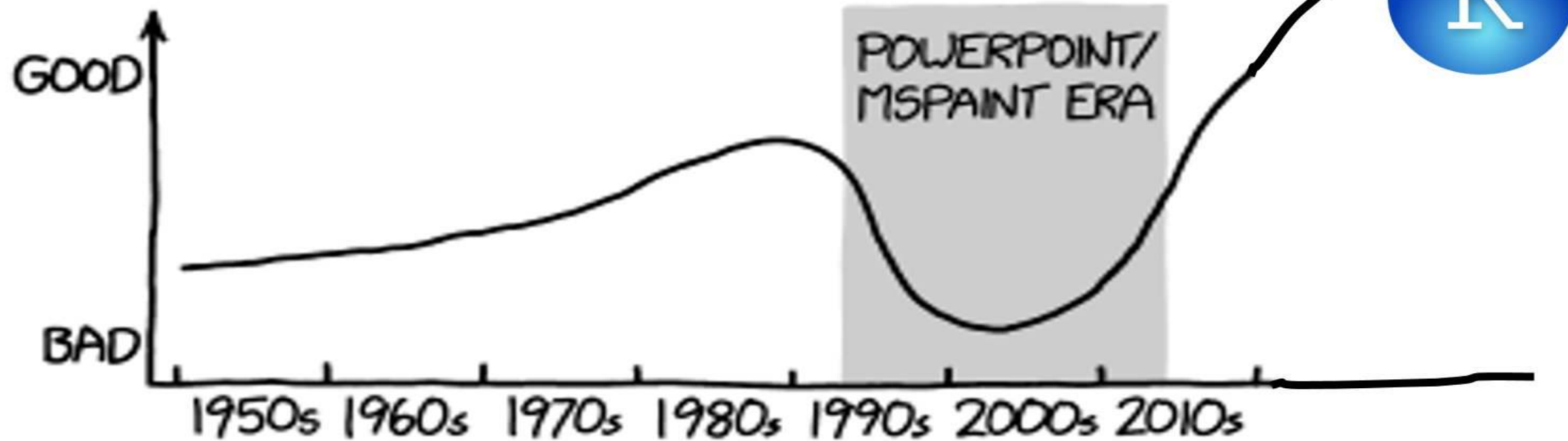
📅 October 07, 2026

References:

- Data Viz “Cheat Sheet”
- Data Viz Reference Page



GENERAL QUALITY OF CHARTS AND GRAPHS IN SCIENTIFIC PAPERS



From [here](#)

*“Having word processing software
doesn’t make us great writers.”*

— Stephen Few

We don't write paragraphs like this

People sometimes do this [use poor graphic choices] because they've seen similar charts in newspapers or on the web and they're naively following a bad example. People who know better sometimes do this because they care more about the visual impact than the clarity of communication. If we wanted to tell the truth in a way people can easily understand, this is not an effective approach.

Image from Few (2012, pg. 227)

So don't make graphs like this

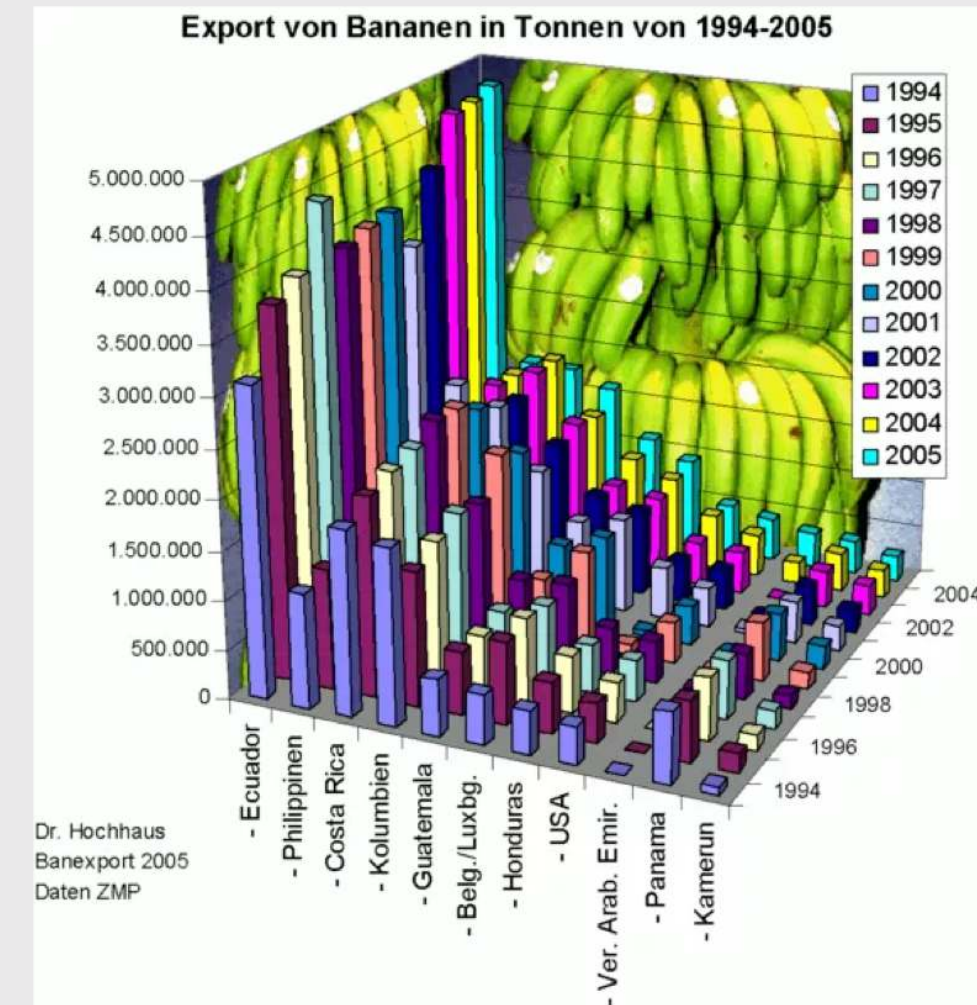


Image from excelcharts.com

Week 7: *Visualizing Information*

1. The Human Visual-Memory System
2. The Psychology of Data Viz
3. 10 Data Viz Best Practices
4. Making a (good) ggplot

Week 7: *Visualizing Information*

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Good visualizations optimize for
the human visual-memory system

A (very) simplified model of the visual-memory system

Iconic memory

- < 1 sec.
- “Pre-attentive”

A (very) simplified model of the visual-memory system

Iconic memory

- < 1 sec.
- “Pre-attentive”

Attention “grabbed”

Working memory

- < few hours
- Limited (3-4 chunks)

A (very) simplified model of the visual-memory system

Effective charts focus on this

Iconic memory

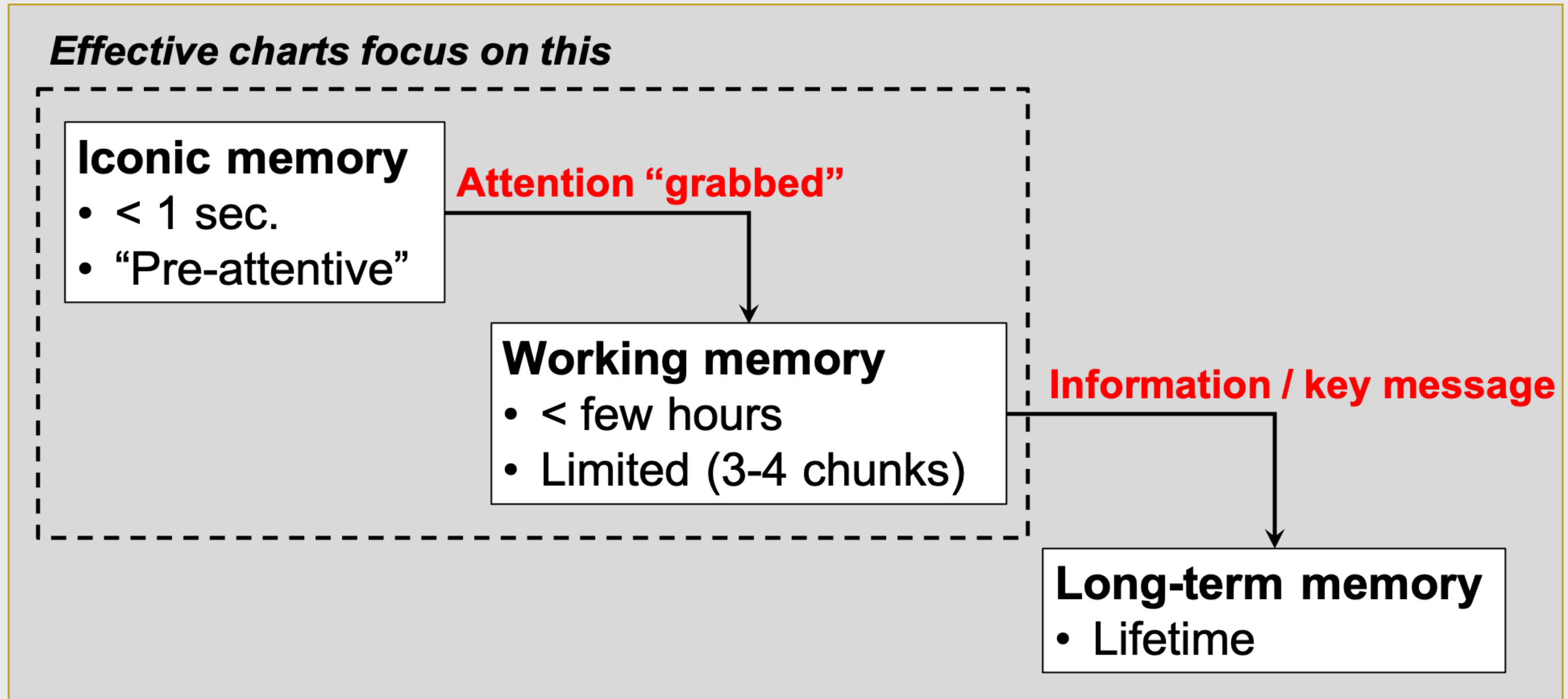
- < 1 sec.
- “Pre-attentive”

Attention “grabbed”

Working memory

- < few hours
- Limited (3-4 chunks)

A (very) simplified model of the visual-memory system



Two objectives of effective charts:

1. Grab & direct attention (iconic memory)
2. Reduce processing demands (working memory)

The power of pre-attentive processing

Count all the “5”s

821134907856412043612
304589640981709812734
123450986124790812734
029860192837401489363
123479827961203459816
234009816256908127634
123459087162342015237
123894789237498230192

The power of pre-attentive processing

Count all the “5”s

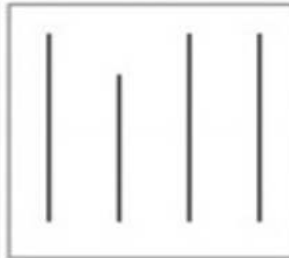
8211349078**5**6412043612
304**5**89640981709812734
1234**5**0986124790812734
029860192837401489363
1234798279612034**5**9816
2340098162**5**6908127634
1234**5**908716234201**5**237
123894789237498230192

Form

Orientation



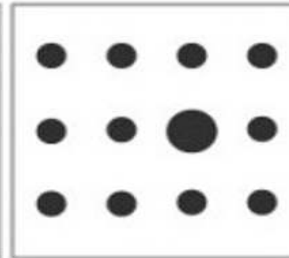
Line Length



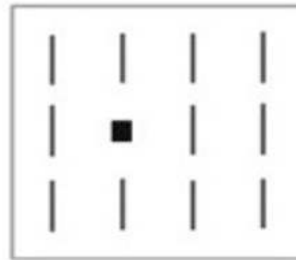
Line Width



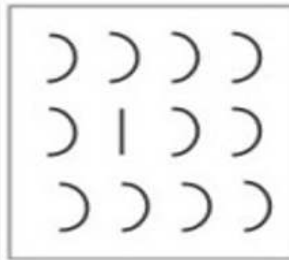
Size



Shape



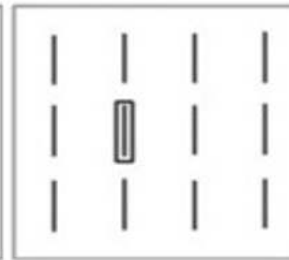
Curvature



Added Marks

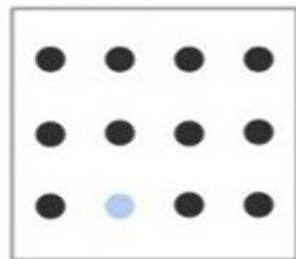


Enclosure

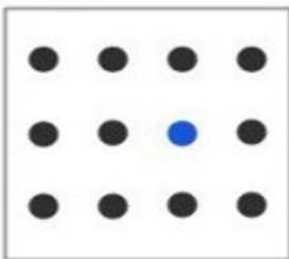


Color

Intensity

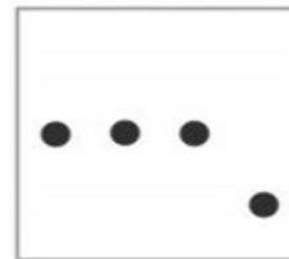


Hue

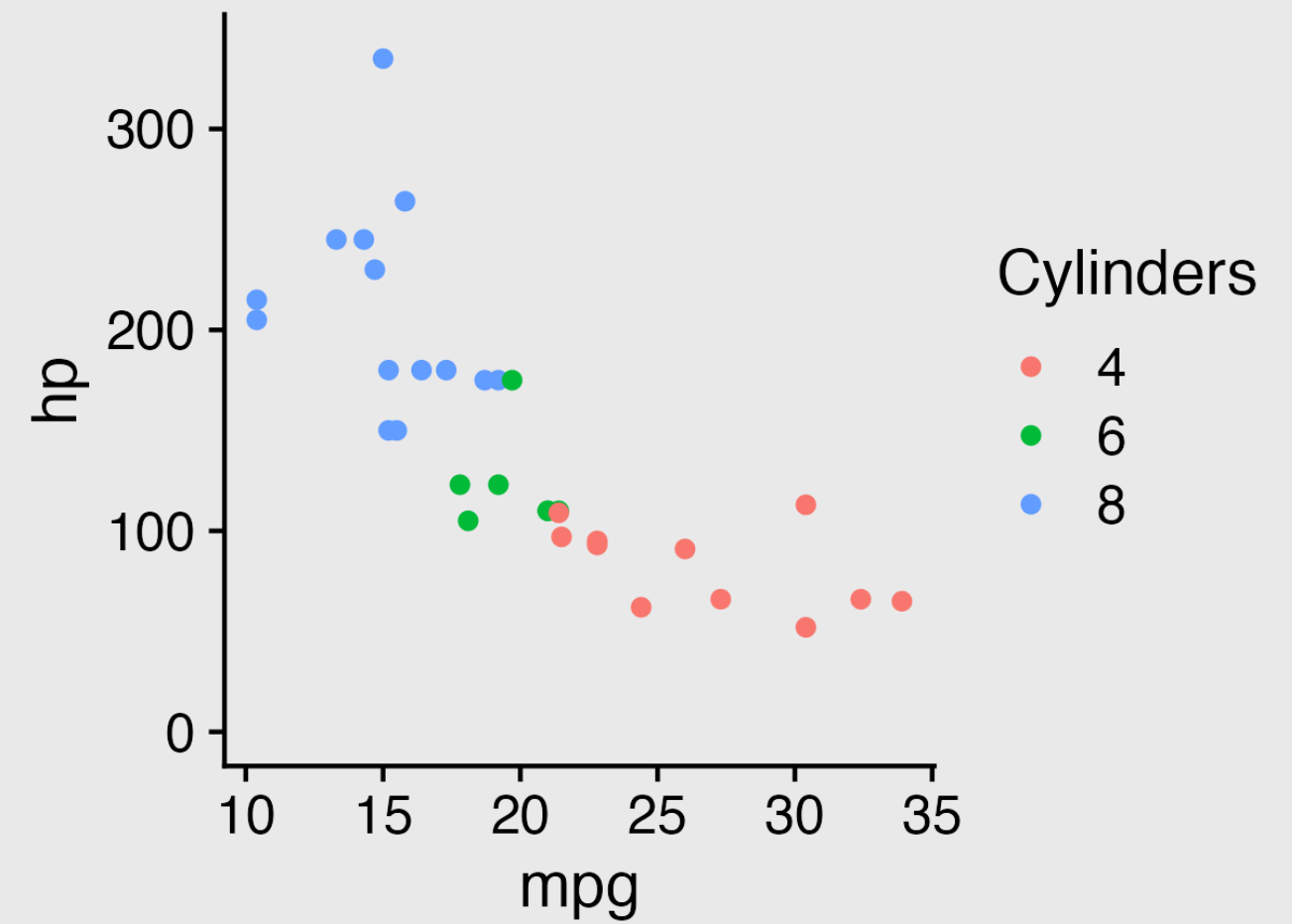


Spatial Position

2-D Position

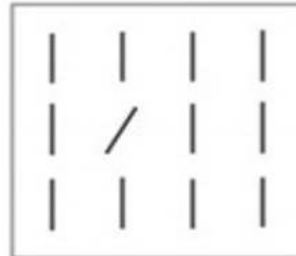


Pre-attentive attributes

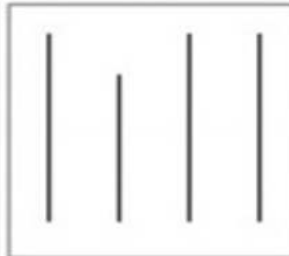


Form

Orientation



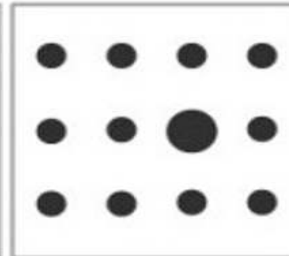
Line Length



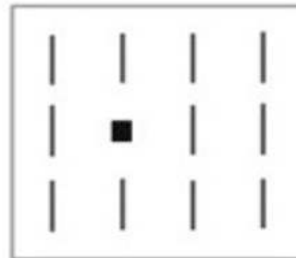
Line Width



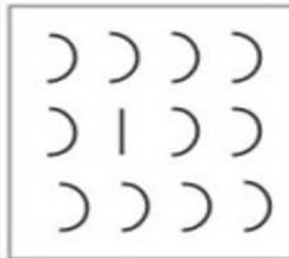
Size



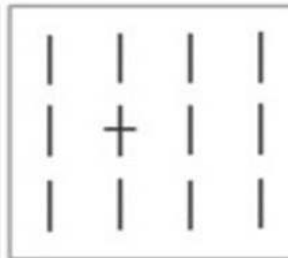
Shape



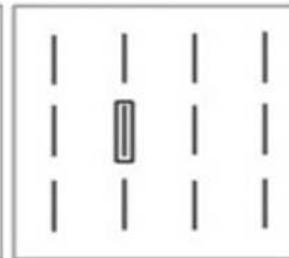
Curvature



Added Marks

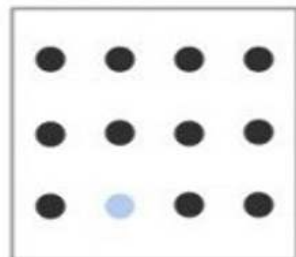


Enclosure

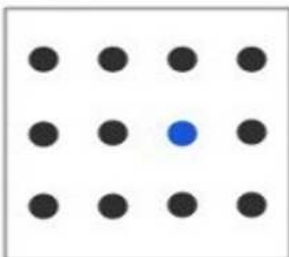


Color

Intensity

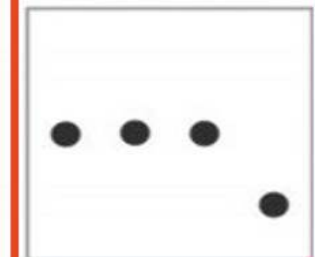


Hue



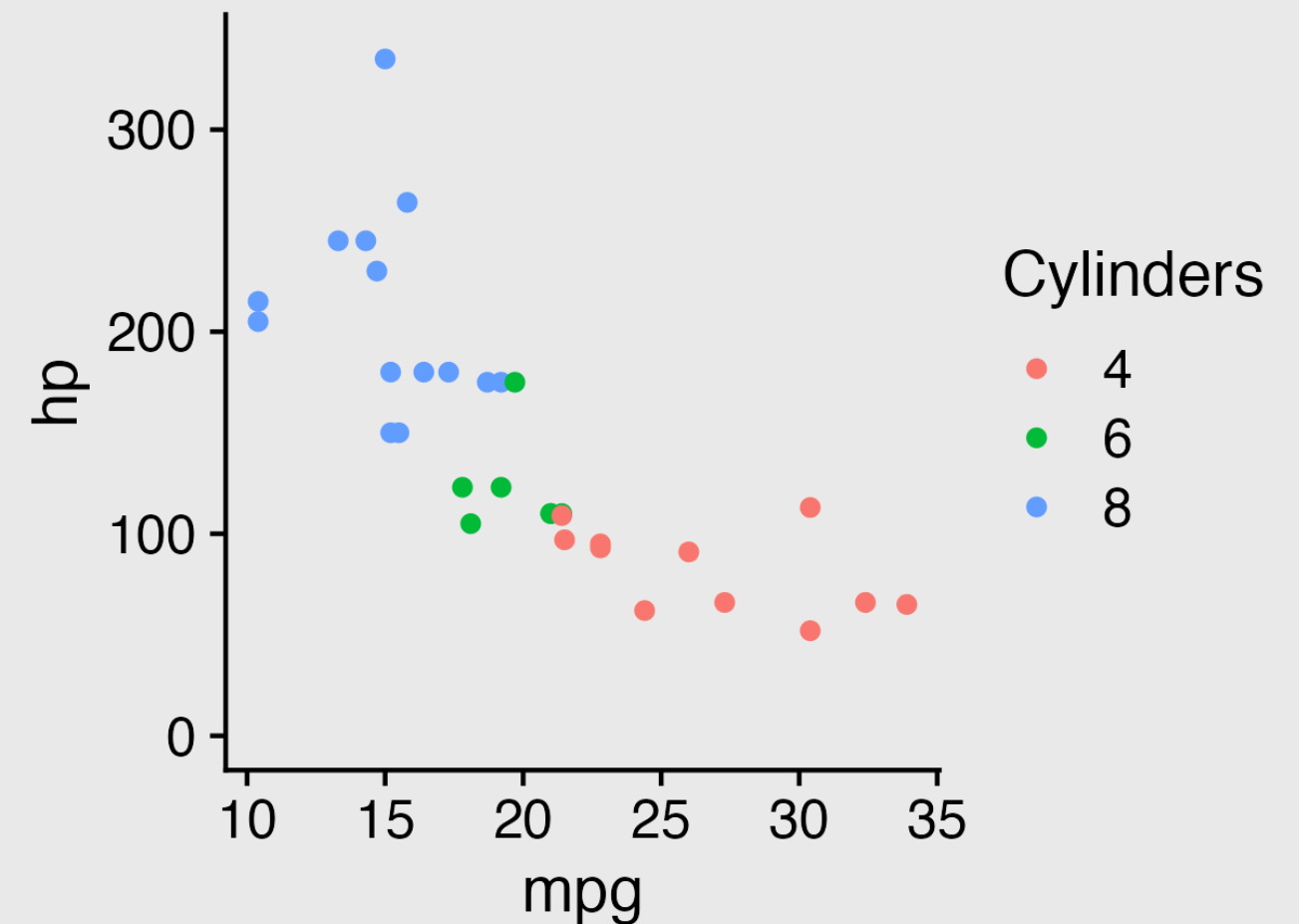
Spatial Position

2-D Position



Pre-attentive attributes

Numerical (ratio) data

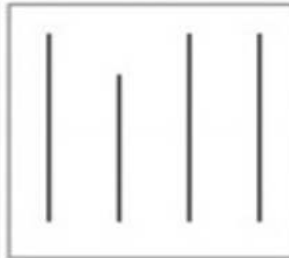


Form

Orientation



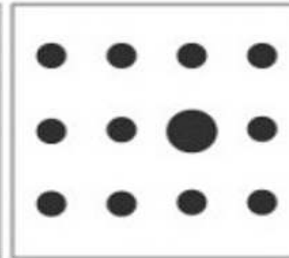
Line Length



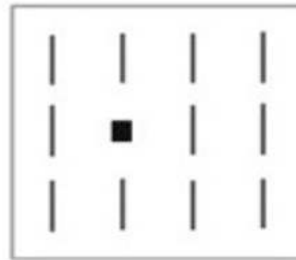
Line Width



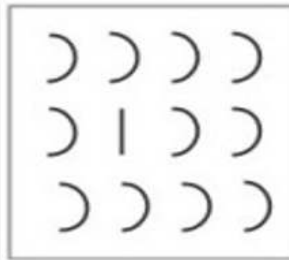
Size



Shape



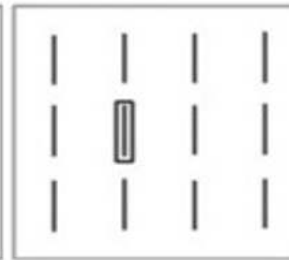
Curvature



Added Marks

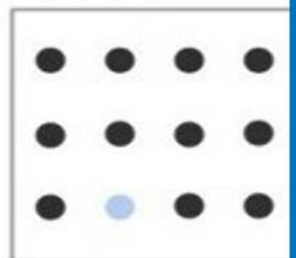


Enclosure

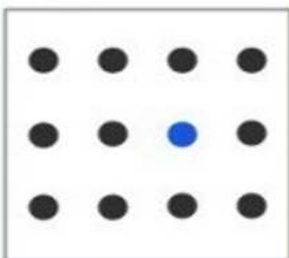


Color

Intensity

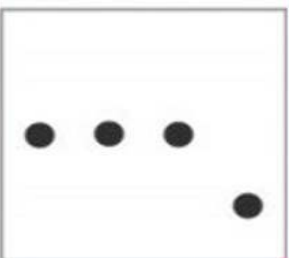


Hue



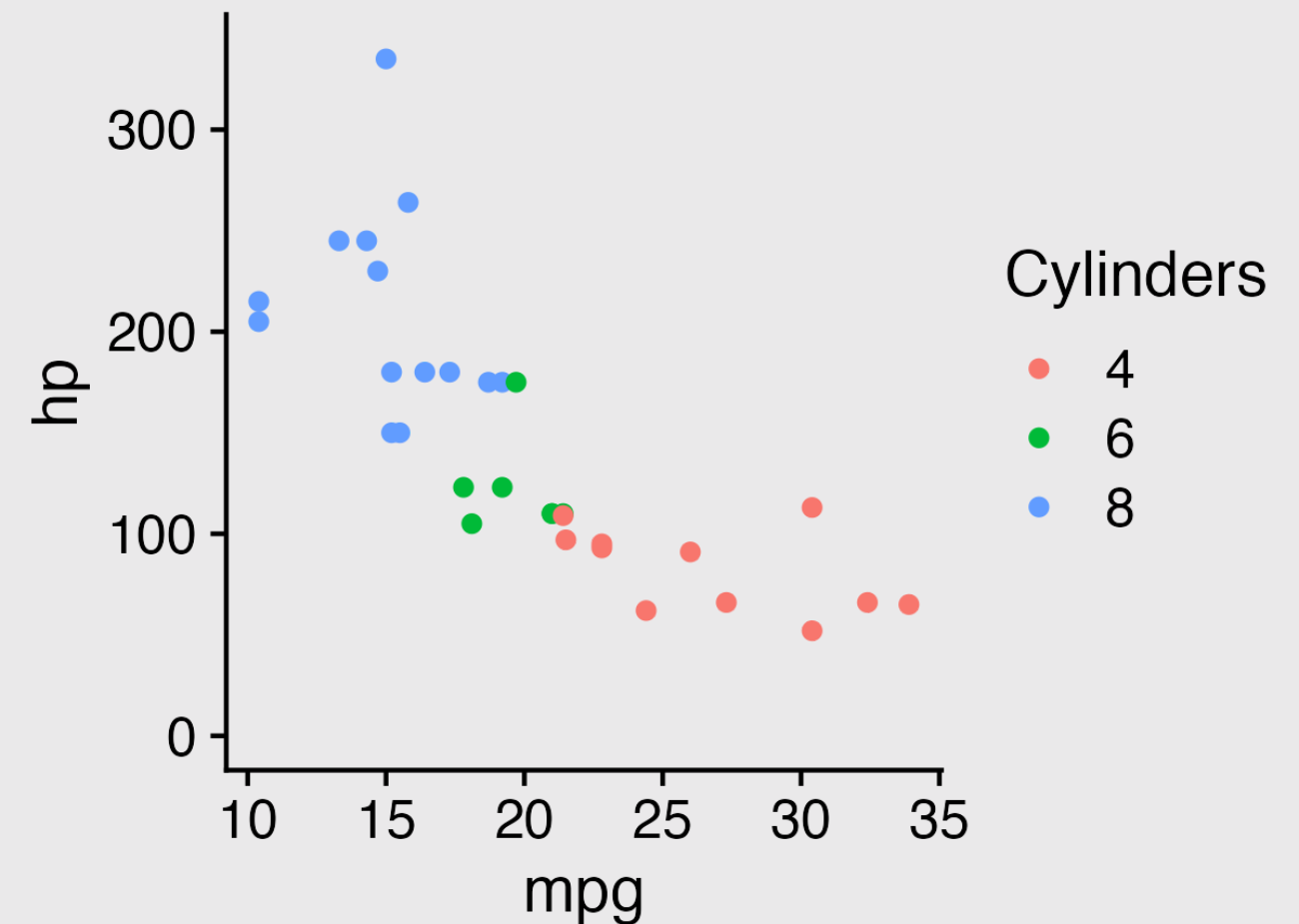
Spatial Position

2-D Position



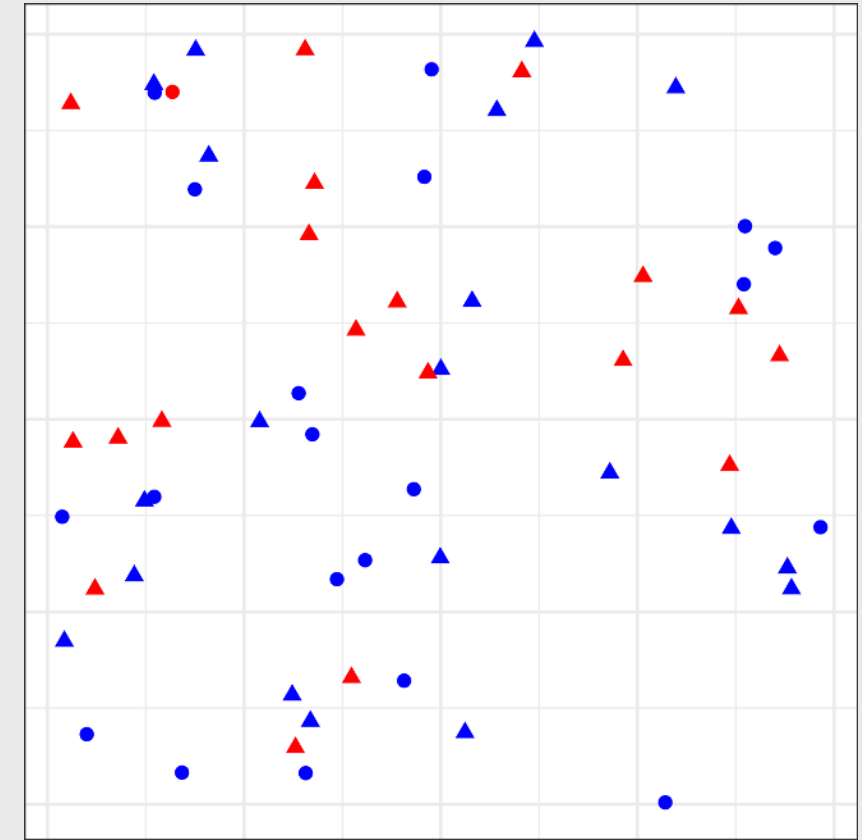
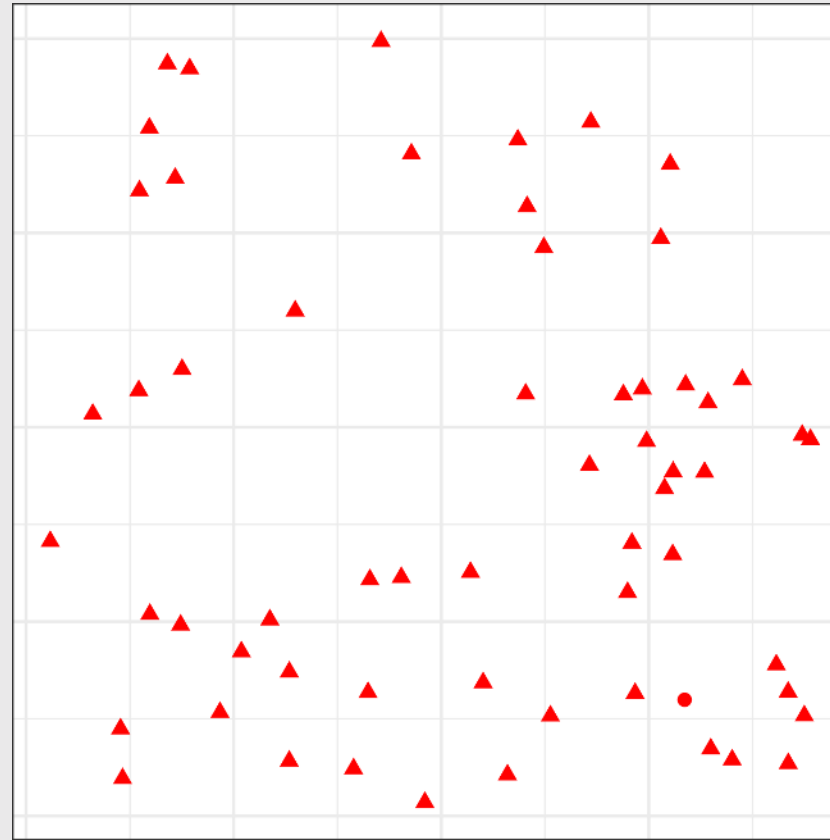
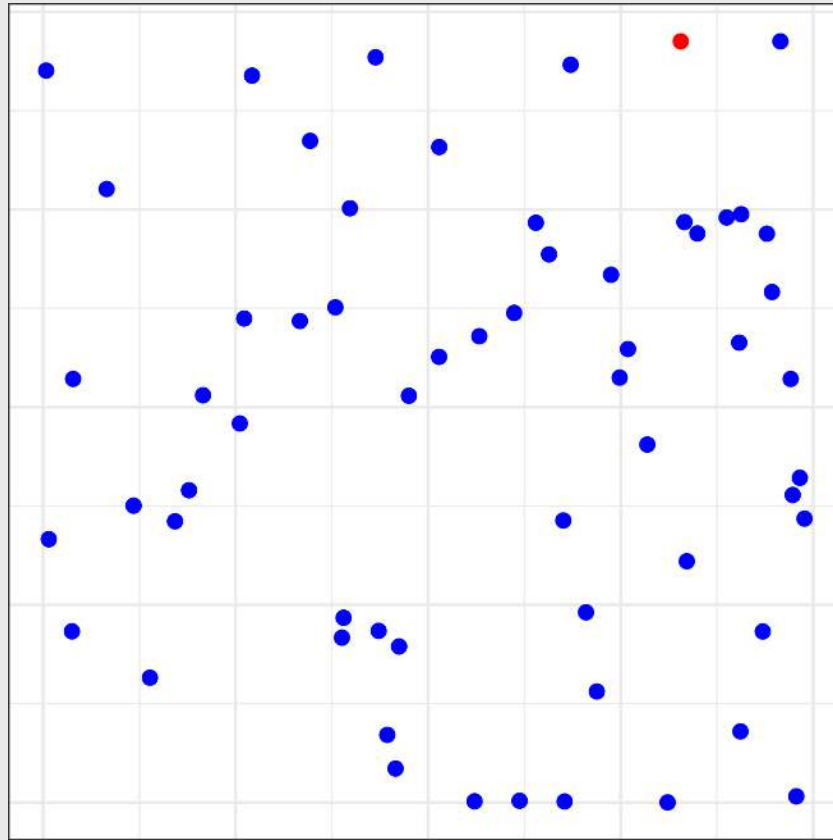
Pre-attentive attributes

Numerical (ratio) data
Categorical (ordinal) data



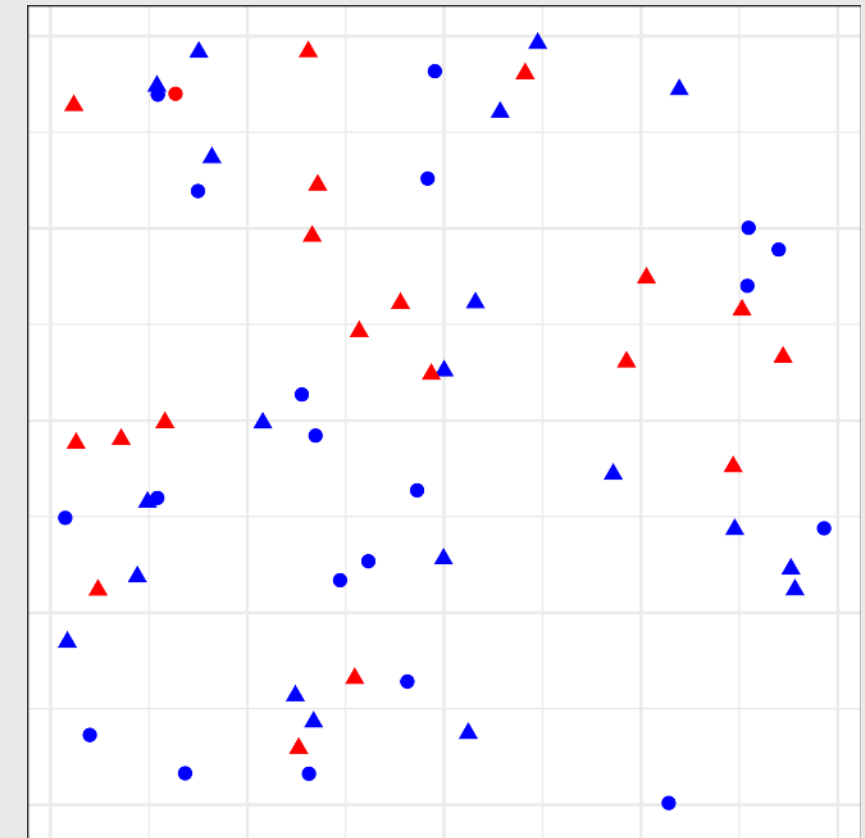
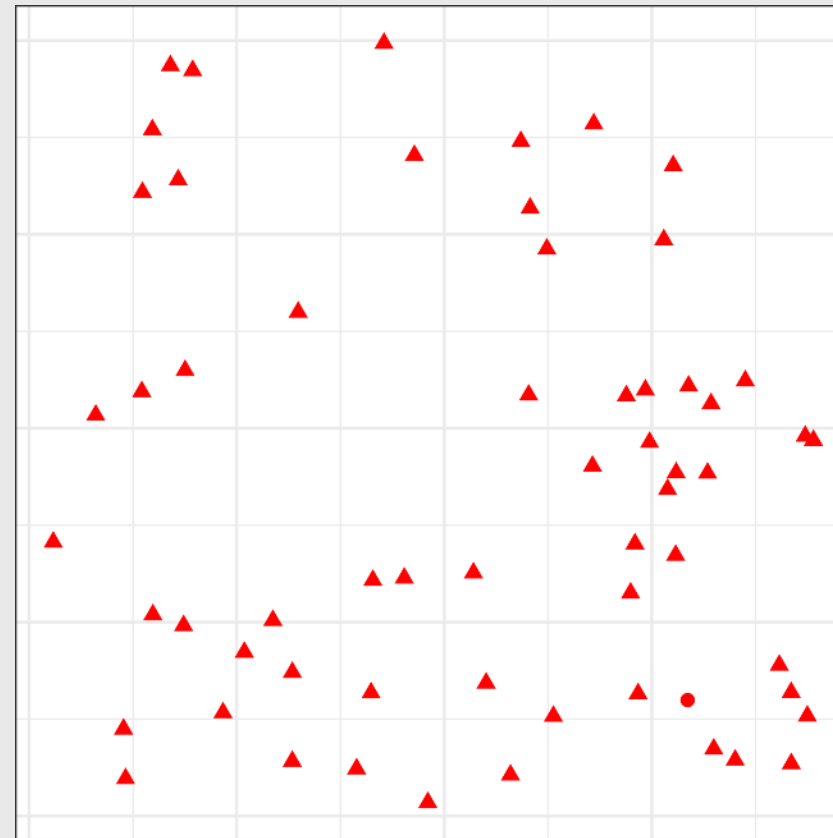
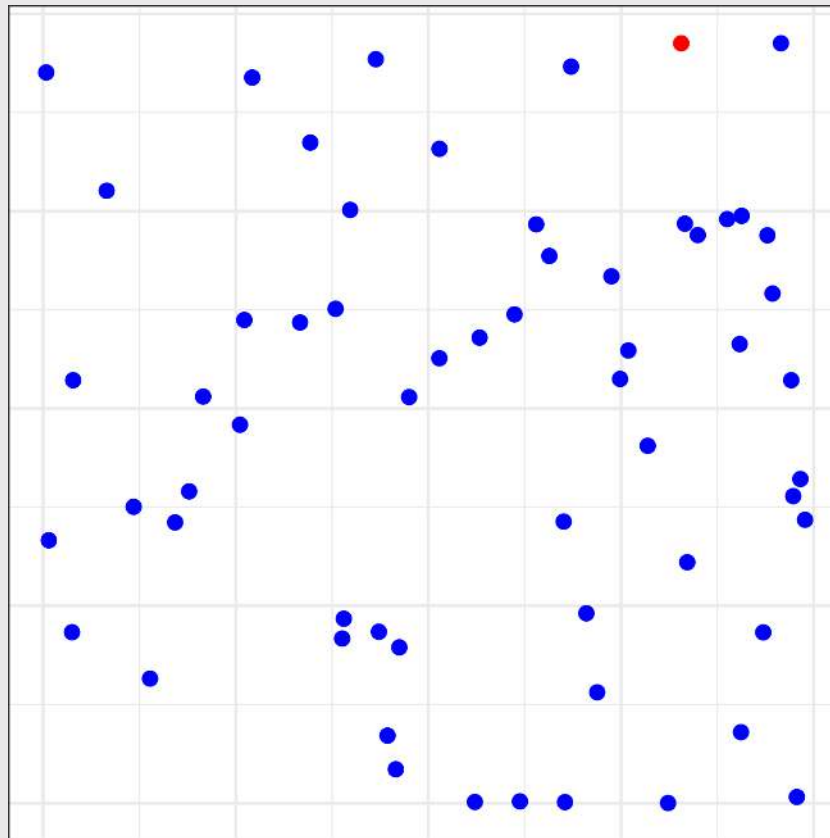
Not all pre-attentive attributes are equal

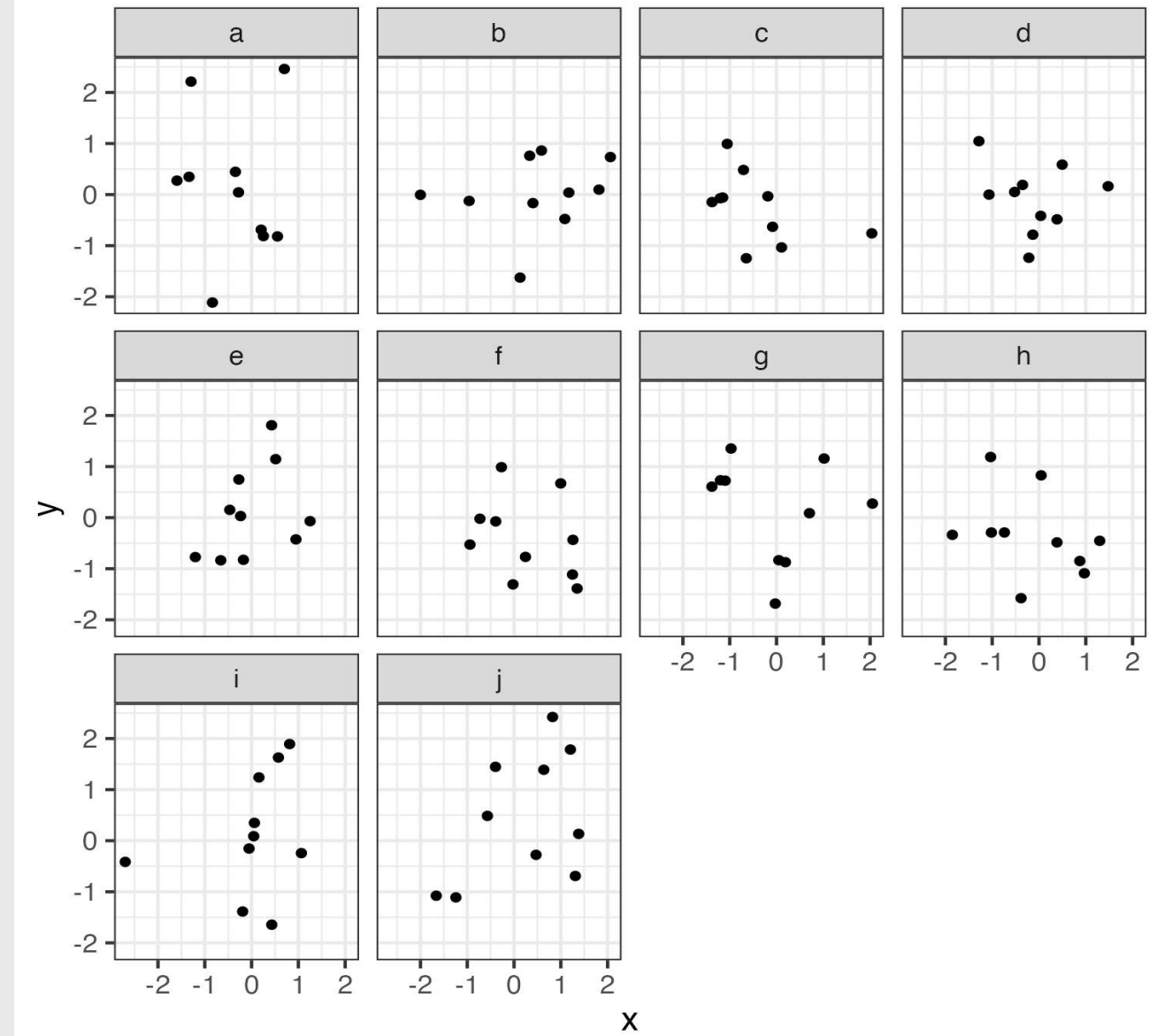
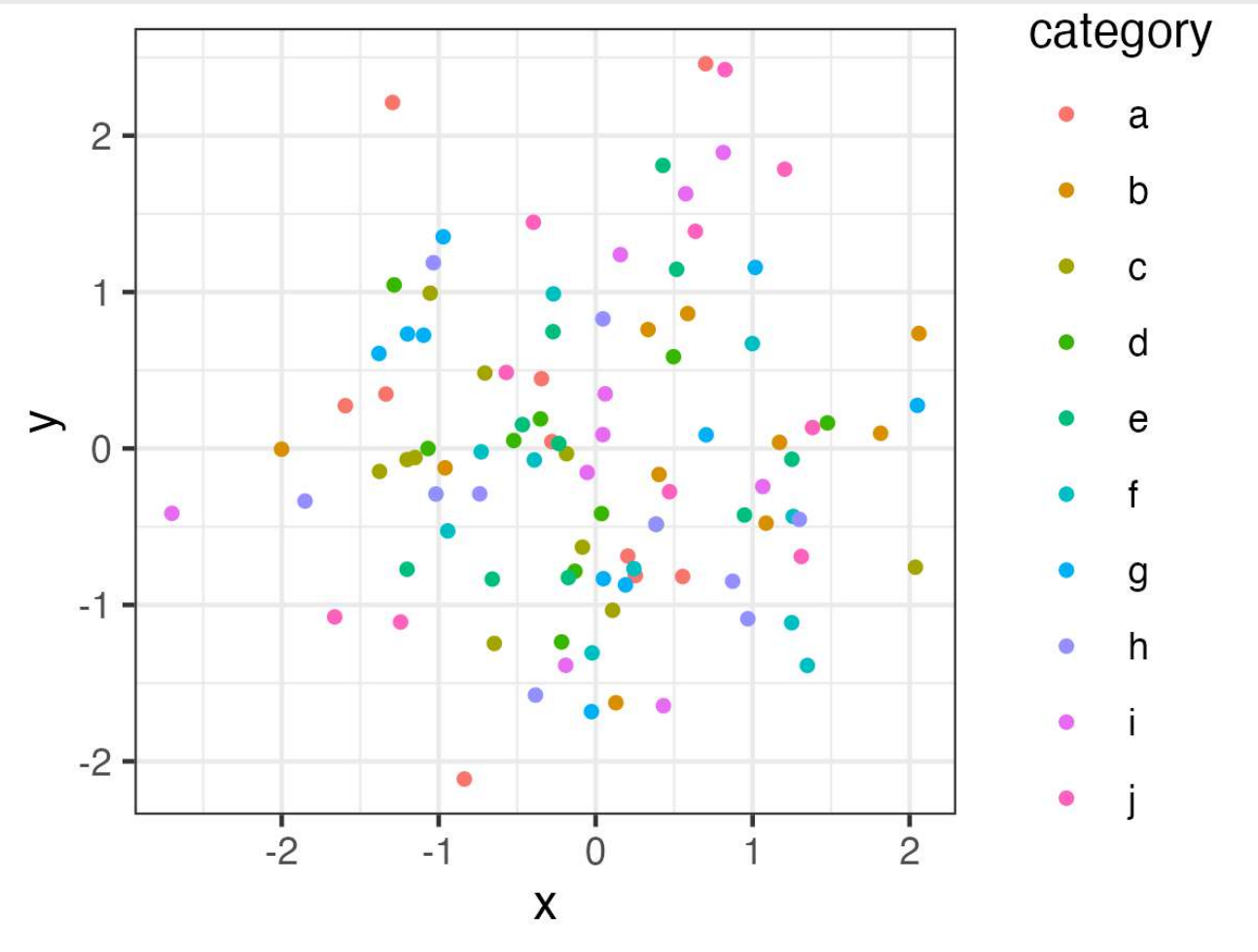
Where is the red dot?

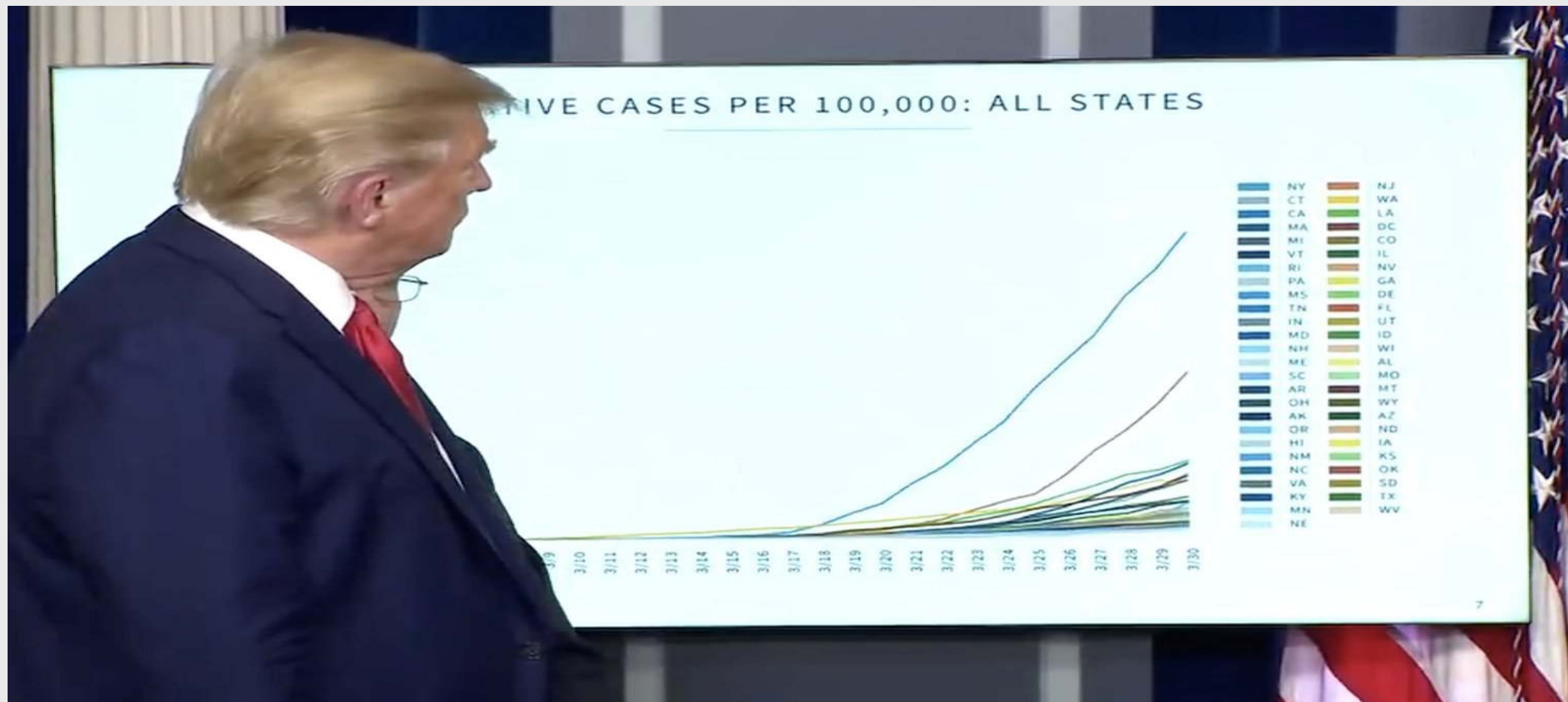


For categorical data:

1. Hue (color) > shape
2. Less is more (stay in working memory!)







Week 7: *Visualizing Information*

1. The Human Visual-Memory System
2. **The Psychology of Data Viz**
3. 10 Data Viz Best Practices
4. Making a (good) ggplot

Much of the content in this section is from
John Rauser's [talk](#) on YouTube

(Always cite your sources)

Graphical Perception and Graphical Methods for Analyzing Scientific Data

William S. Cleveland and Robert McGill

Graphs provide powerful tools both for analyzing scientific data and for communicating quantitative information. The computer graphics revolution, which began in the 1960's and has intensified during the past several years, stimulated the invention of graphical meth-

ods: types of graphs and types of quantitative information to be shown on graphs (1-4). One purpose of this article is to describe and illustrate several of these new methods.

What has been missing, until recently, in this period of rapid graphical invention and deployment is the study of graphs and the human visual system. When a graph is constructed, quantitative and categorical information is encoded, chiefly through position, shape, size, symbols, and color. When a person looks at a graph, the information is visually decoded by the person's visual sys-

tem. The first step is to identify elementary graphical-perception tasks that are used to visually extract quantitative information from a graph. (By "quantitative information" we mean numerical values

formed with greater accuracy. This is illustrated by several examples in which some much-used graphical forms are presented, set aside, and replaced by new methods.

Study of Graphical Perception: Theory

Figure 2 provides an illustration of theoretical reasoning that borrows some ideas from the field of computational vision (8). Suppose that the goal is to judge the ratio, r , of the slope of line segment BC to the slope of line segment AB in each of the three panels. Our visual system tells us that r is greater than 1 in each panel, which is correct. Our visual system also tells us that r is closer to 1 in the two rectangular panels than in the square panel; that is, the slope of BC appears closer to the slope of AB in the two rectangular panels than in the square panel. This, however, is incorrect; r is the same in all three panels.

The reason for the distortion in judging Fig. 2 is that our visual system is geared to judging angle rather than slope. In their work on computational theories of vision in artificial intelligence, Marr (8) and Stevens (9) have investigated how people judge the slant and tilt (10) of the surfaces of three-dimensional objects. They argue that we judge slant and tilt as

Cleveland, W. S., & McGill, R. (1985). Graphical perception and graphical methods for analyzing scientific data. *Science, New Series*, 229(4716), 828-833.

Cleveland's operations of pattern perception:

1. Estimation
2. Assembly
3. Detection

Cleveland's operations of pattern perception:

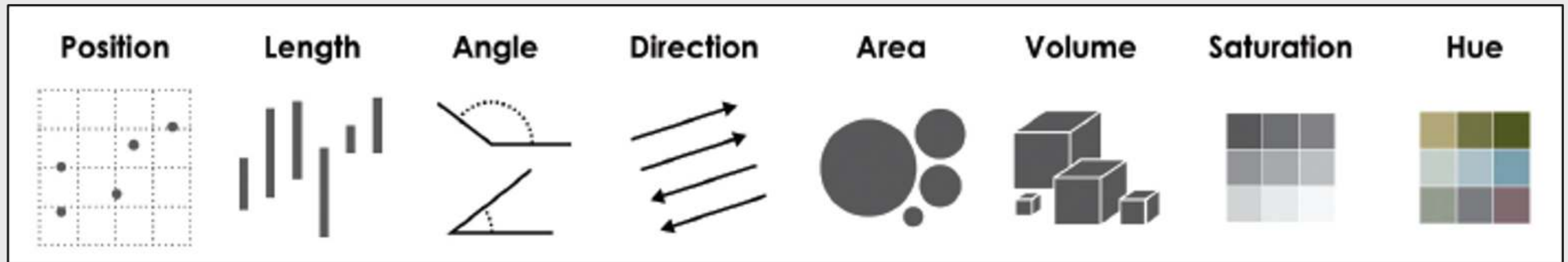
1. **Estimation** ———>

2. Assembly

3. Detection

- **Discrimination** (X equal to Y?)
- **Ranking** (X greater than Y?)
- **Ratioing** (X double Y?)

Estimation: Hierarchy for *numerical* data



More Accurate

Less Accurate

Example: Life expectancy in countries in Asia

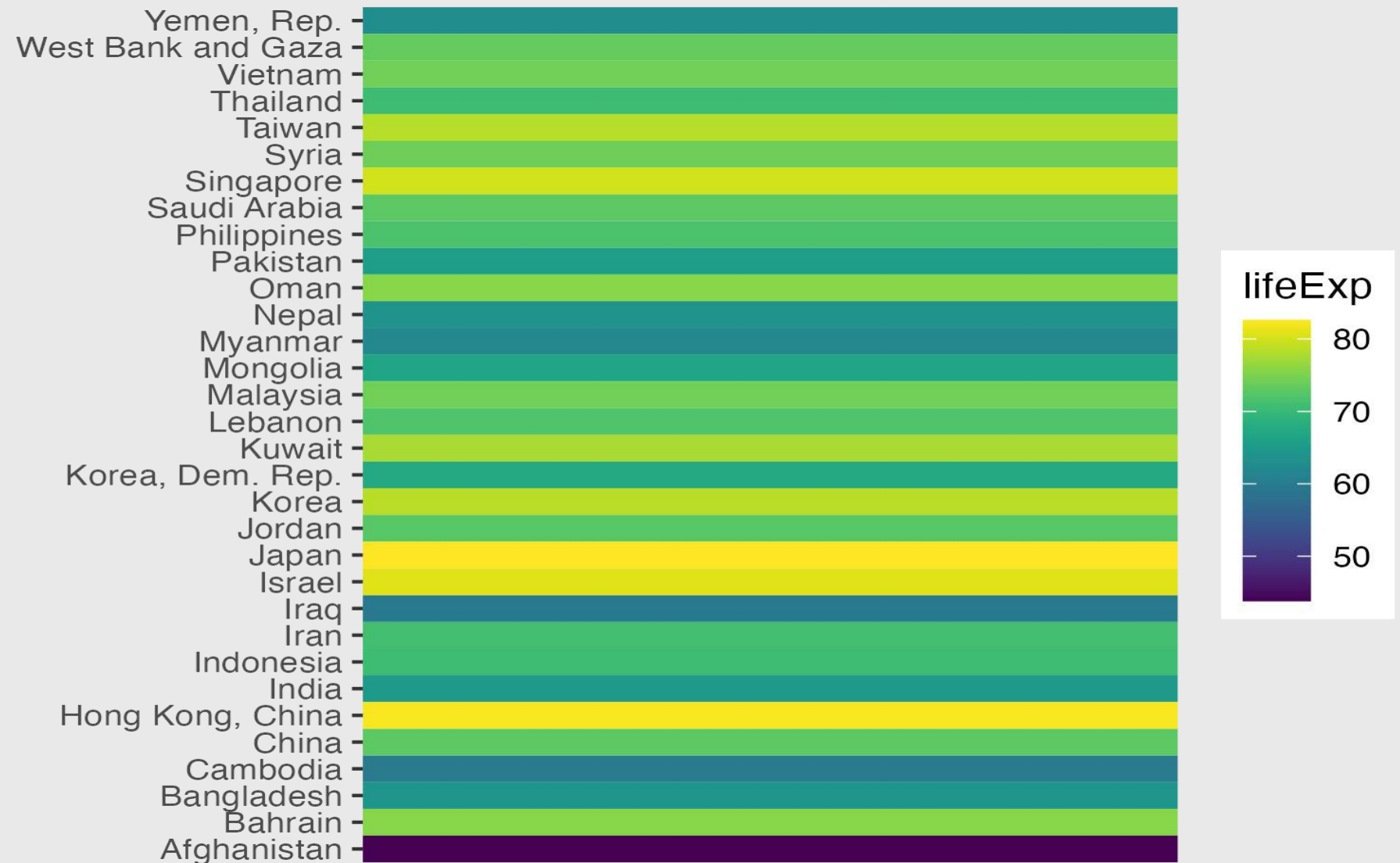
```
#>      country lifeExp
#> 1  Afghanistan 43.828
#> 2      Iraq 59.545
#> 3  Cambodia 59.723
#> 4  Myanmar 62.069
#> 5  Yemen, Rep. 62.698
#> 6      Nepal 63.785
#> 7  Bangladesh 64.062
#> 8      India 64.698
#> 9  Pakistan 65.483
#> 10  Mongolia 66.803
#> 11 Korea, Dem. Rep. 67.297
#> 12  Thailand 70.616
```

1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. Angle
5. Area
6. Color saturation
7. **Color hue**



1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. Angle
5. Area
6. Color saturation
7. **Color hue**

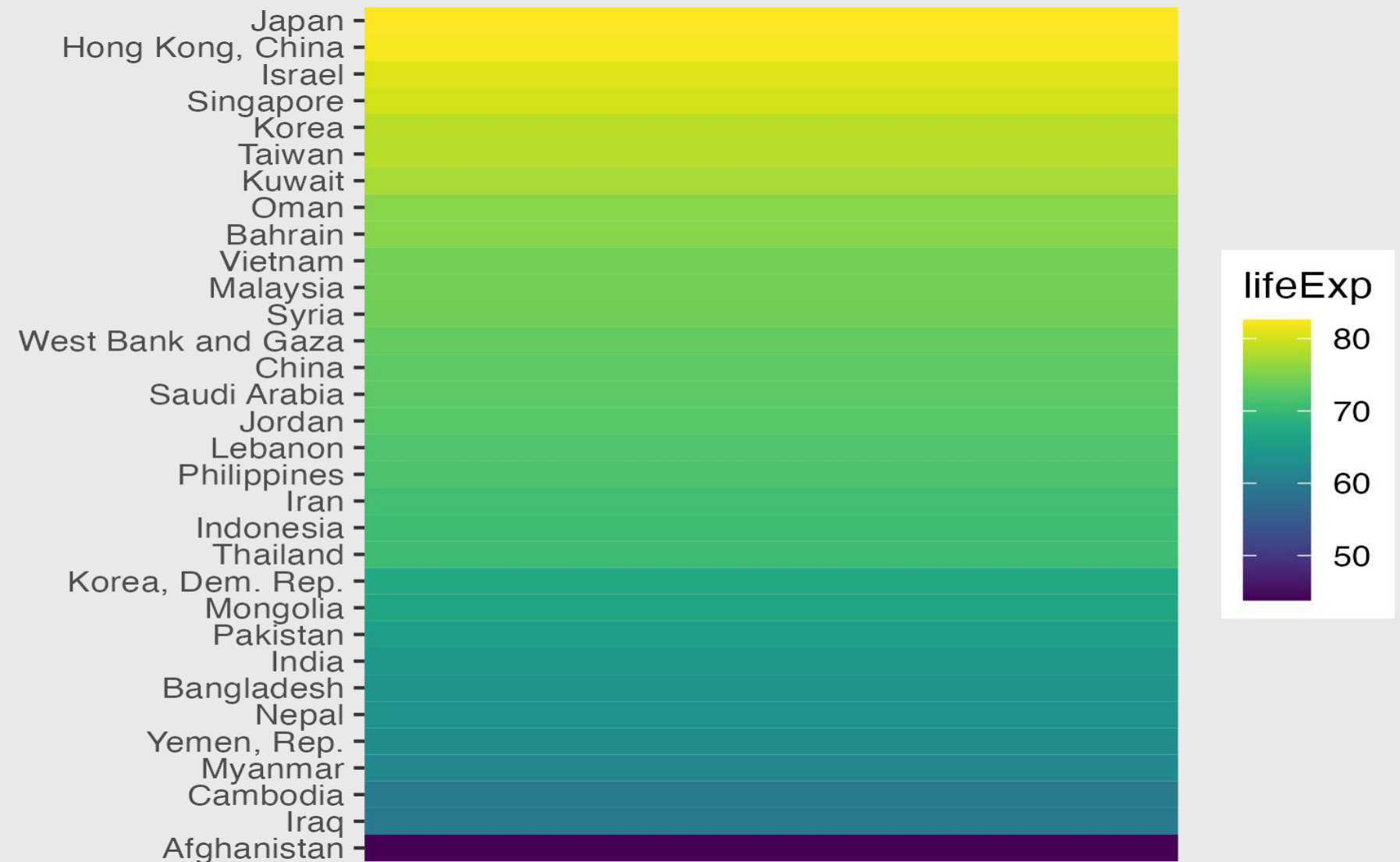
-  /  **Discriminate**
-  /  **Rank**
-  **Ratio**



1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. Angle
5. Area
6. Color saturation
7. **Color hue**

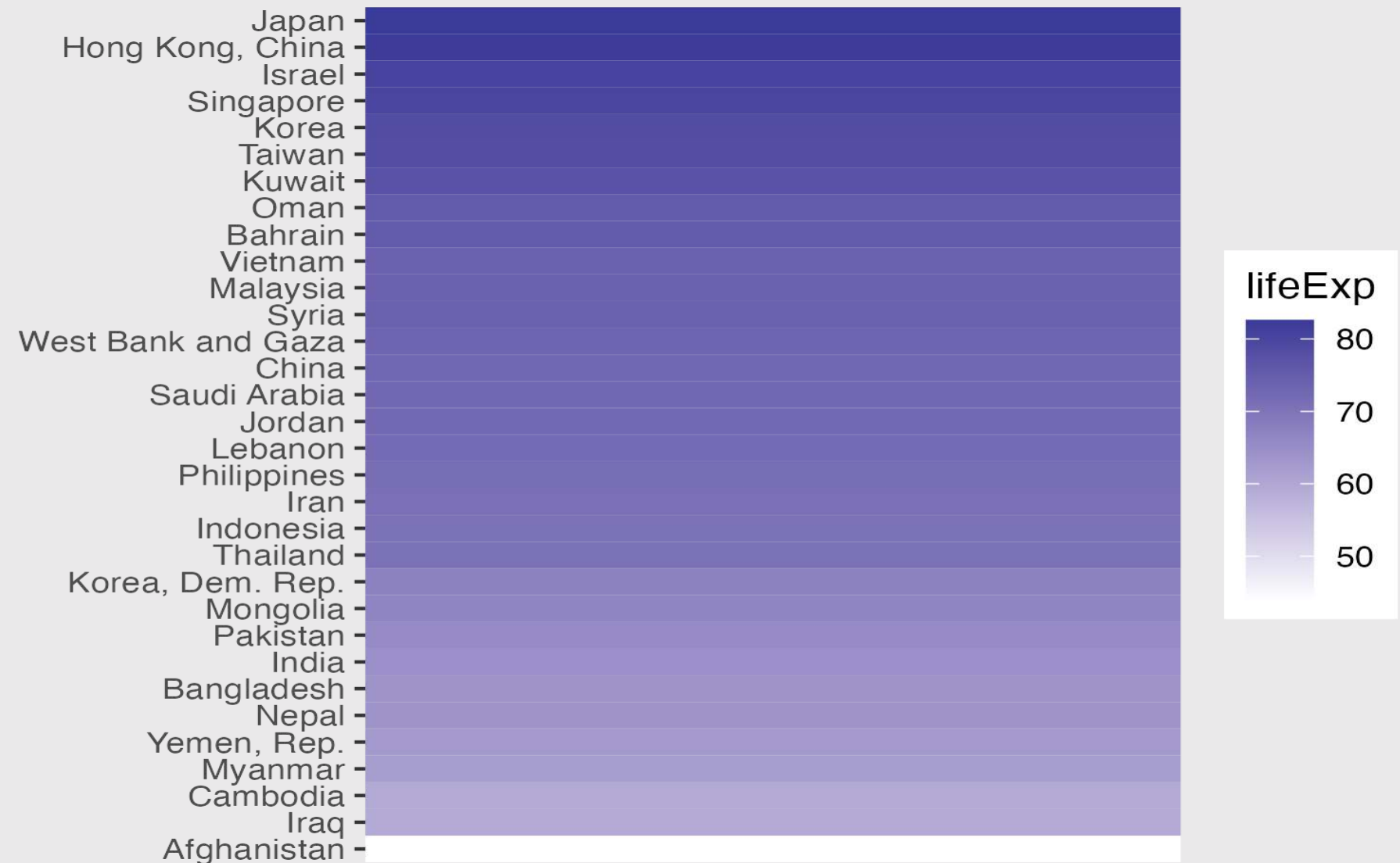
Sorting helps a bit...

-  /  **Discriminate**
-  /  **Rank**
-  **Ratio**



1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. Angle
5. Area
6. **Color saturation**
7. Color hue

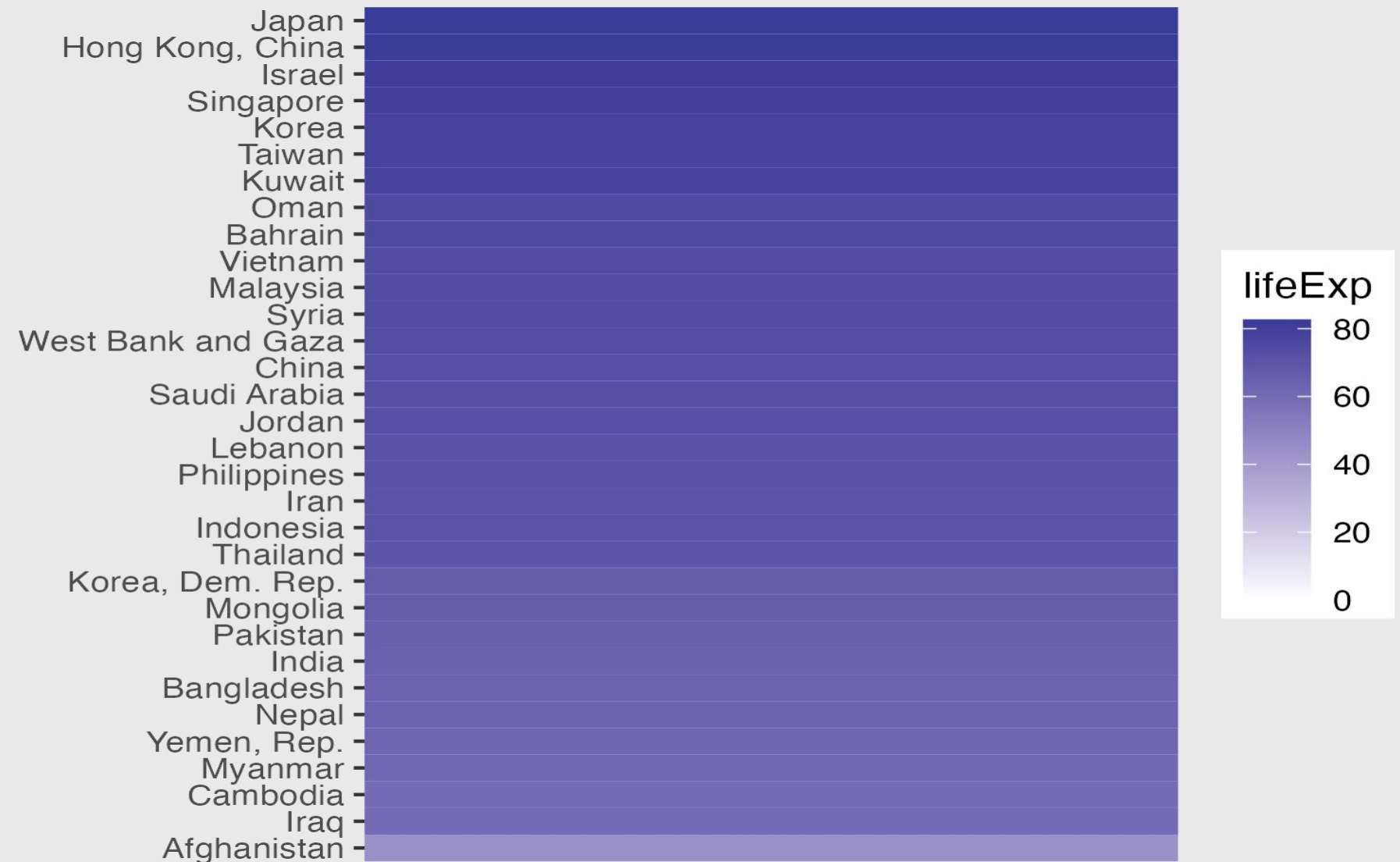
-  /  **Discriminate**
-  /  **Rank**
-  **Ratio**



1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. Angle
5. Area
6. **Color saturation**
7. Color hue

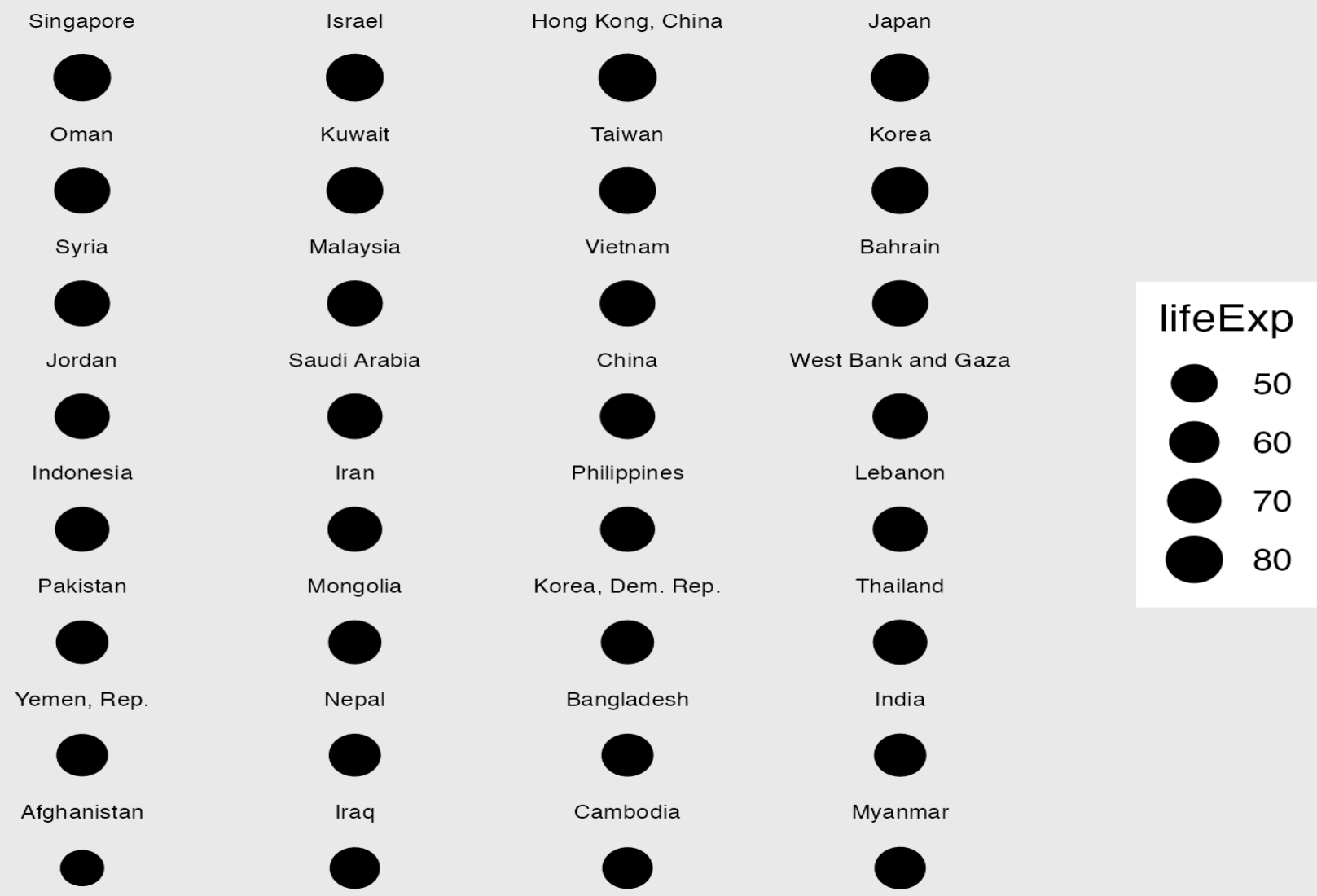
Align to 0 scale:

- ☒ / ☒ **Discriminate**
- ☒ / ☒ **Rank**
- ☒ / ☒ **Ratio**



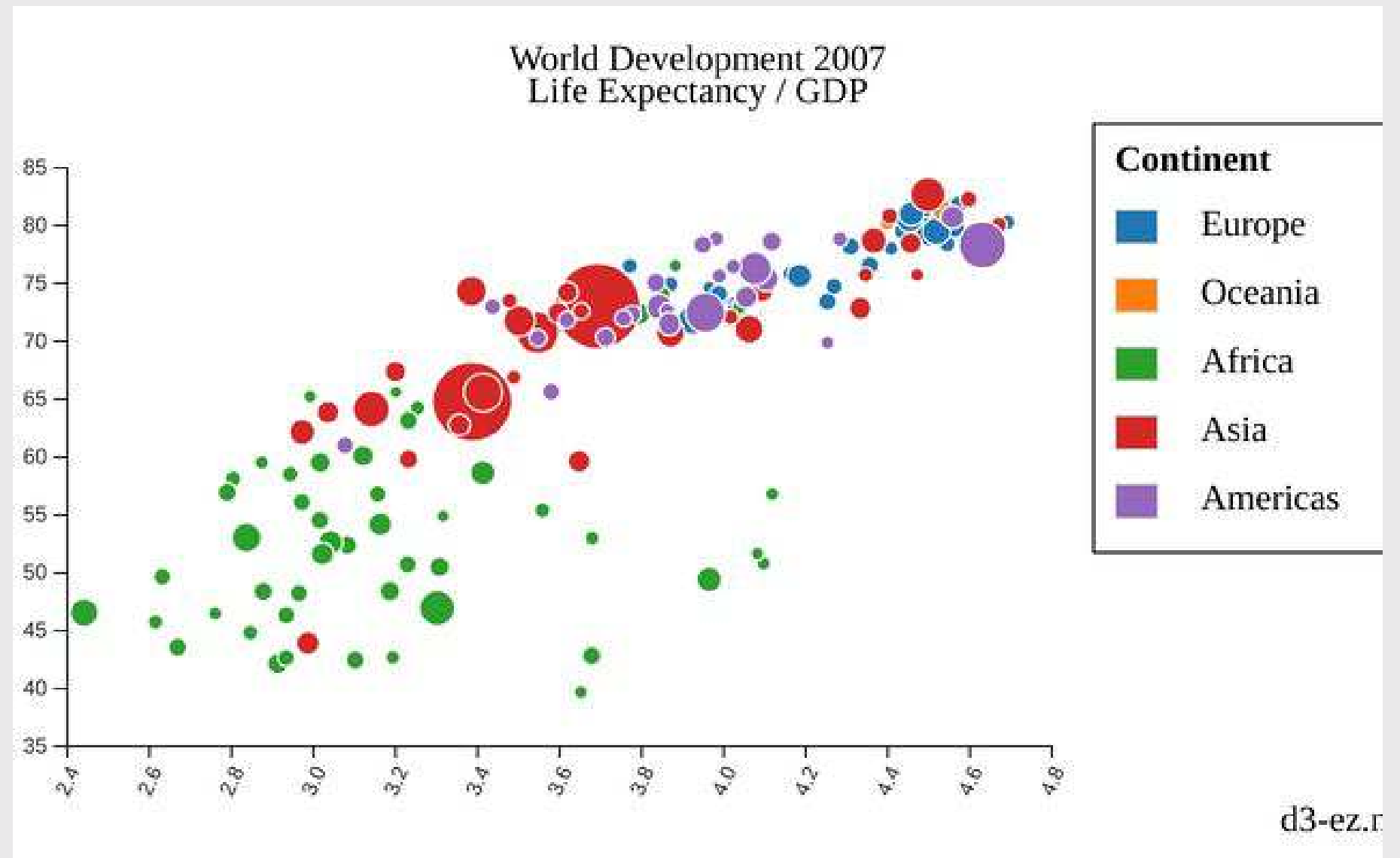
- 1. Position on a common scale
- 2. Position on non-aligned scales
- 3. Length
- 4. Angle
- 5. **Area**
- 6. Color saturation
- 7. Color hue

-  /  **Discriminate**
-  /  **Rank**
-  /  **Ratio**



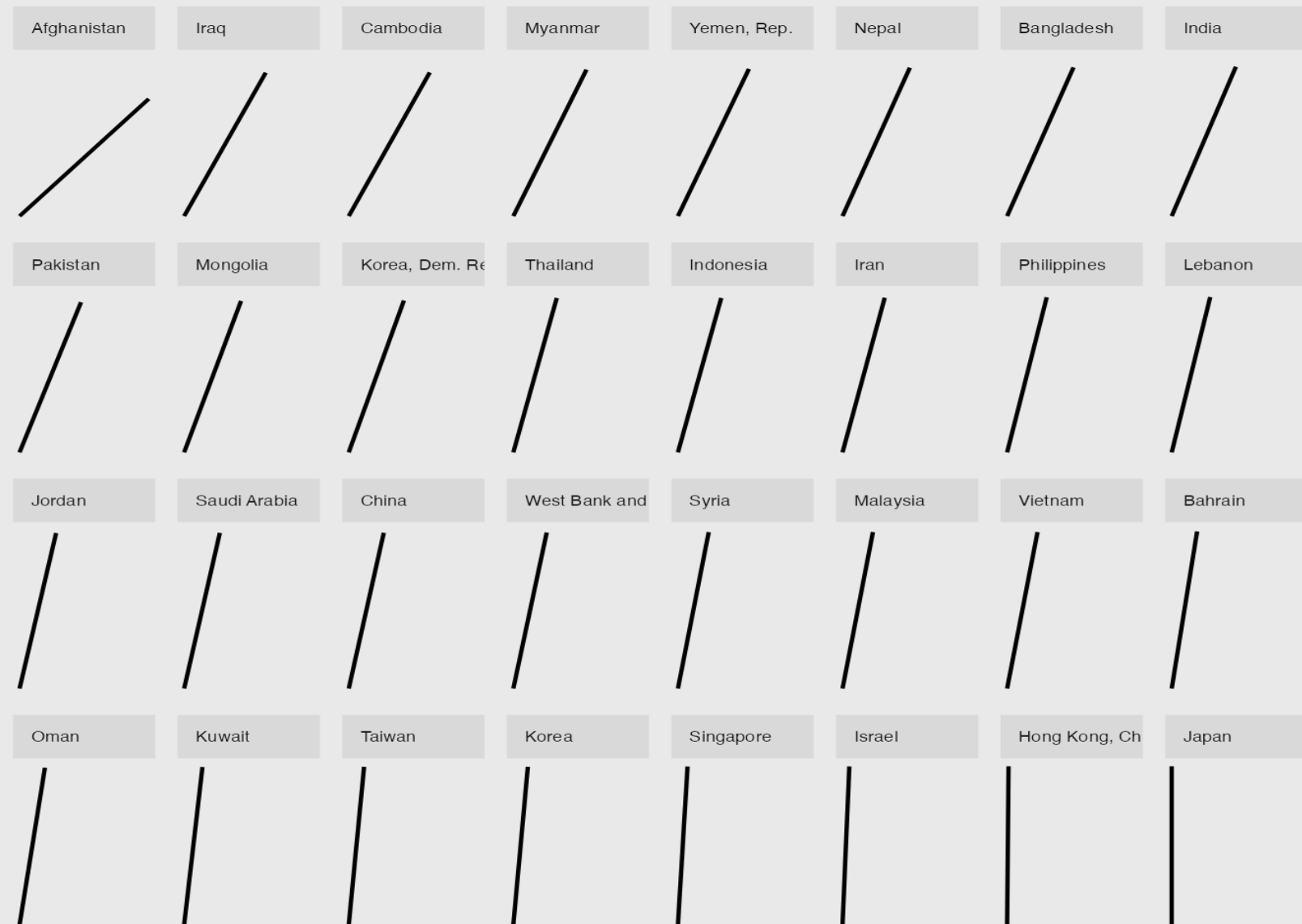
1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. Angle
5. **Area**
6. Color saturation
7. Color hue

Area works okay for “bubble” charts

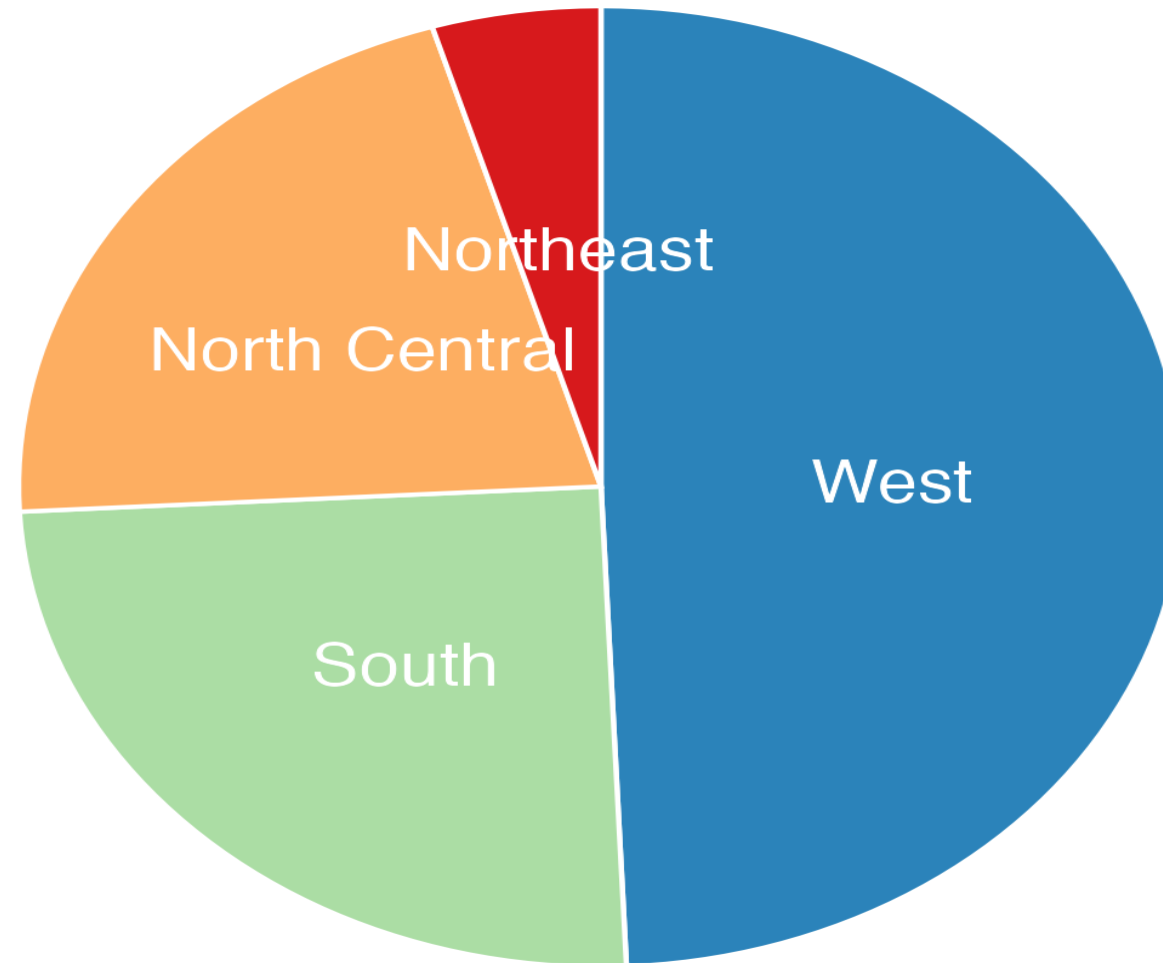


1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. **Angle**
5. Area
6. Color saturation
7. Color hue

- ☒ / ☒ **Discriminate**
- ☒ **Rank**
- ☒ / ☒ **Ratio**



1. Position on a common scale
2. Position on non-aligned scales
3. Length
4. **Angle**
5. Area
6. Color saturation
7. Color hue



1. Position on a common scale
2. Position on non-aligned scales
3. **Length**
4. Angle
5. Area
6. Color saturation
7. Color hue

-  /  **Discriminate**
-  /  **Rank**
-  **Ratio**



1. Position on a common scale

2. **Position on non-aligned scales**

3. Length

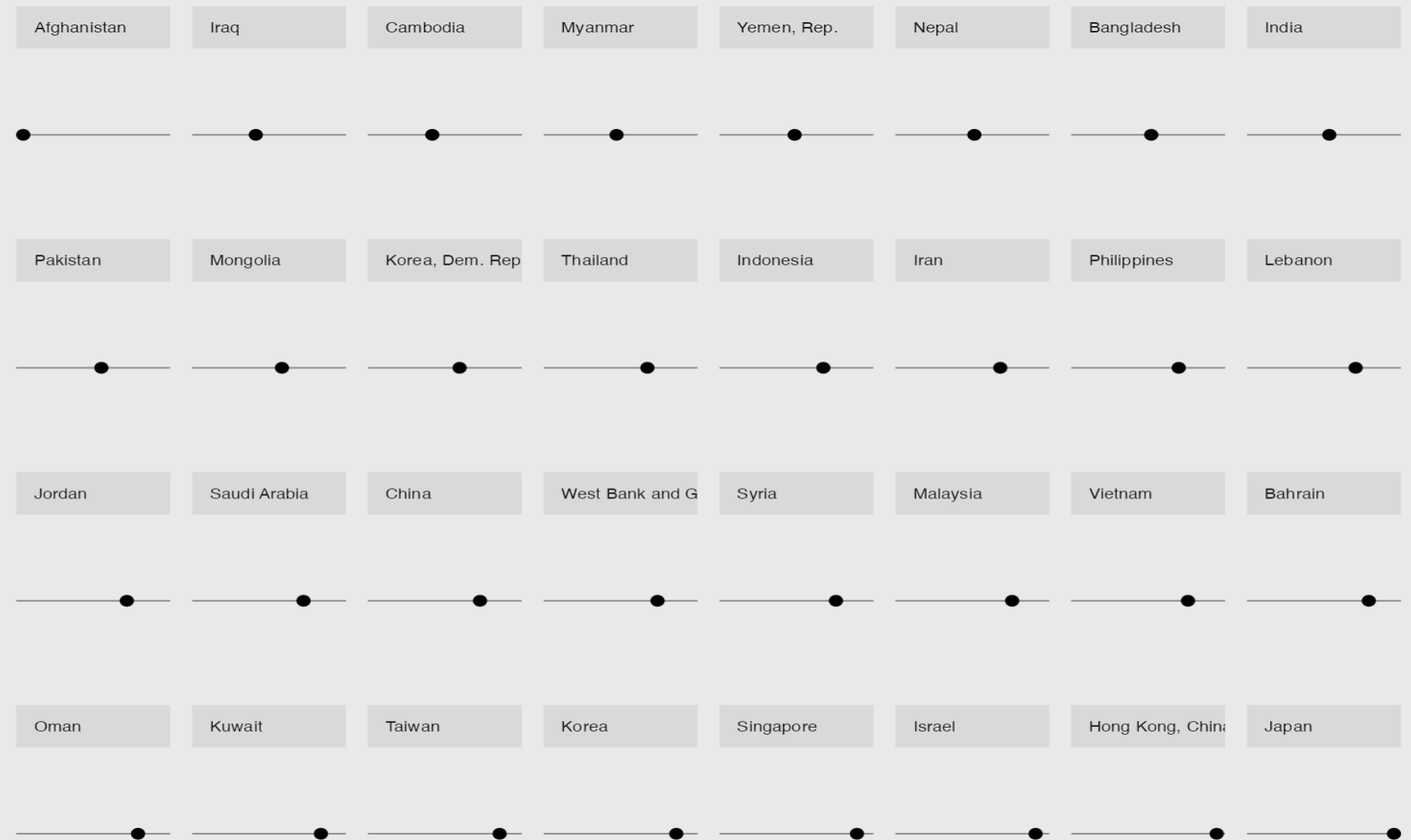
4. Angle

5. Area

6. Color saturation

7. Color hue

- ☒ / ☒ **Discriminate**
- ☒ / ☒ **Rank**
- ☒ **Ratio**



1. **Position on a common scale**

2. Position on non-aligned scales

3. Length

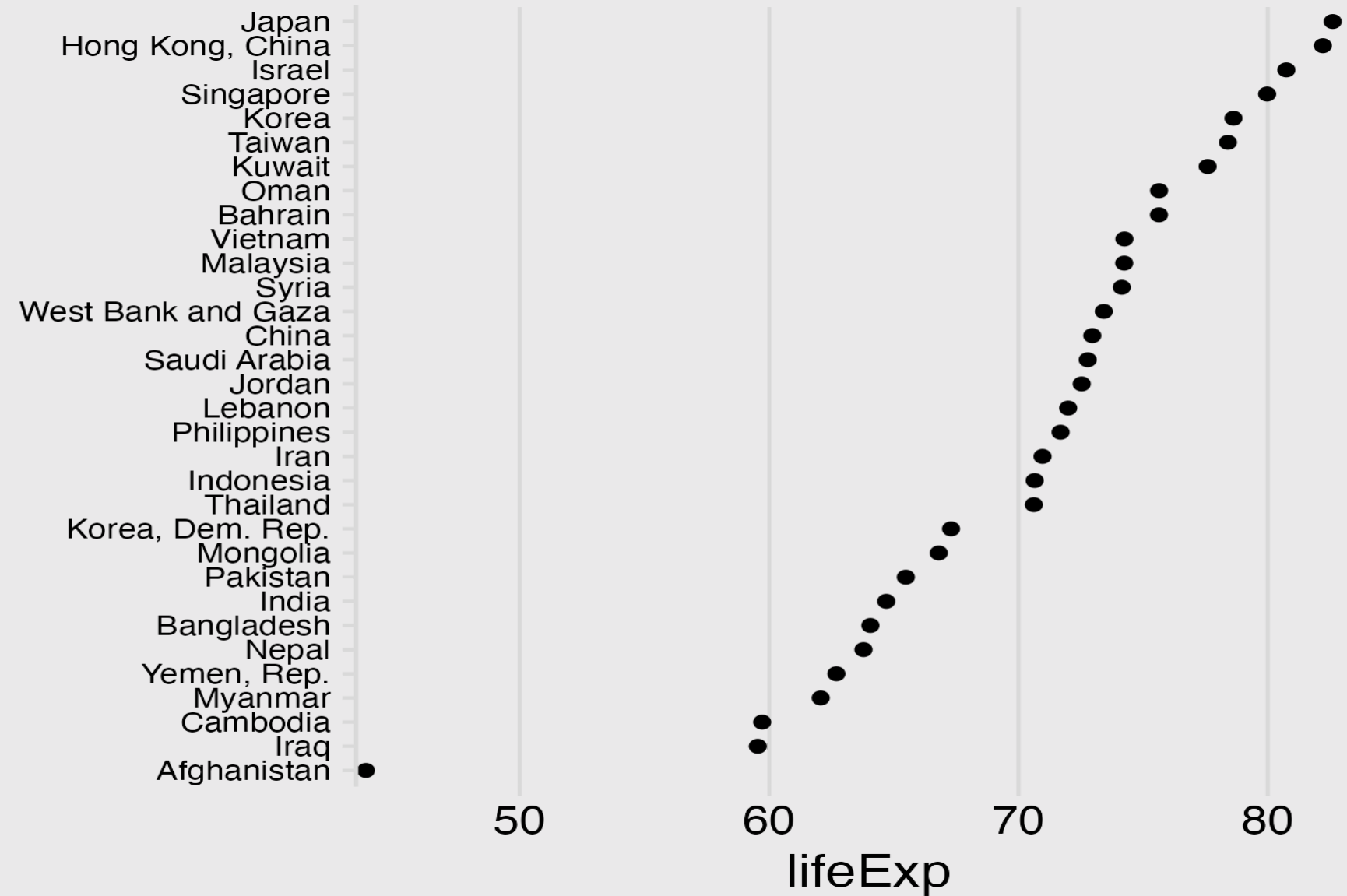
4. Angle

5. Area

6. Color saturation

7. Color hue

- ☒ **Discriminate**
- ☒ **Rank**
- ☒ **Ratio**



1. **Position on a common scale**

2. Position on non-aligned scales

3. Length

4. Angle

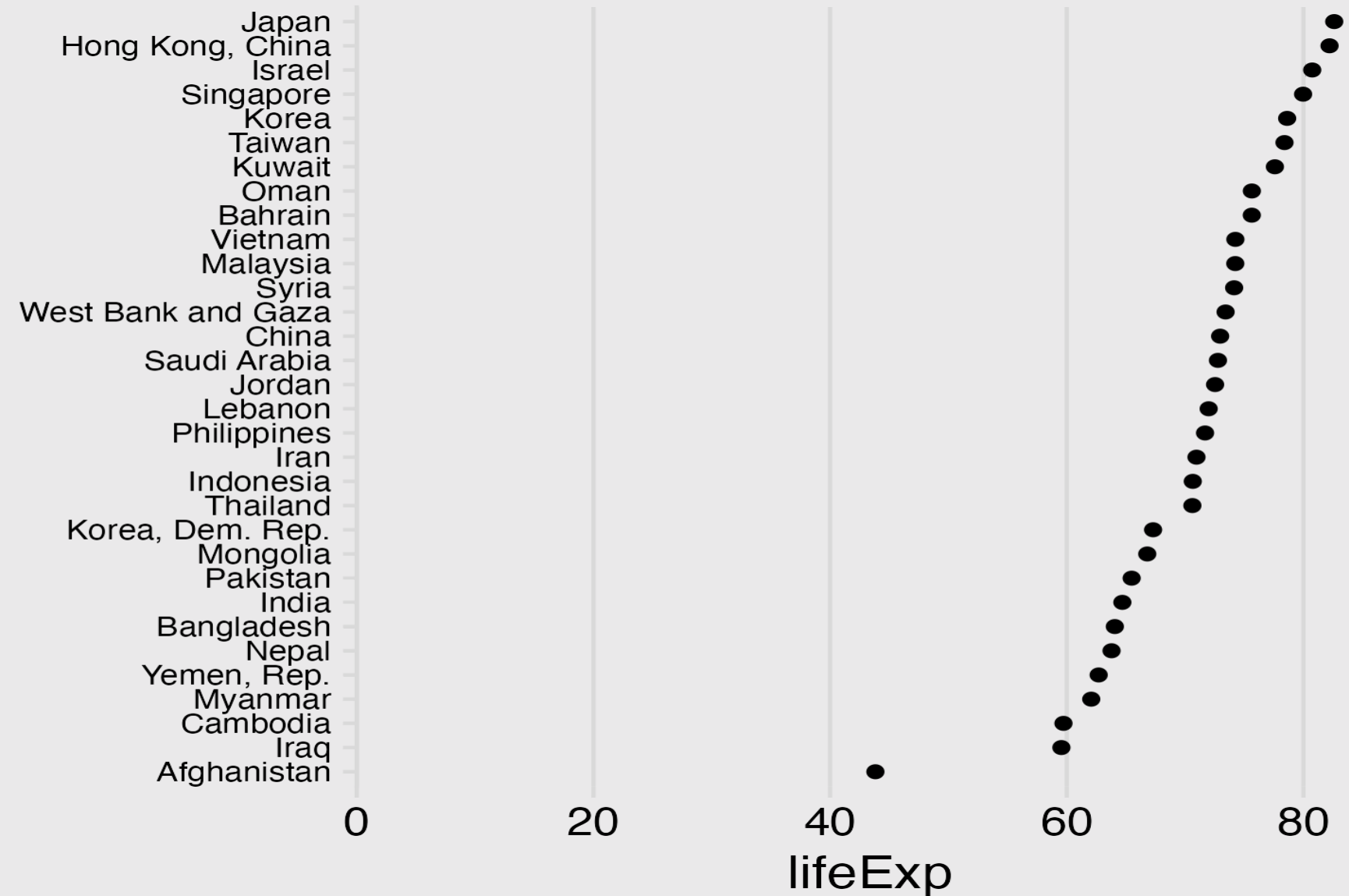
5. Area

6. Color saturation

7. Color hue

No need to scale to 0:

- Lowers resolution
- Isn't needed for accurate ratioing



1. **Position on a common scale**

2. Position on
non-aligned scales

3. Length

4. Angle

5. Area

6. Color saturation

7. Color hue

Sorting still matters!

Yemen, Rep.
West Bank and Gaza
Vietnam
Thailand
Taiwan
Syria
Singapore
Saudi Arabia
Philippines
Pakistan
Oman
Nepal
Myanmar
Mongolia
Malaysia
Lebanon
Kuwait
Korea, Dem. Rep.
Korea
Jordan
Japan
Israel
Iraq
Iran
Indonesia
India
Hong Kong, China
China
Cambodia
Bangladesh
Bahrain
Afghanistan

50

60

70

80

lifeExp

Cleveland's operations of pattern perception:

1. Estimation
2. Assembly
3. Detection

Cleveland's operations of pattern perception:

1. Estimation

2. **Assembly** ———> The grouping of graphical elements

3. Detection

Assembly: Gestalt Psychology

The whole has a reality that is entirely separate from the parts



WWF

Reification

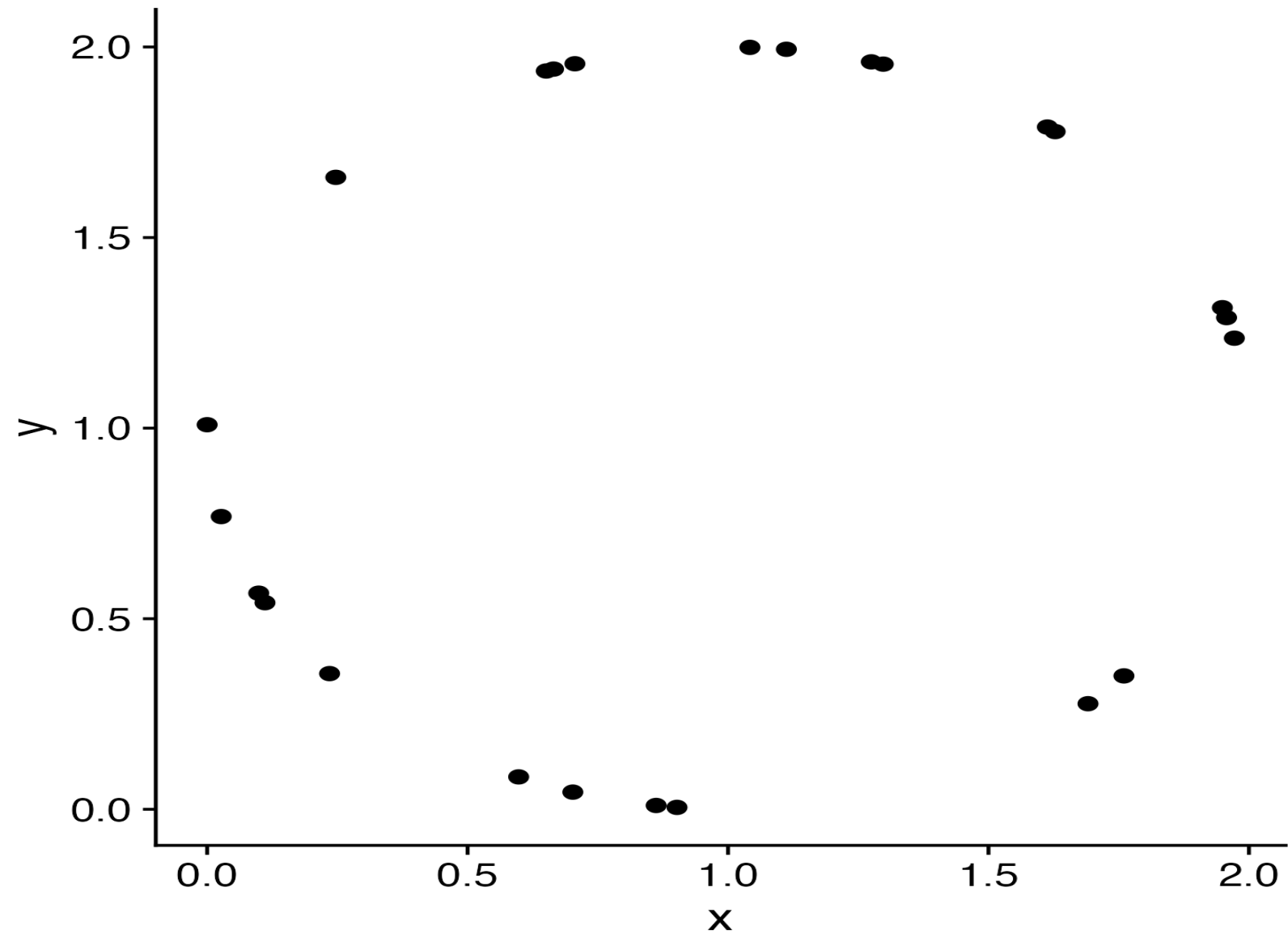


Emergence



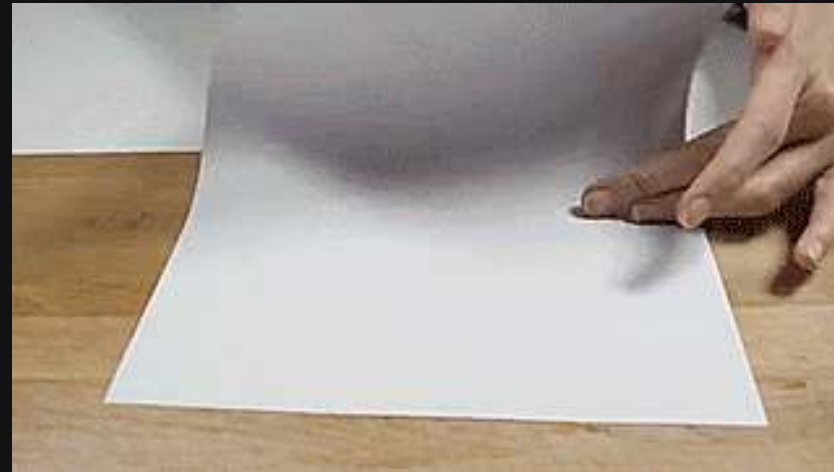
Law of Closure

Our minds fill in the missing information



Prägnanz

We strongly prefer to interpret stimuli as regular, simple, and orderly



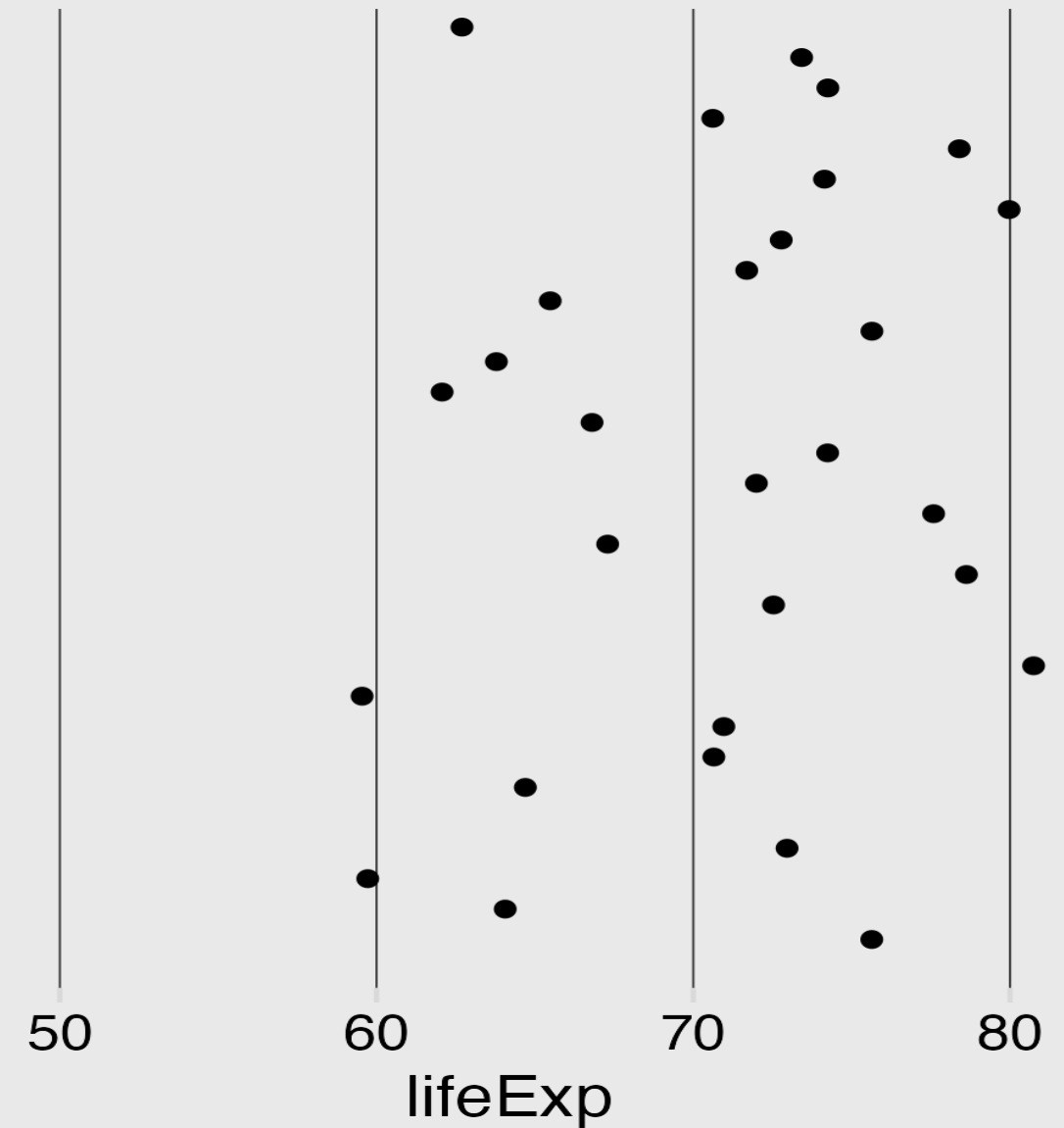
Prägnanz

This should cause
you cognitive pain

It's the graphical
equivalent of this:

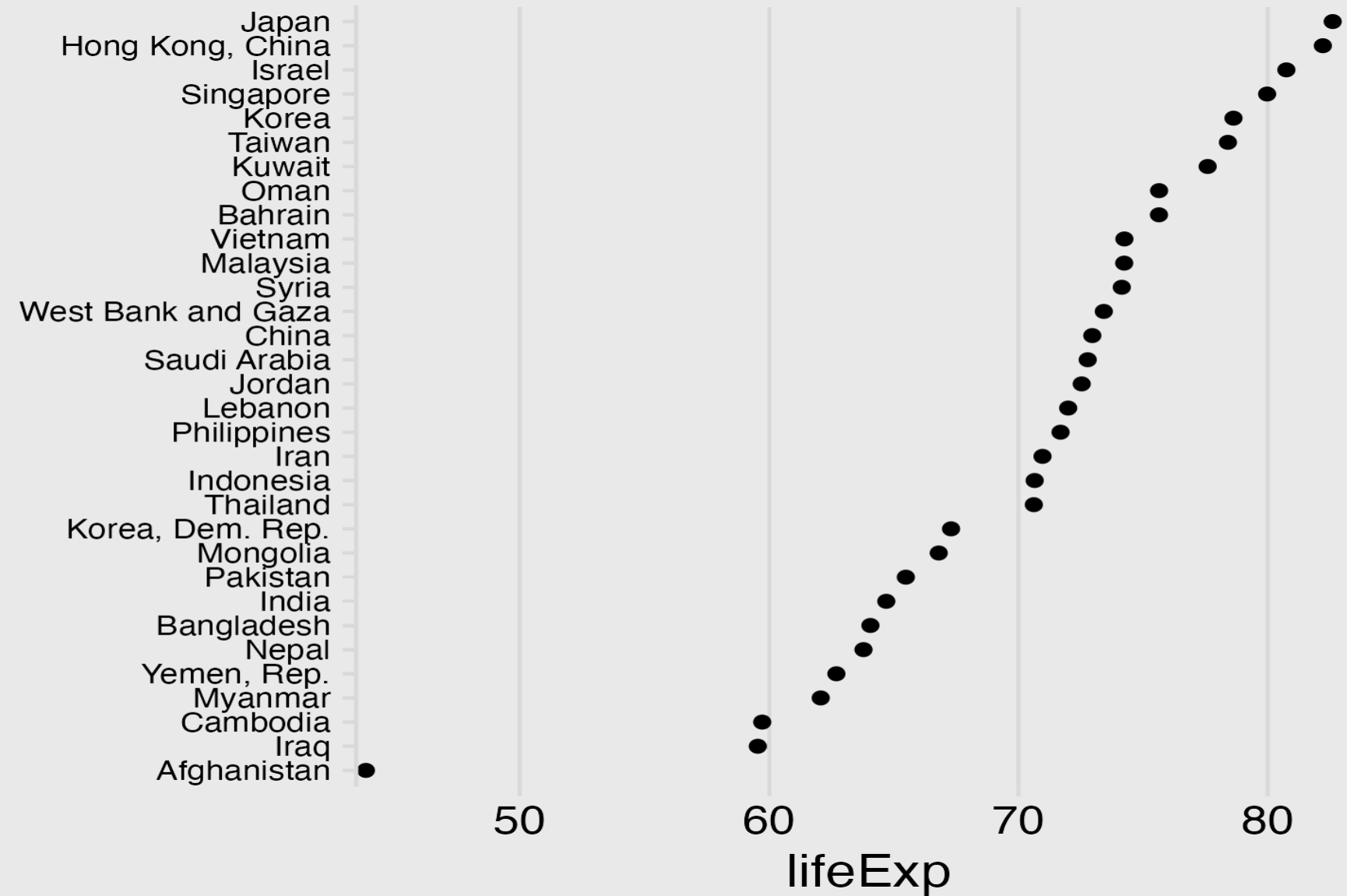


Yemen, Rep.
West Bank and Gaza
Vietnam
Thailand
Taiwan
Syria
Singapore
Saudi Arabia
Philippines
Pakistan
Oman
Nepal
Myanmar
Mongolia
Malaysia
Lebanon
Kuwait
Korea, Dem. Rep.
Korea
Jordan
Japan
Israel
Iraq
Iran
Indonesia
India
Hong Kong, China
China
Cambodia
Bangladesh
Bahrain
Afghanistan



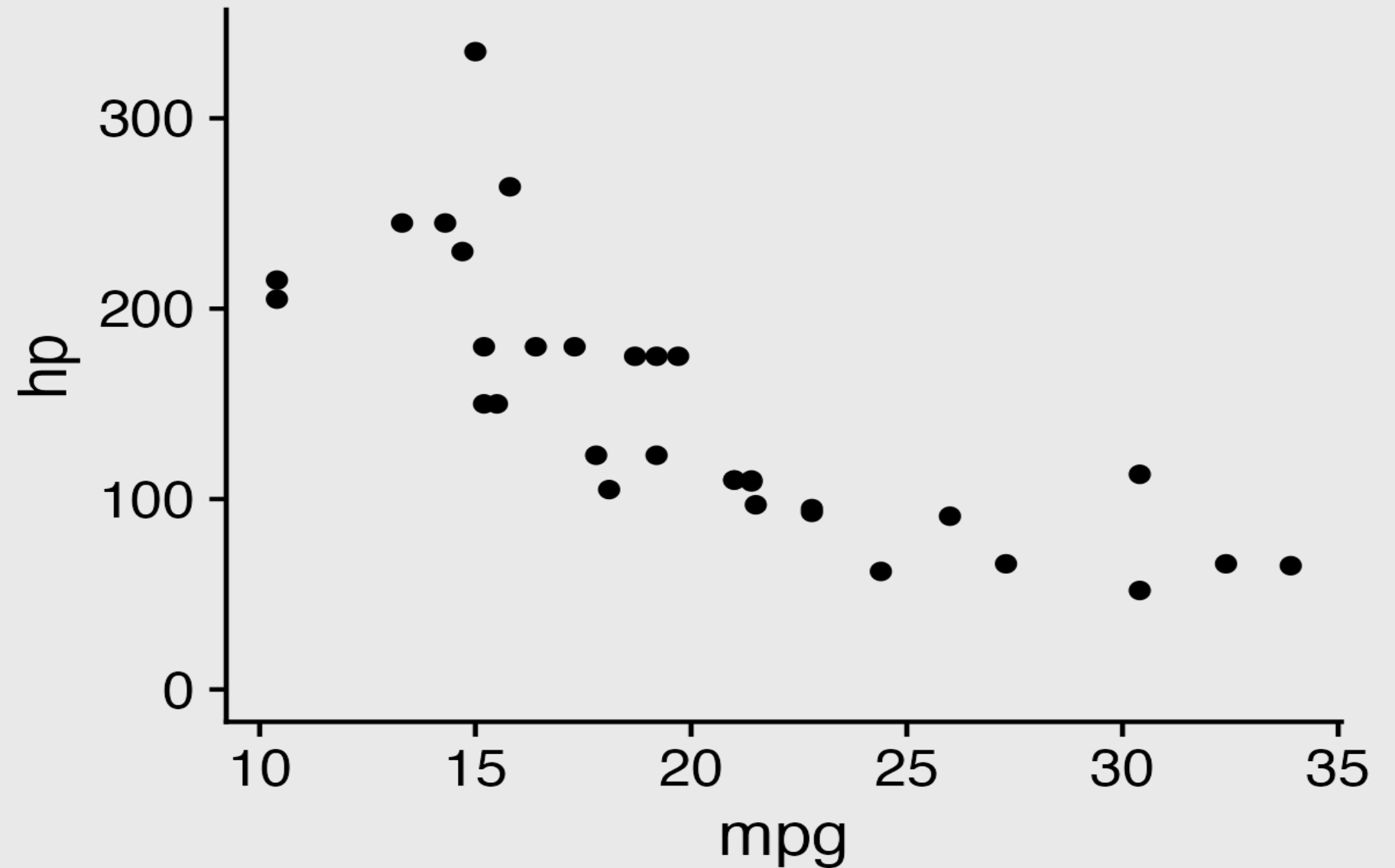
Prägnanz

This makes our
brains happy



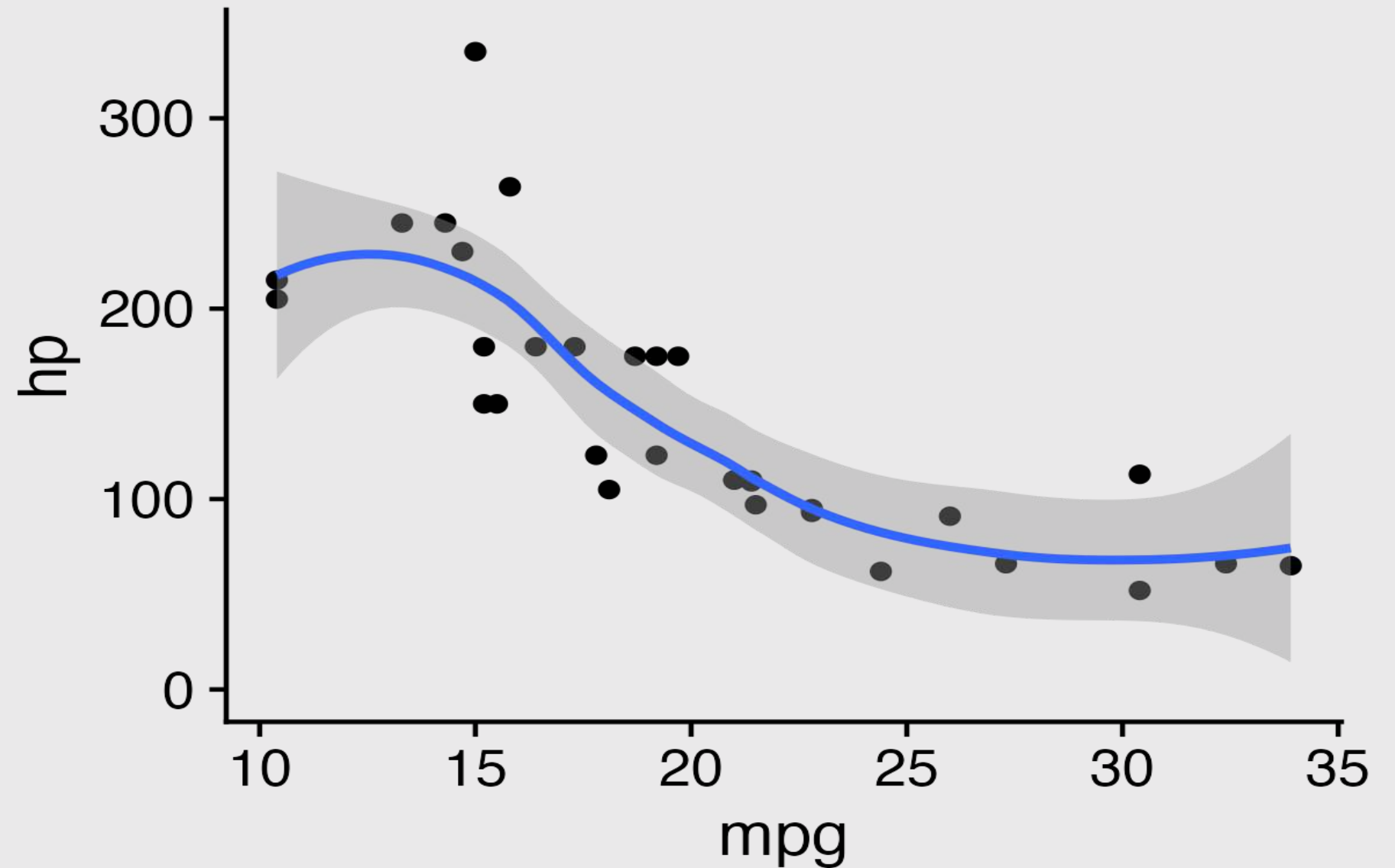
Law of Continuity

We will group together objects that follow an established direction



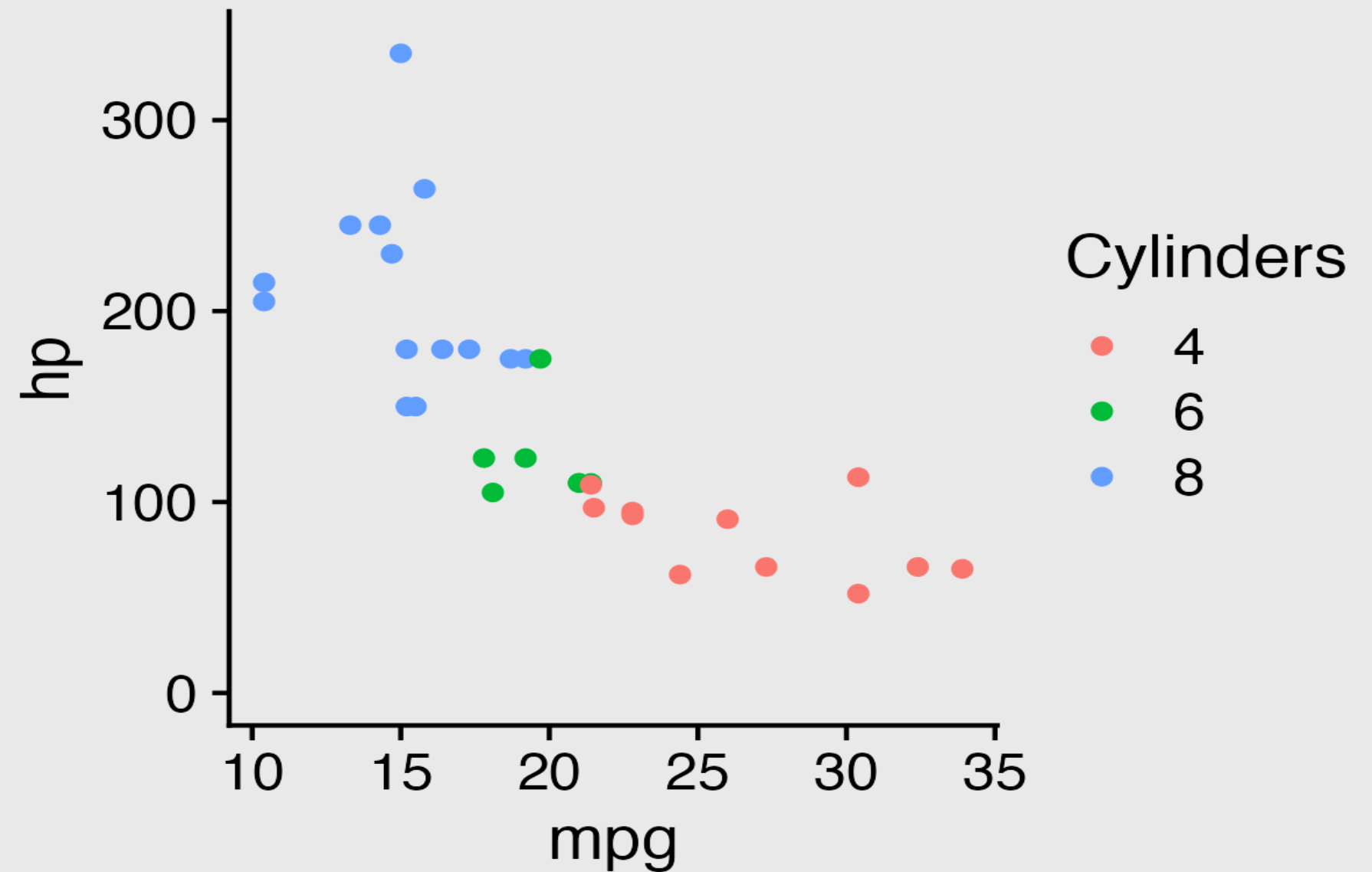
Law of Continuity

We will group together objects that follow an established direction



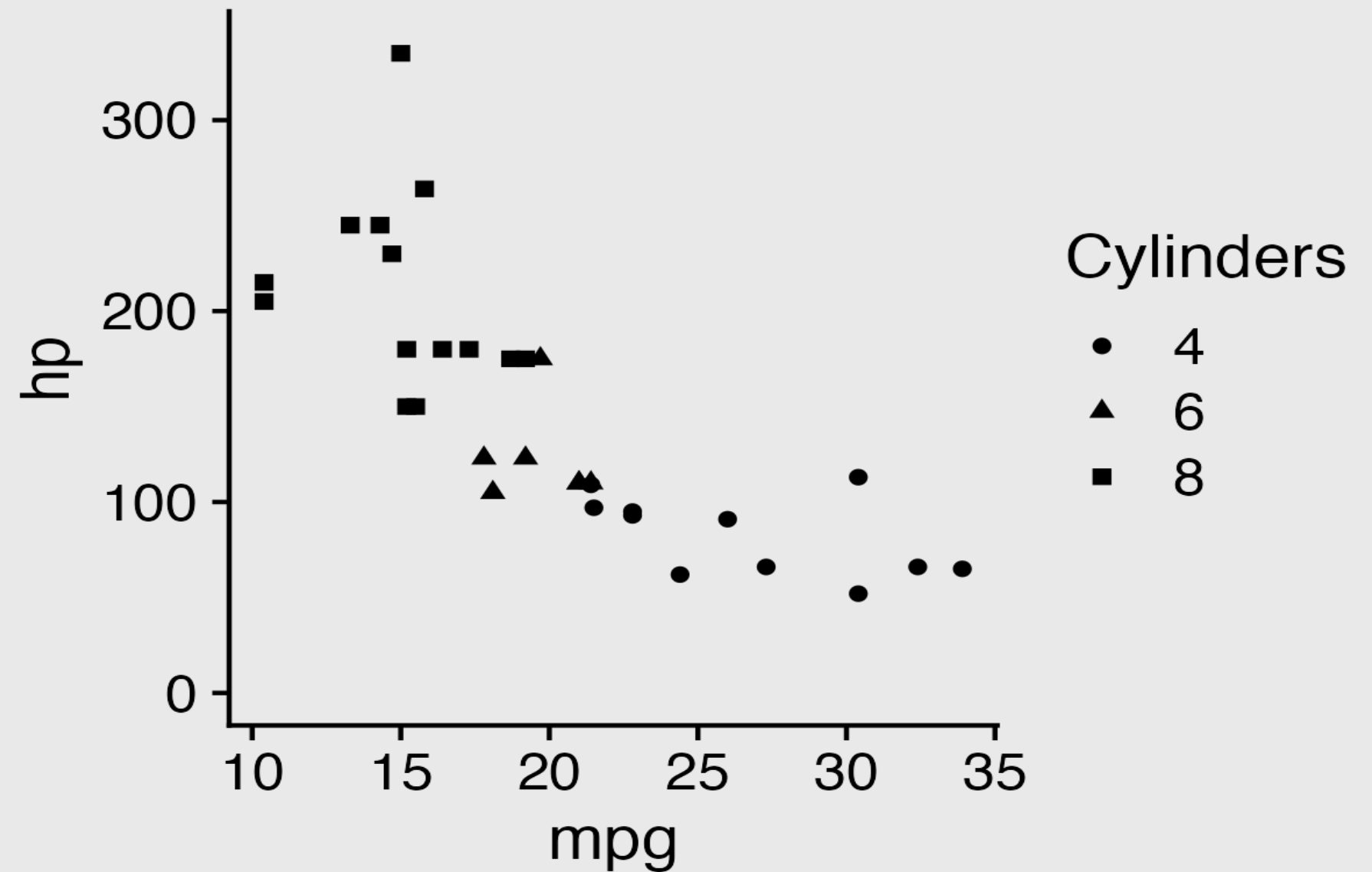
Law of Similarity

We see elements that are *physically similar* as part of the same object



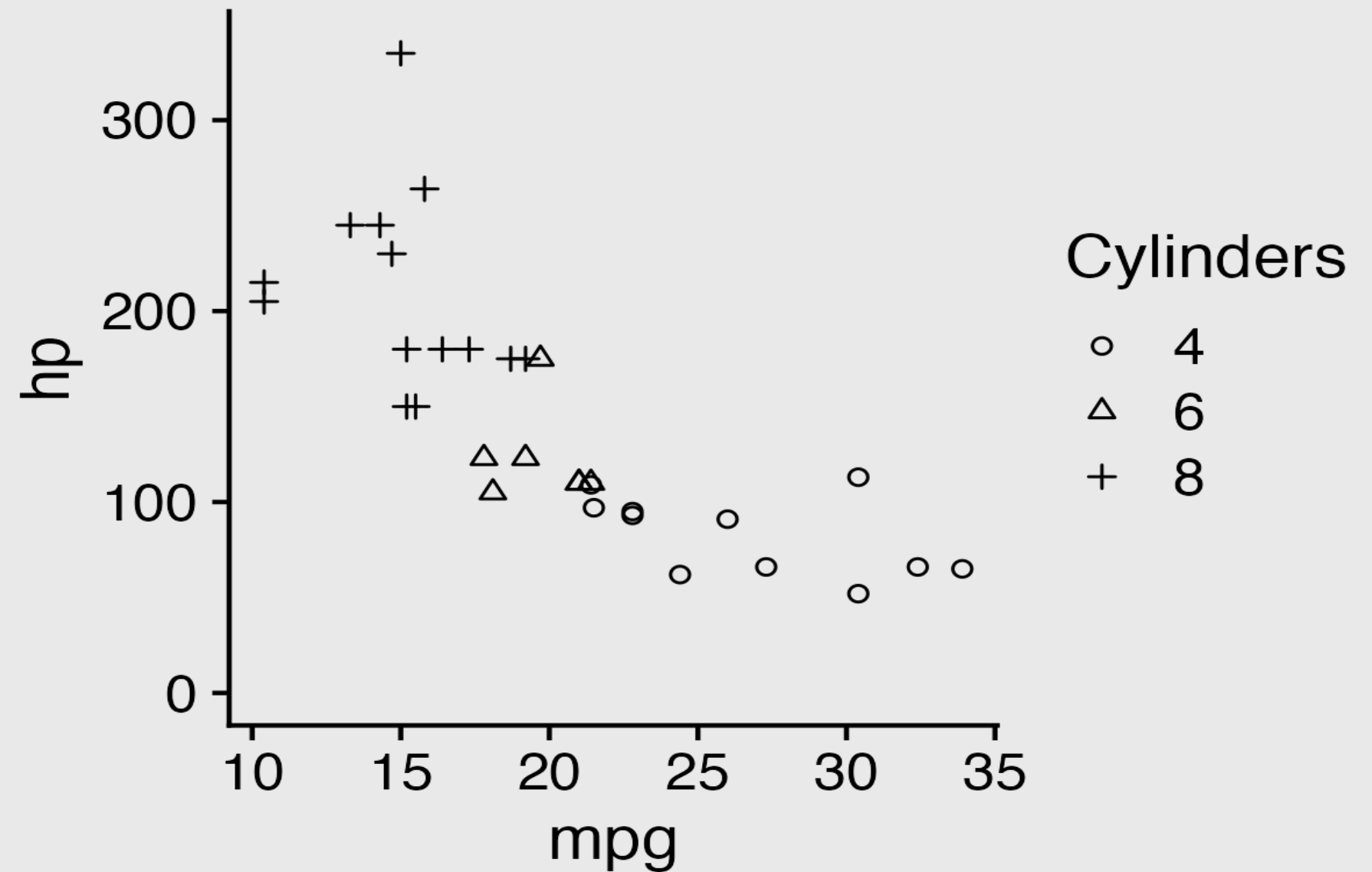
Law of Similarity

We see elements that are *physically similar* as part of the same object



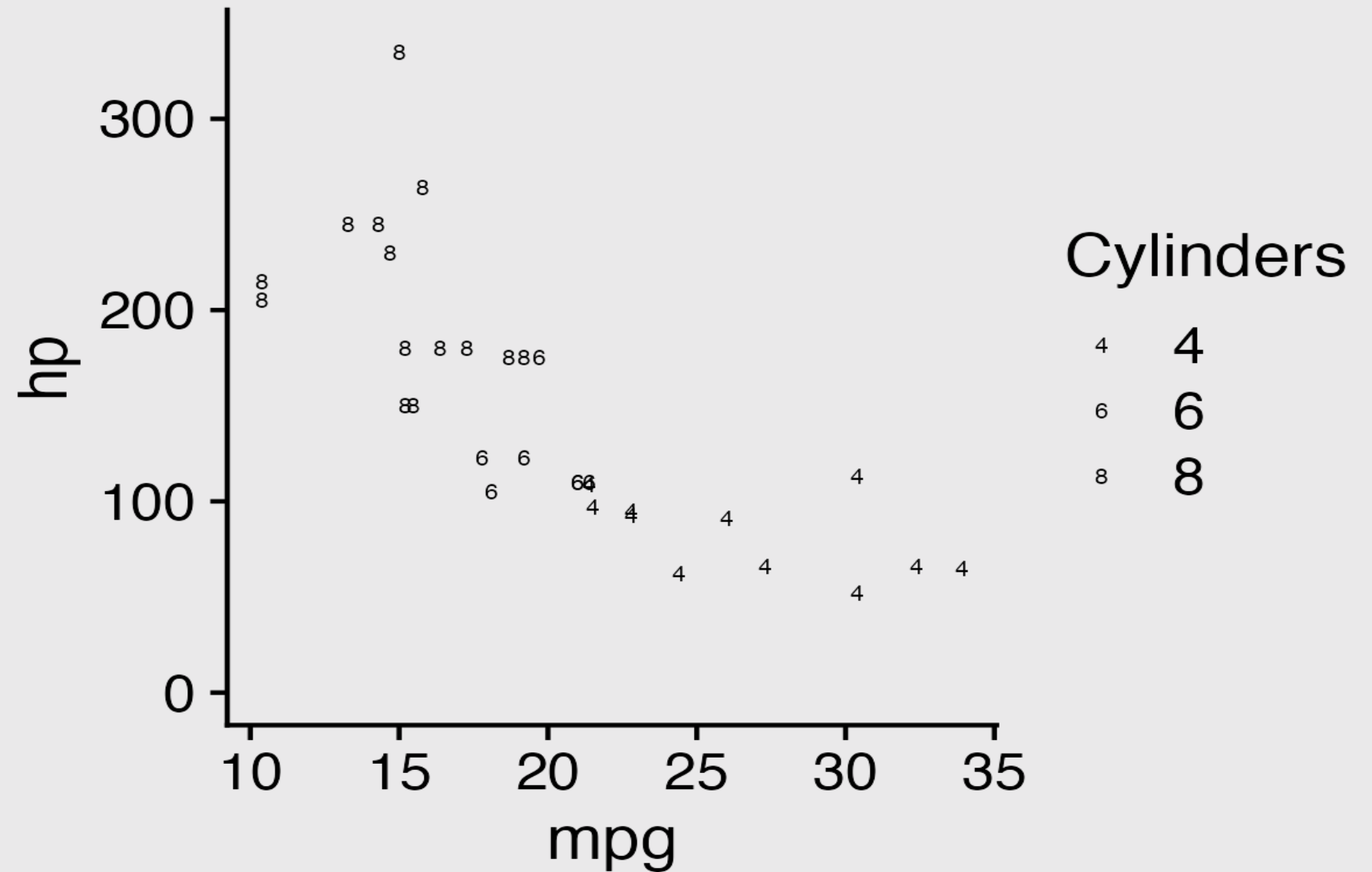
Law of Similarity

We see elements that are *physically similar* as part of the same object



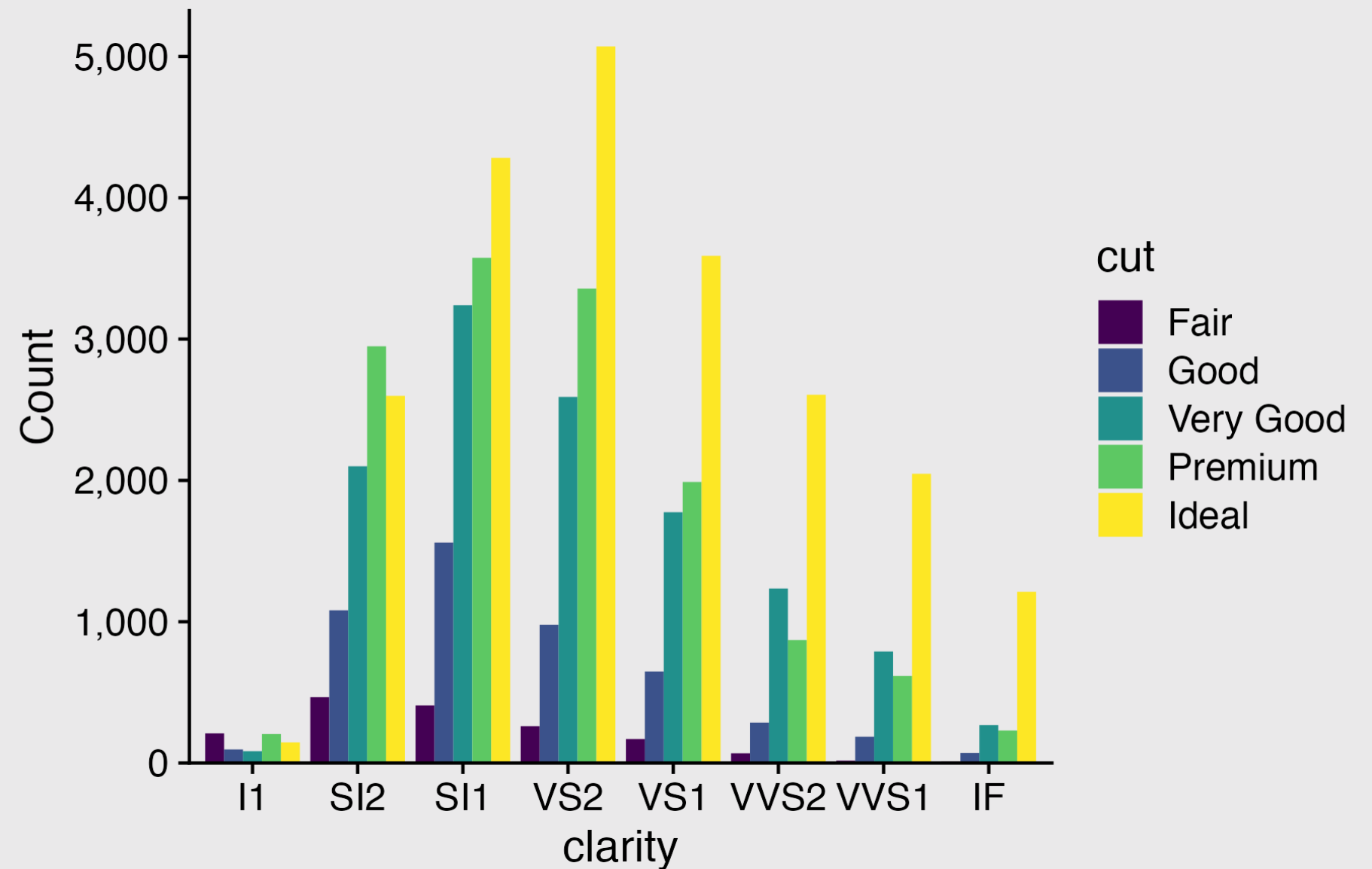
Law of Similarity

We see elements that are *physically similar* as part of the same object



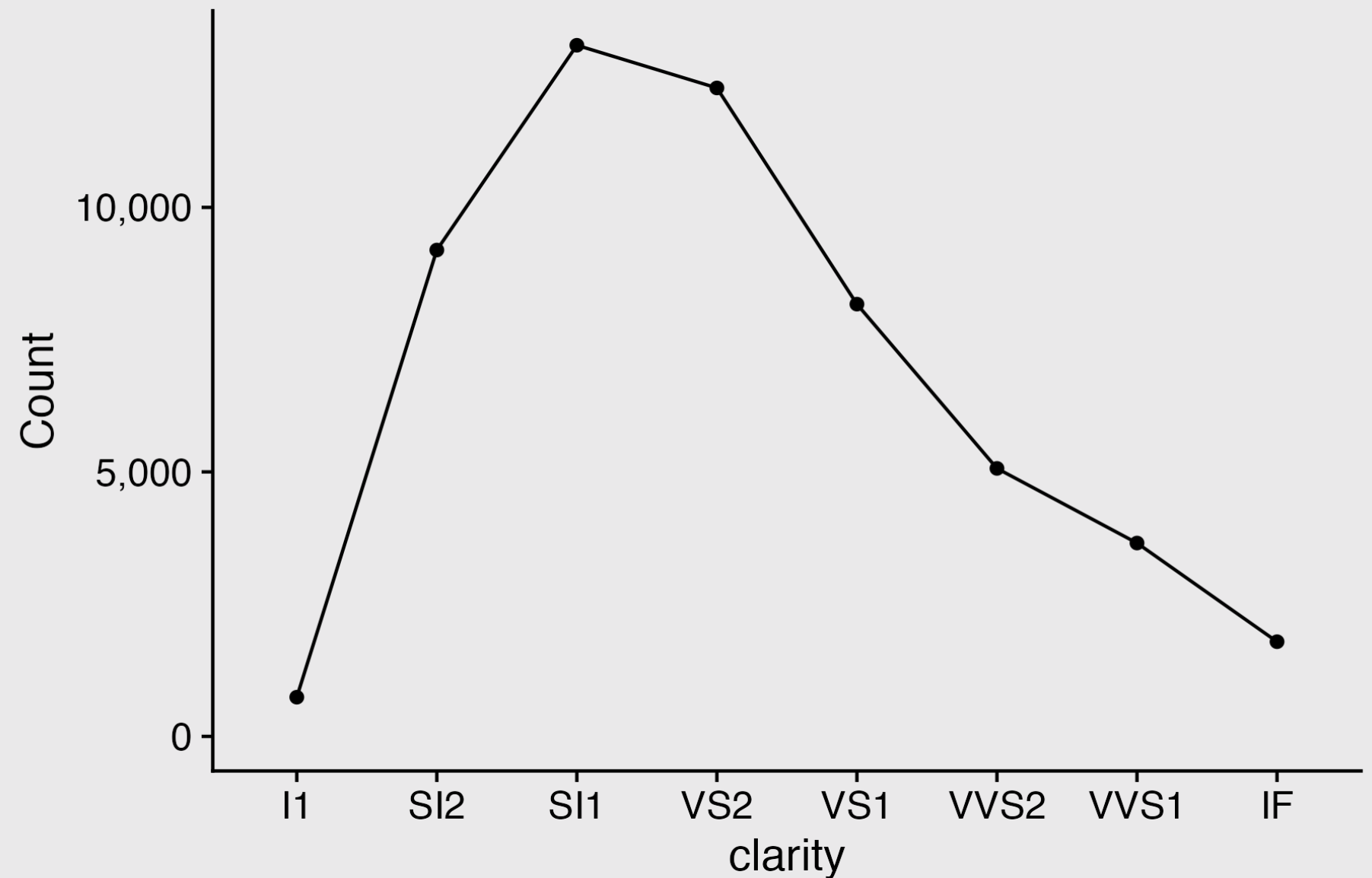
Law of Proximity

We tend to see elements that are *physically near* each other as part of the same object



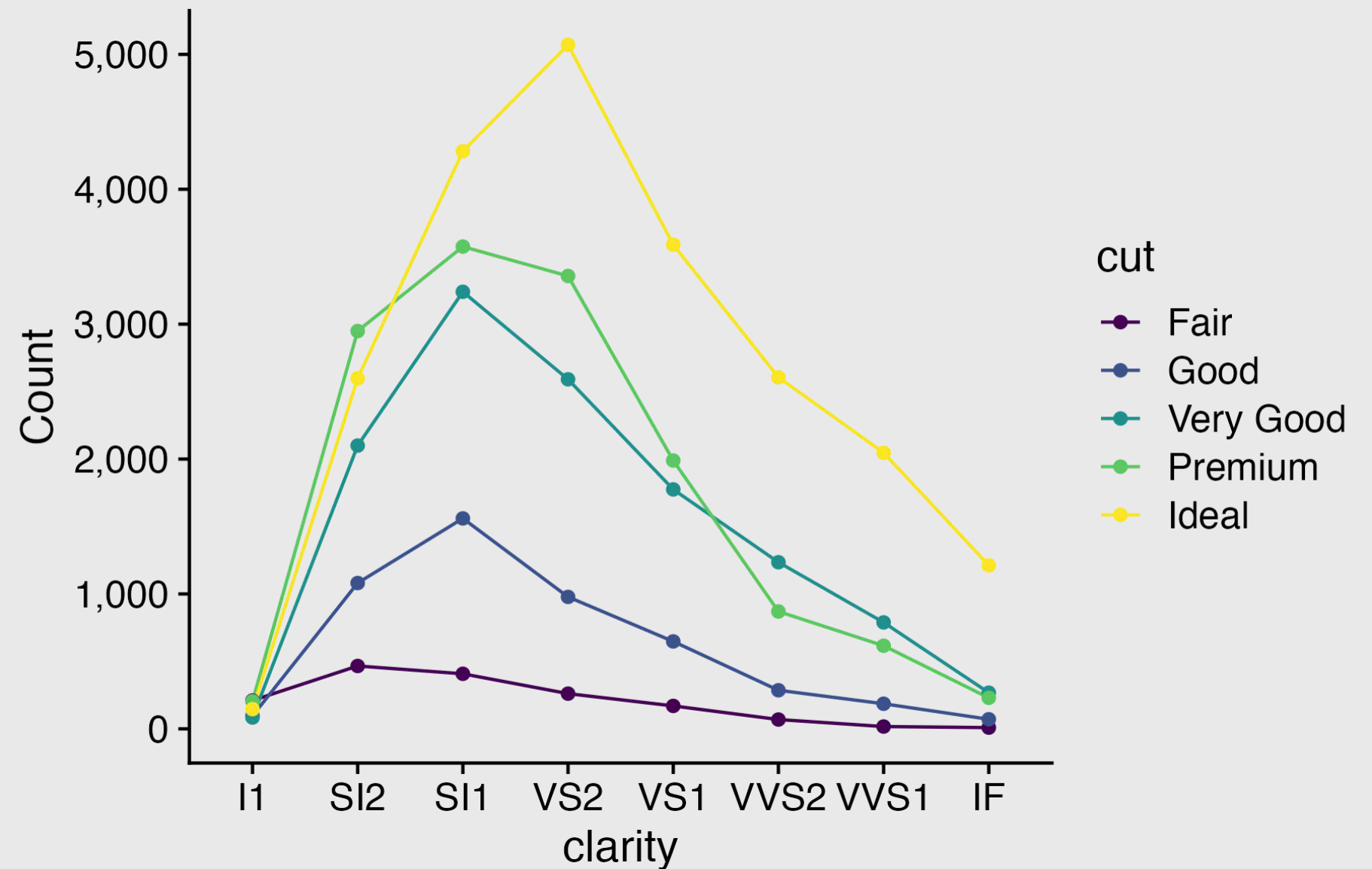
Law of Proximity

We tend to see elements that are *physically near* each other as part of the same object



Law of Proximity

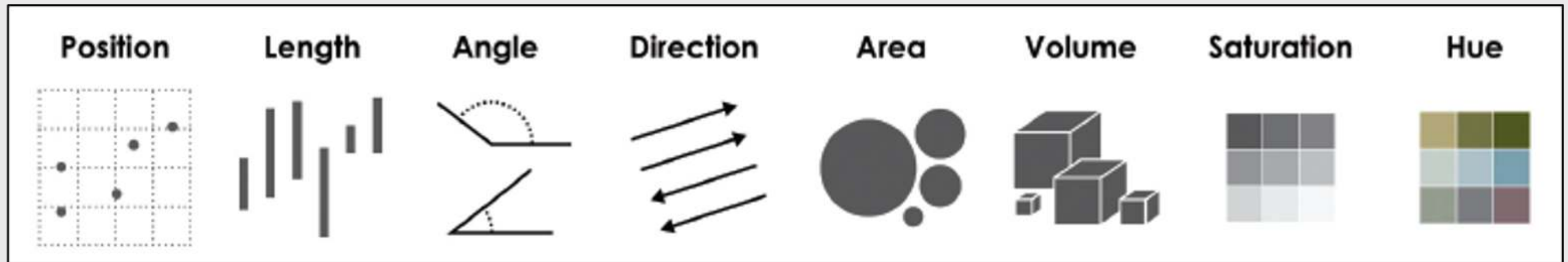
We tend to see elements that are *physically near* each other as part of the same object



Cleveland's operations of pattern perception:

1. Estimation
2. Assembly
3. Detection

Estimation: Hierarchy for *numerical* data



More Accurate

Less Accurate

Assembly: Gestalt Psychology

Law of Closure	Prägnanz	Law of Continuity	Law of Similarity	Law of Proximity
Fill in the missing information	We like regular, simple, and orderly	Group together objects with established direction	Physically <i>similar</i> = same object	Physically <i>near</i> = same object

Cleveland's operations of pattern perception:

1. Estimation

2. Assembly

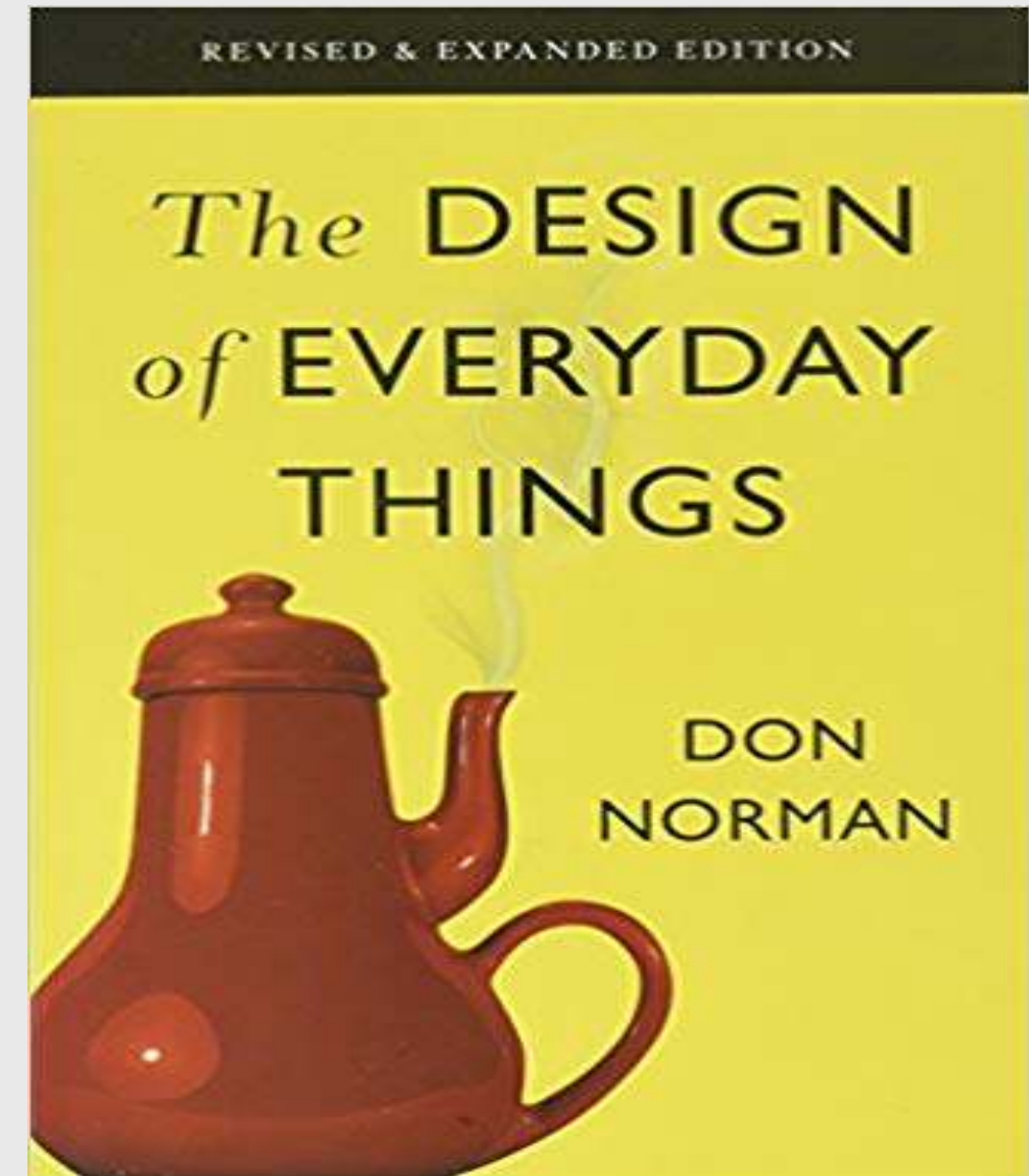
3. **Detection** ———>

Recognizing that a geometric object encodes a physical value



Norman door (n.):

1. A door where the design tells you to do the opposite of what you're actually supposed to do.
2. A door that gives the wrong signal and needs a sign to correct it.

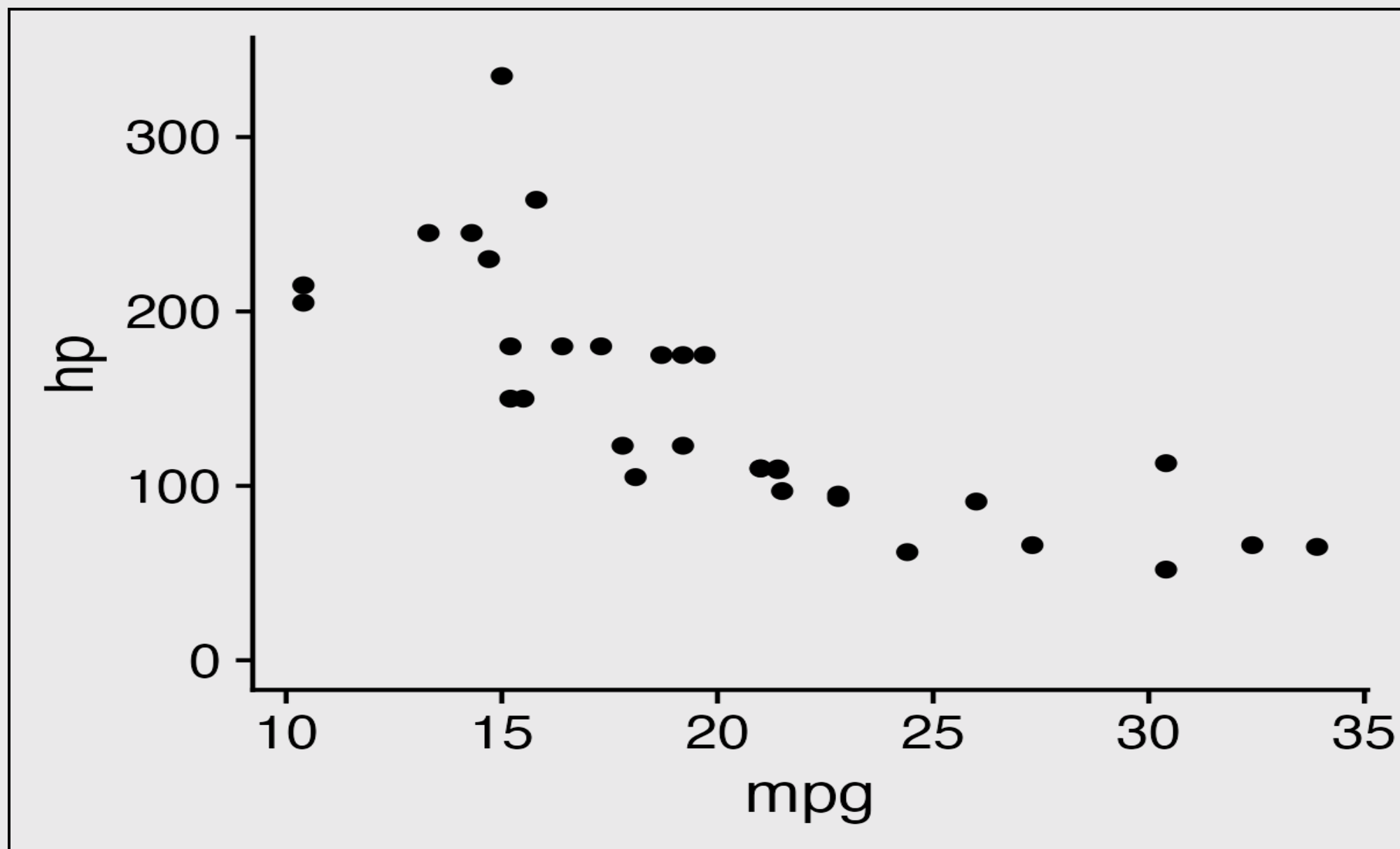


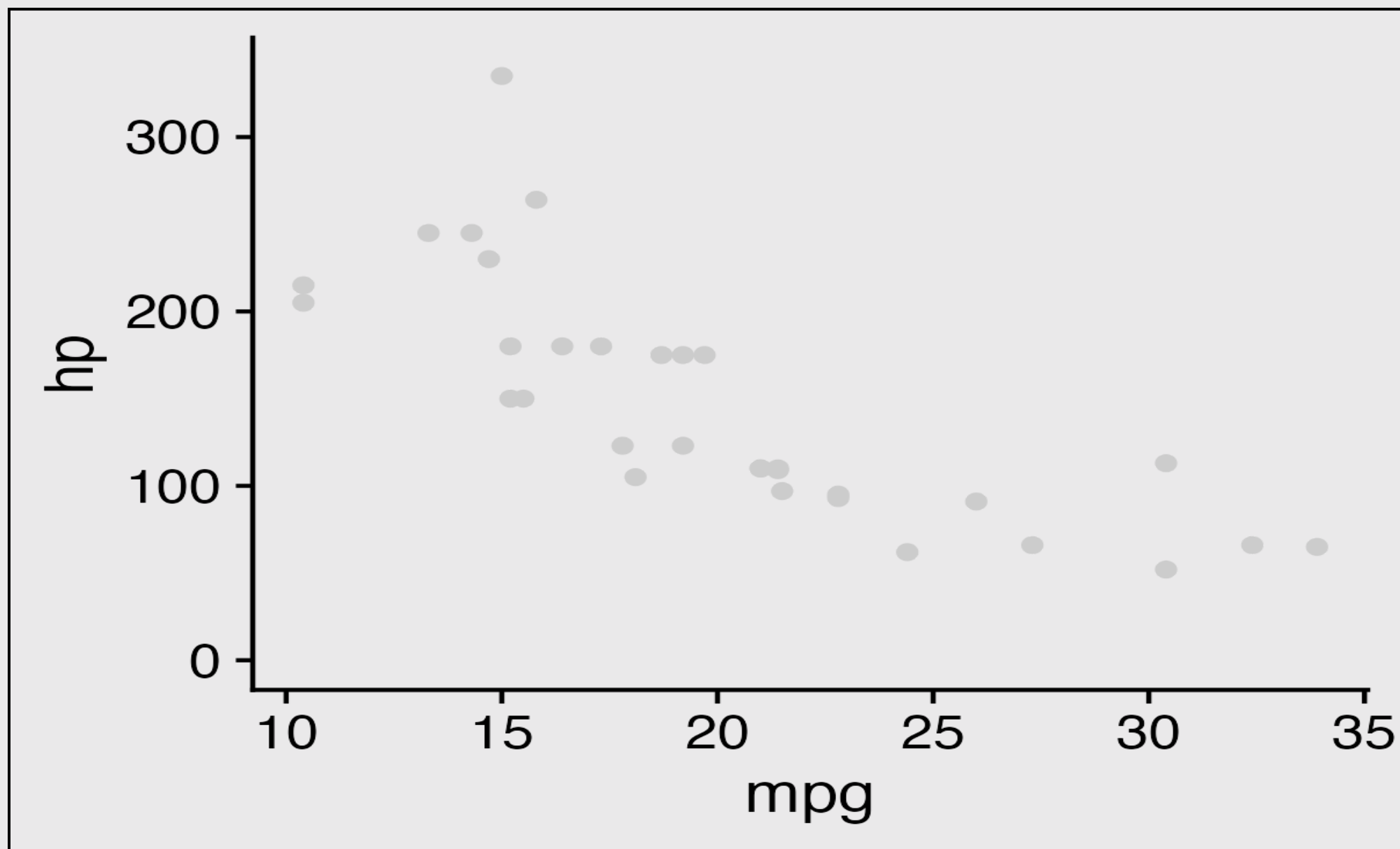
Norman door

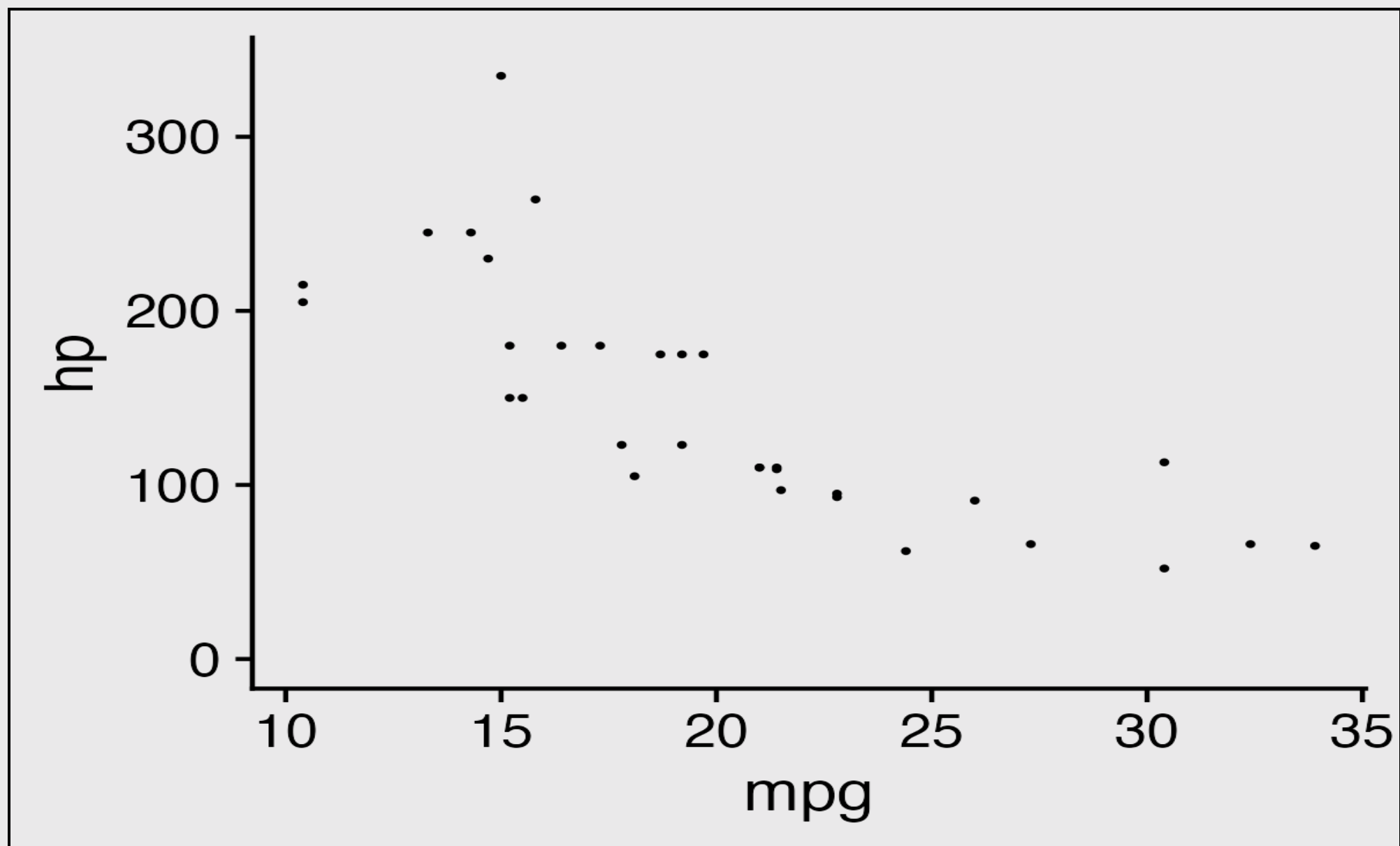


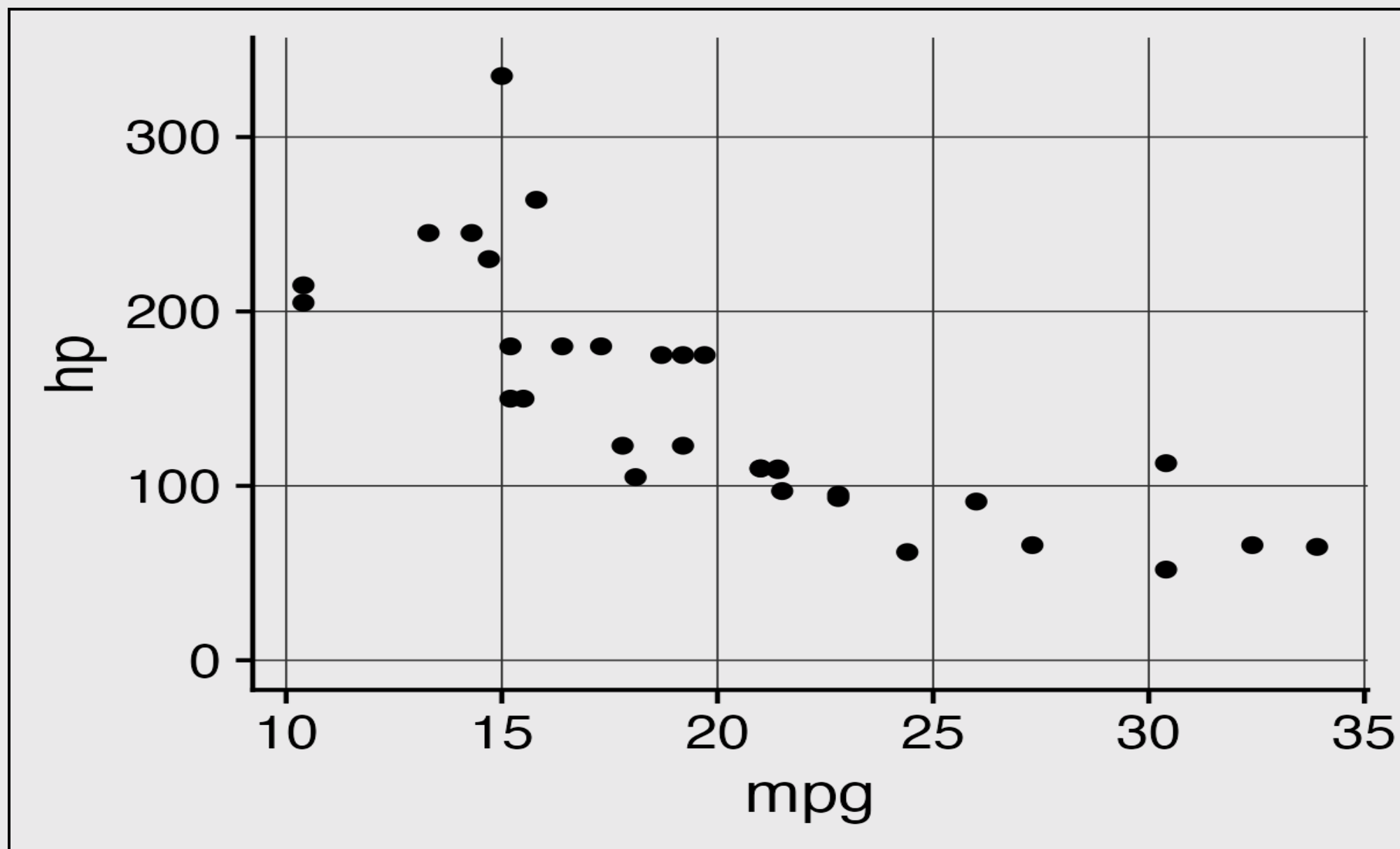
Non-Norman door

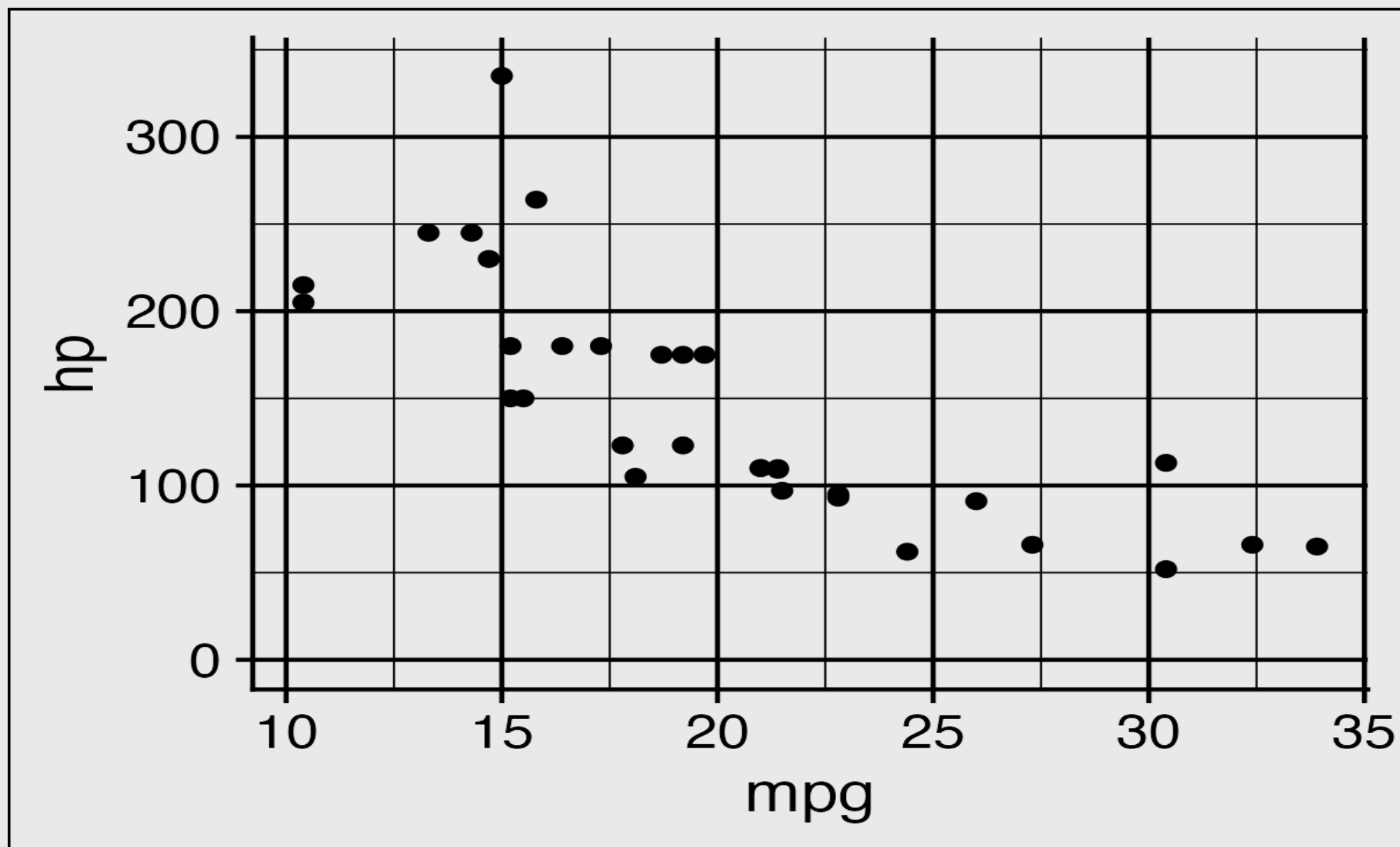


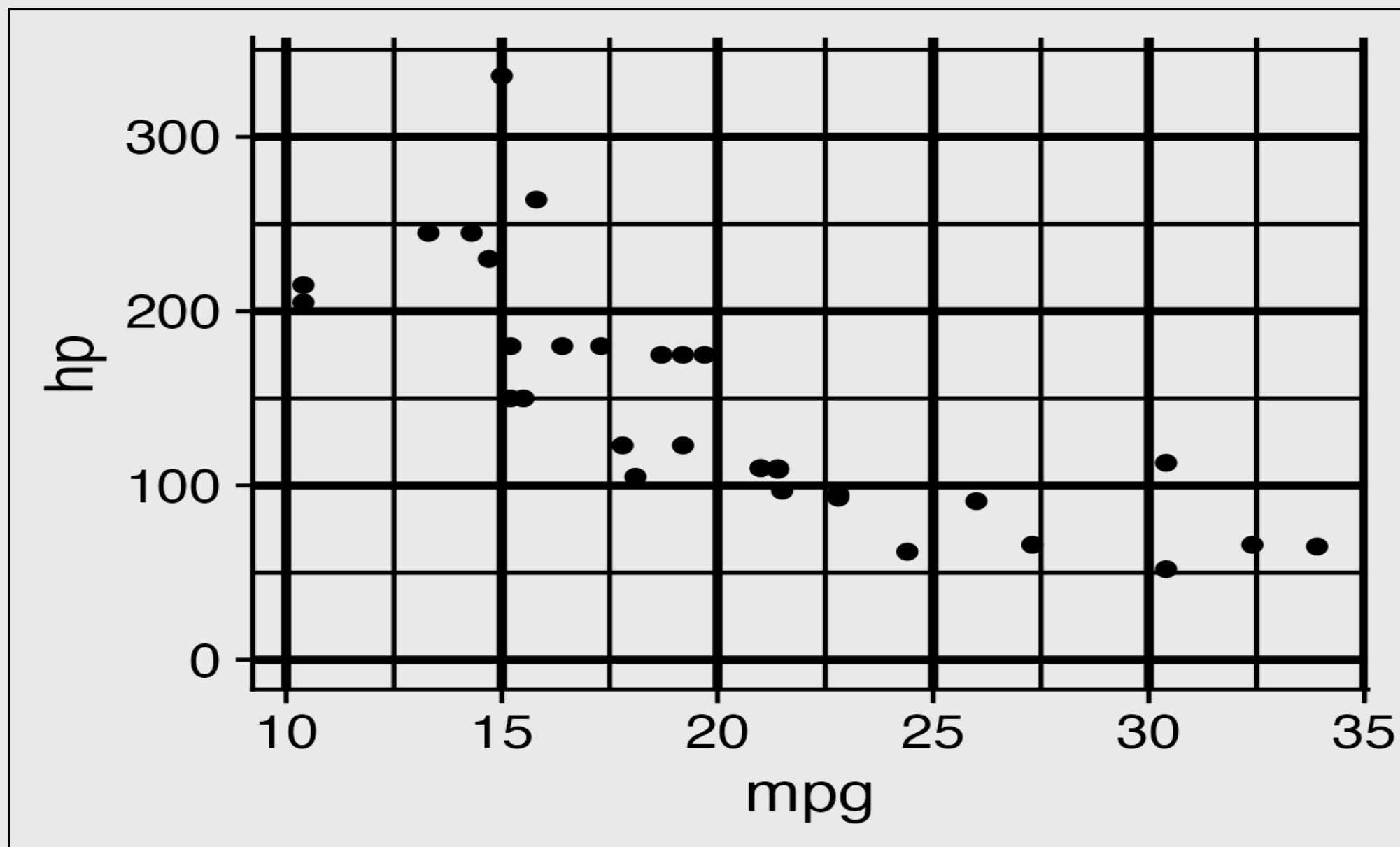




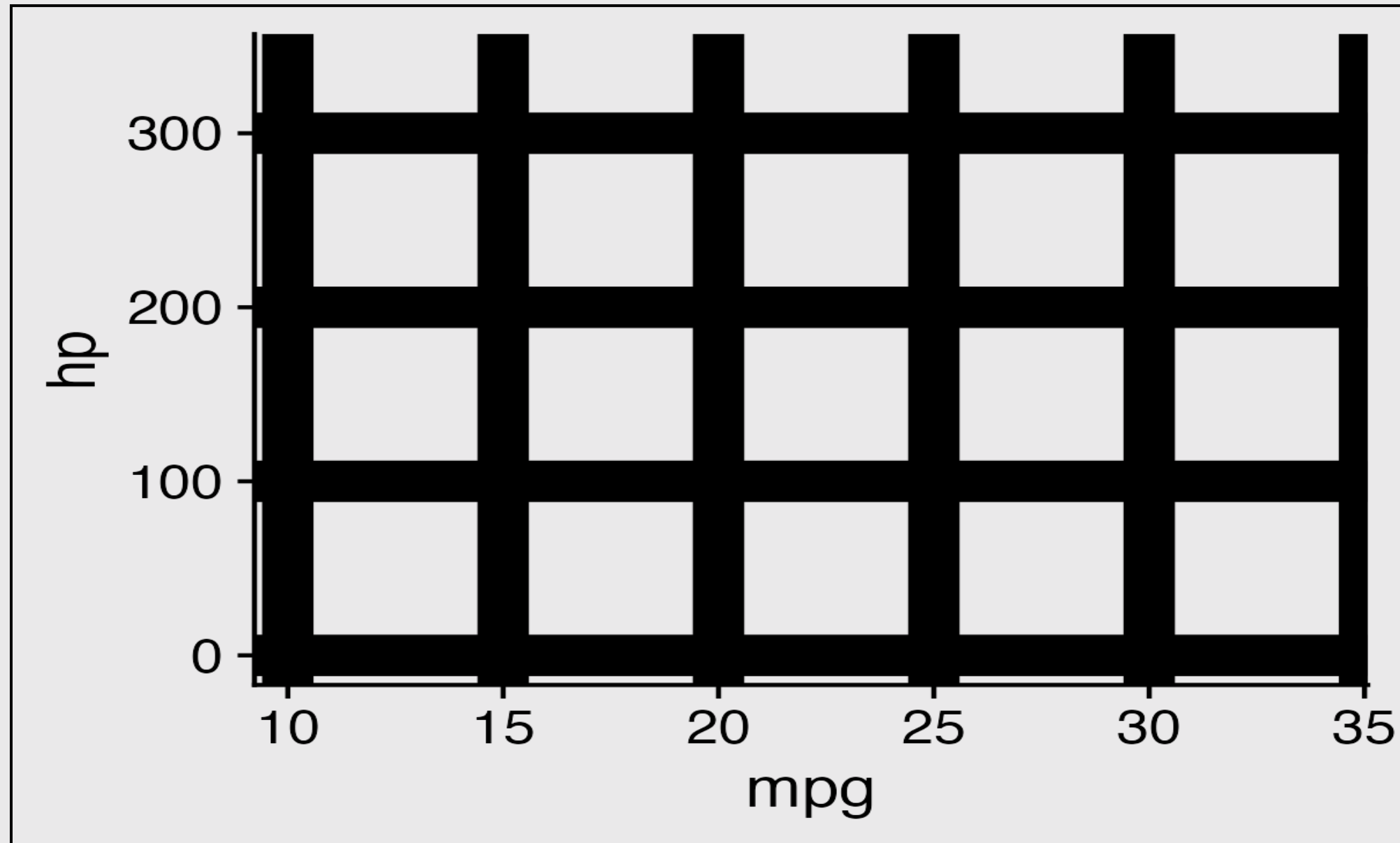


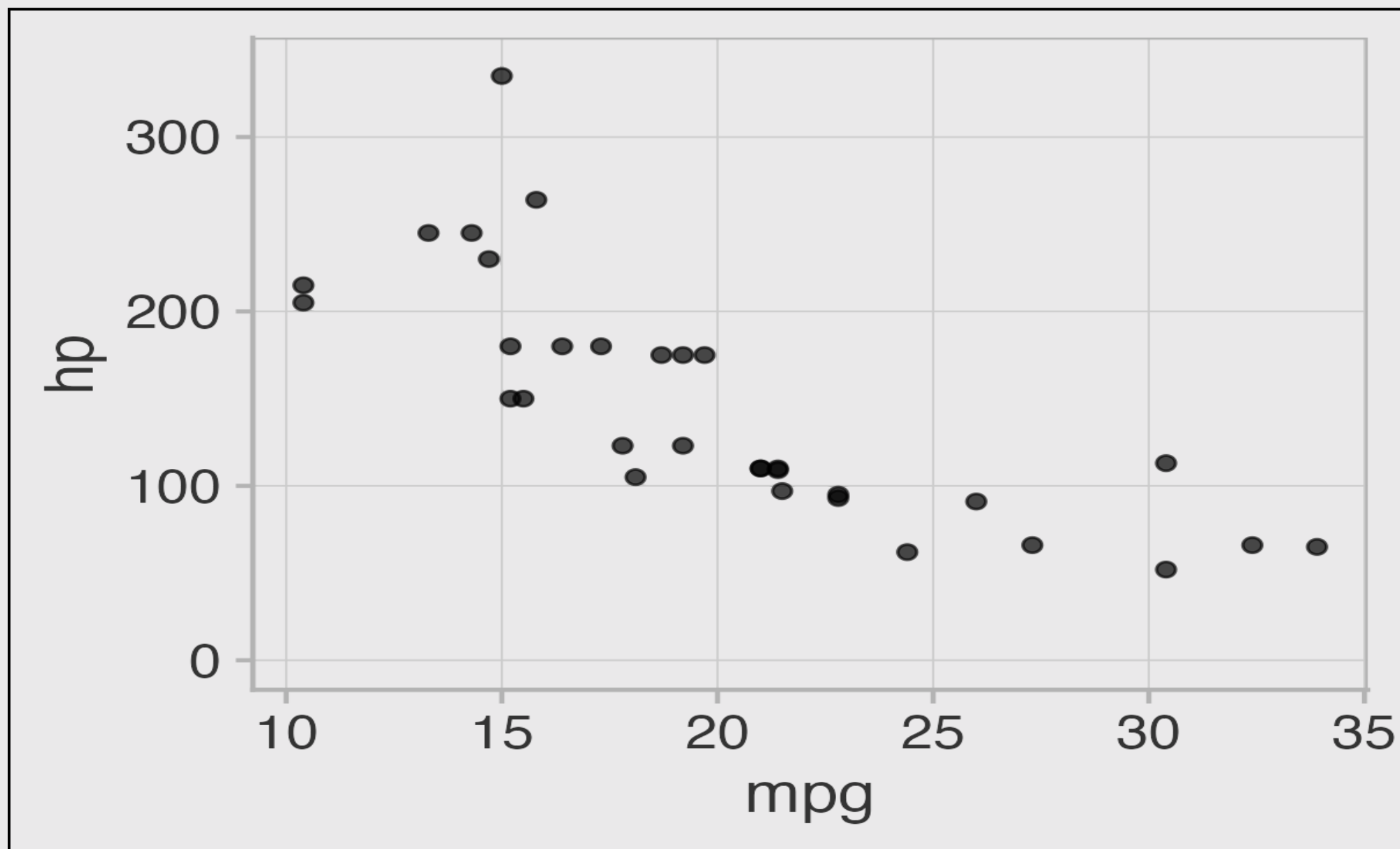






The white circles you see at the intersections is called the **"Hermann Grid illusion"**





Break!

Stand up, Move around, Stretch!

05 : 00

Week 7: *Visualizing Information*

1. The Human Visual-Memory System
2. The Psychology of Data Viz
3. 10 Data Viz Best Practices
4. Making a (good) ggplot

10 Data Viz Best Practices

1. Remove chart chunk
2. Don't make 3D plots*
3. Don't lie
4. Don't use pie charts for proportions*
5. Don't stack bars*
6. Rotate and sort categorical axes*
7. Eliminate legends & directly label geoms*
8. Don't use pattern fills
9. Don't use red & green together
10. Consider tables for small data sets

*most of the time

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8. Don't use pattern fills
9. Don't use red & green together
10. Consider tables for small data sets

*most of the time

“Erase non-data ink.”

— Ed Tufte

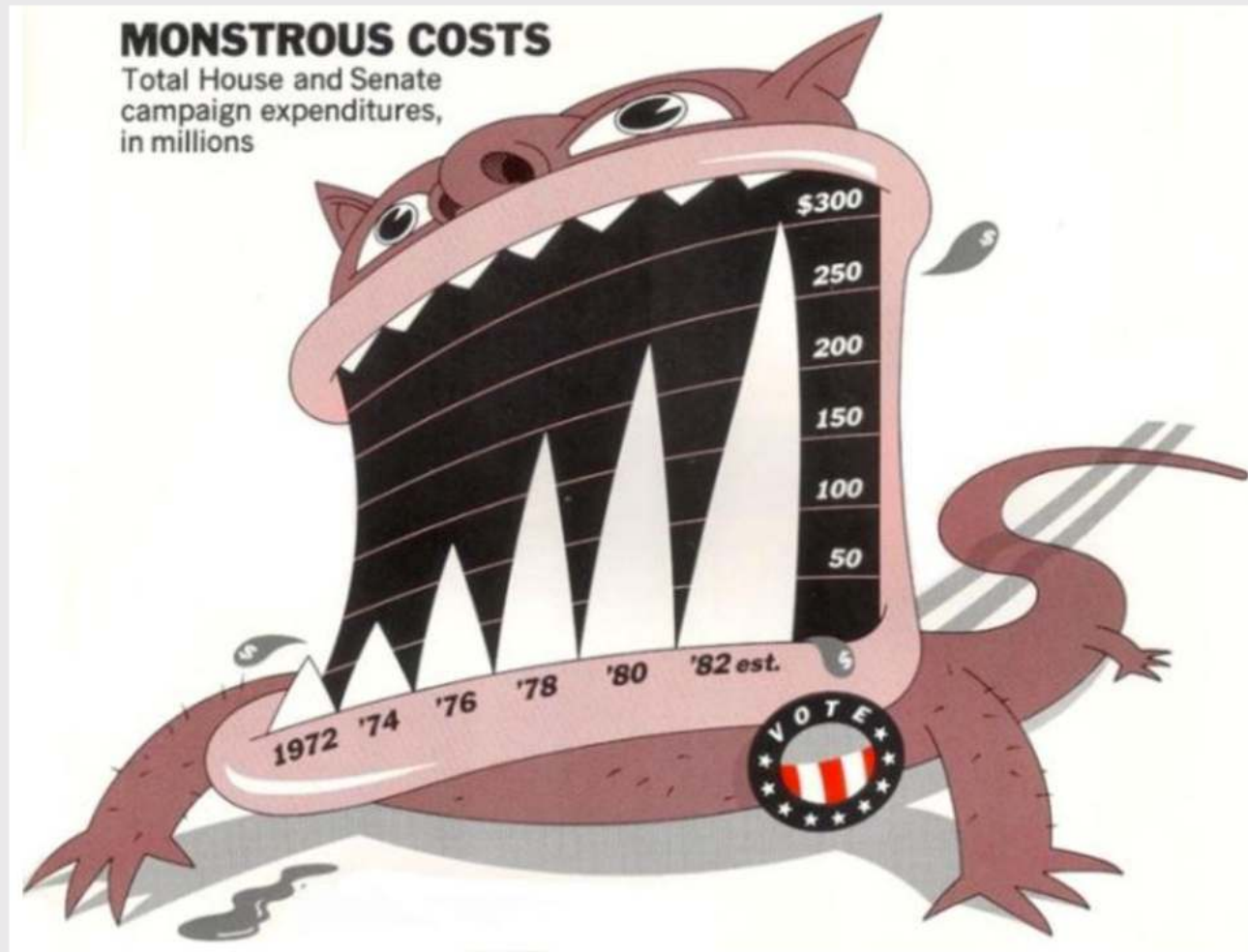
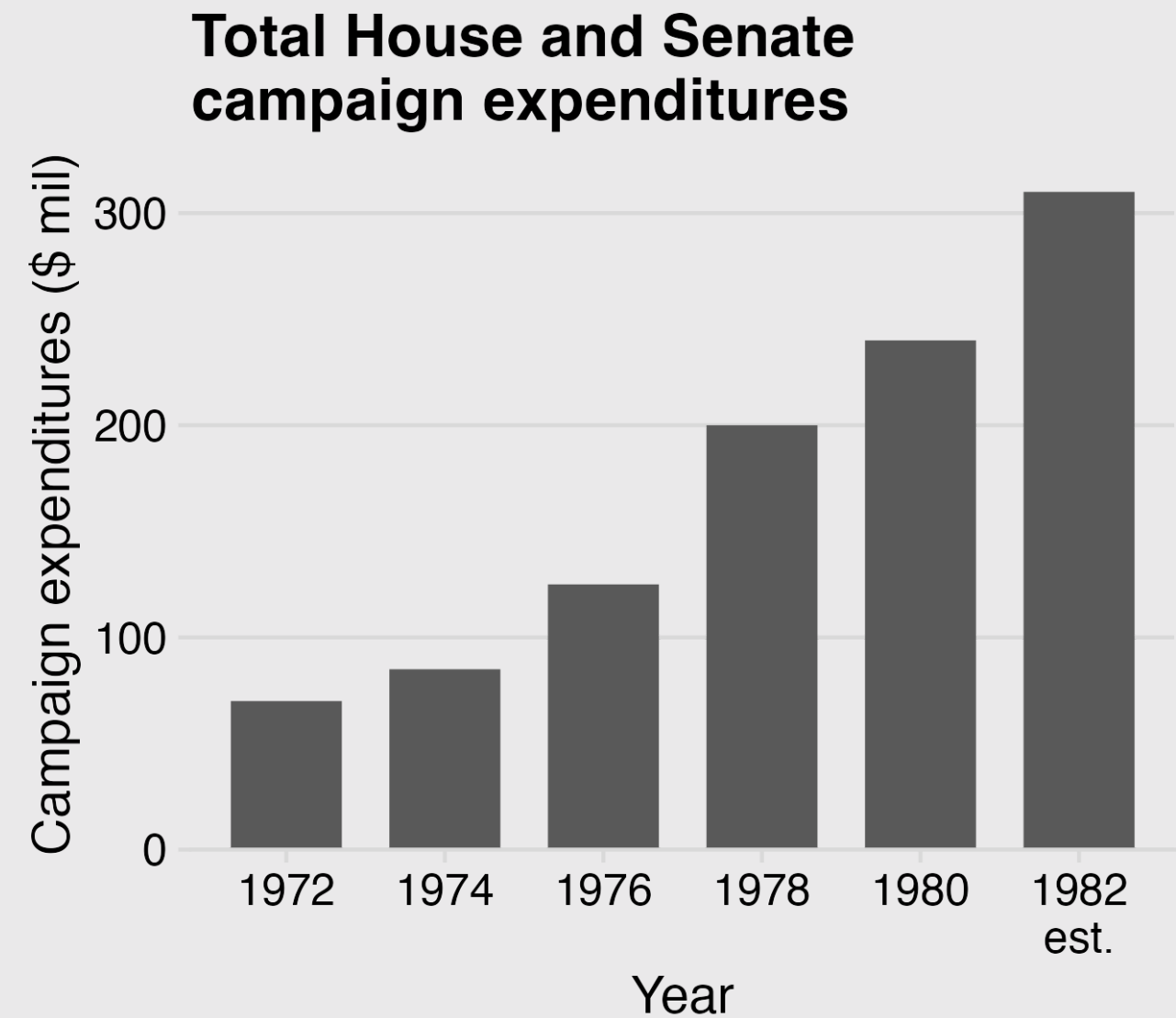


Figure 1.6: 'Monstrous Costs' by Nigel Holmes, in Healy, 2018



Remove
to improve
(the **data-ink** ratio)

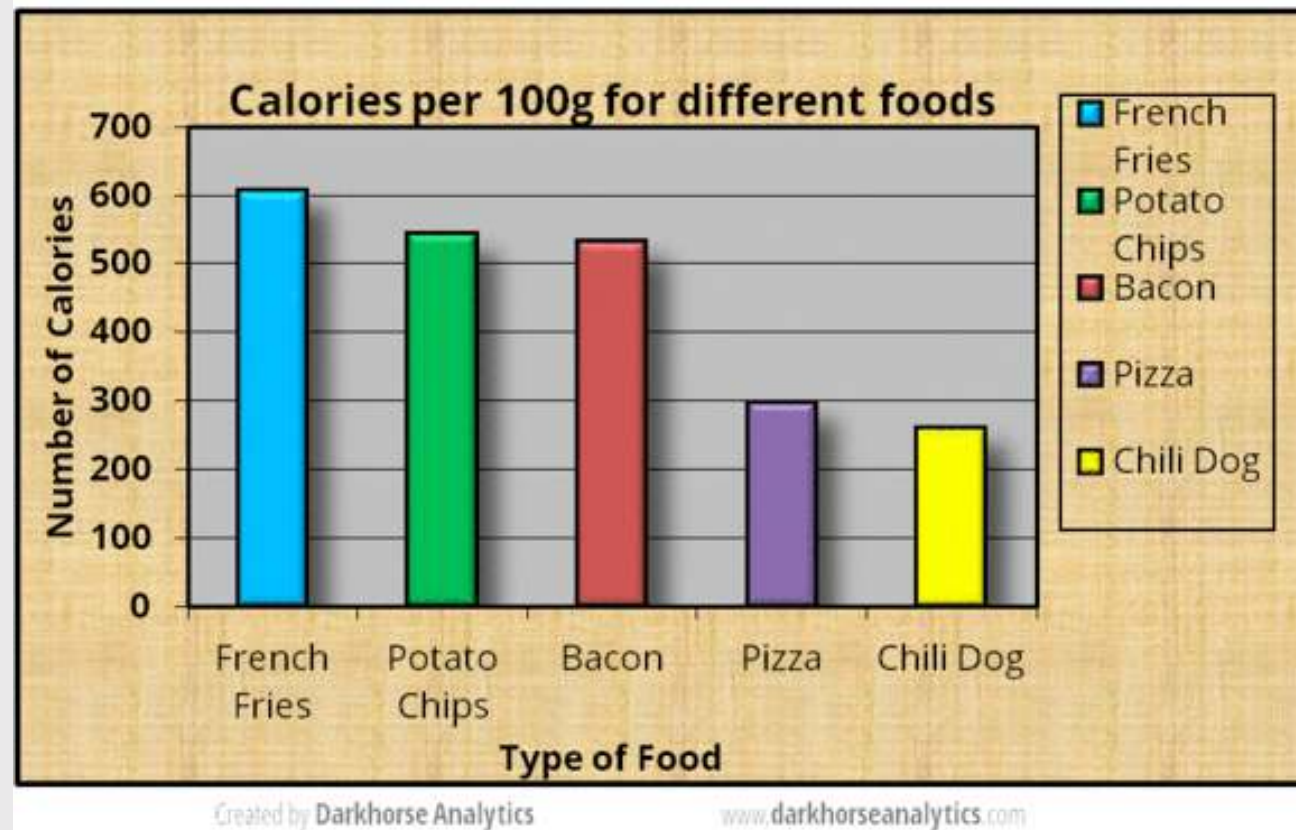
Created by **Darkhorse Analytics**

www.darkhorseanalytics.com

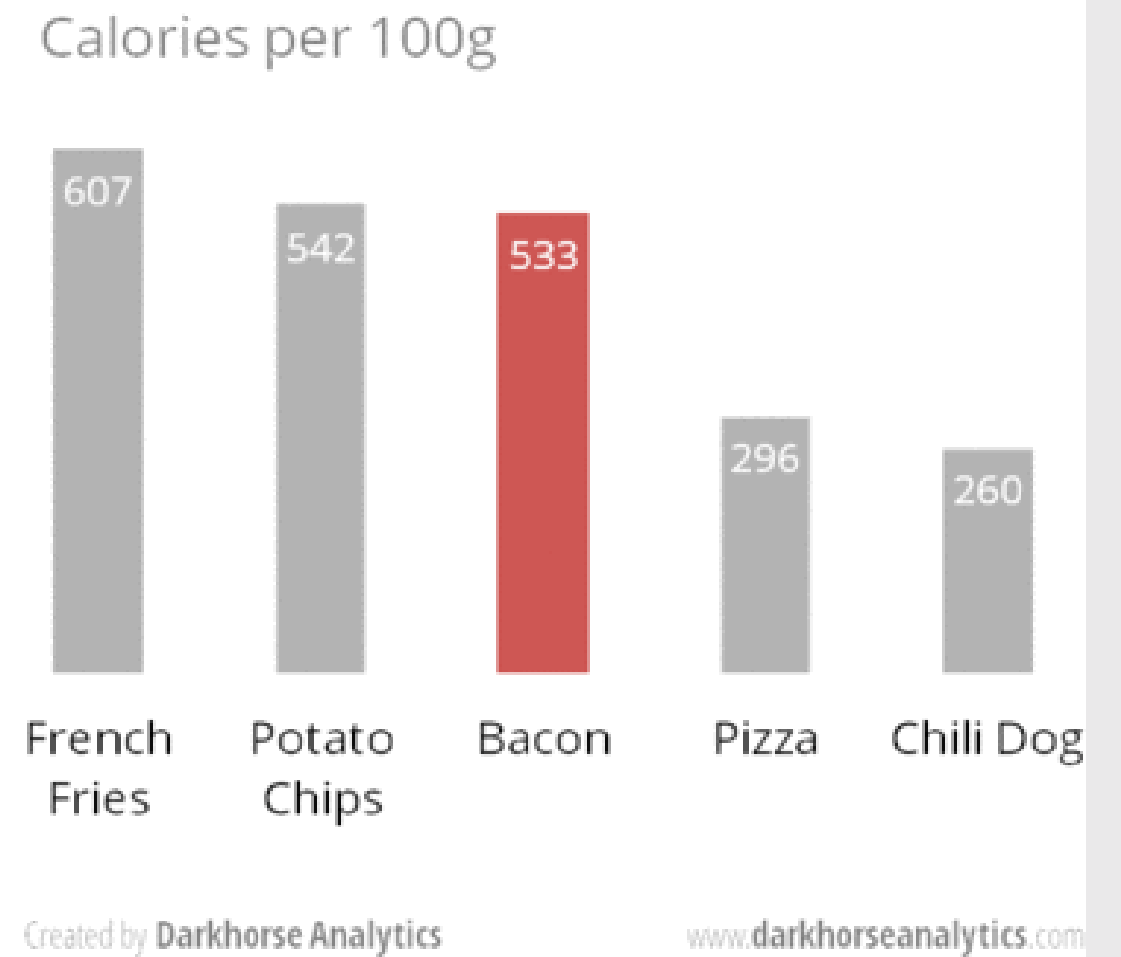
Figure 24.1: From Data Looks Better Naked by Darkhorse Analytics



Before



After

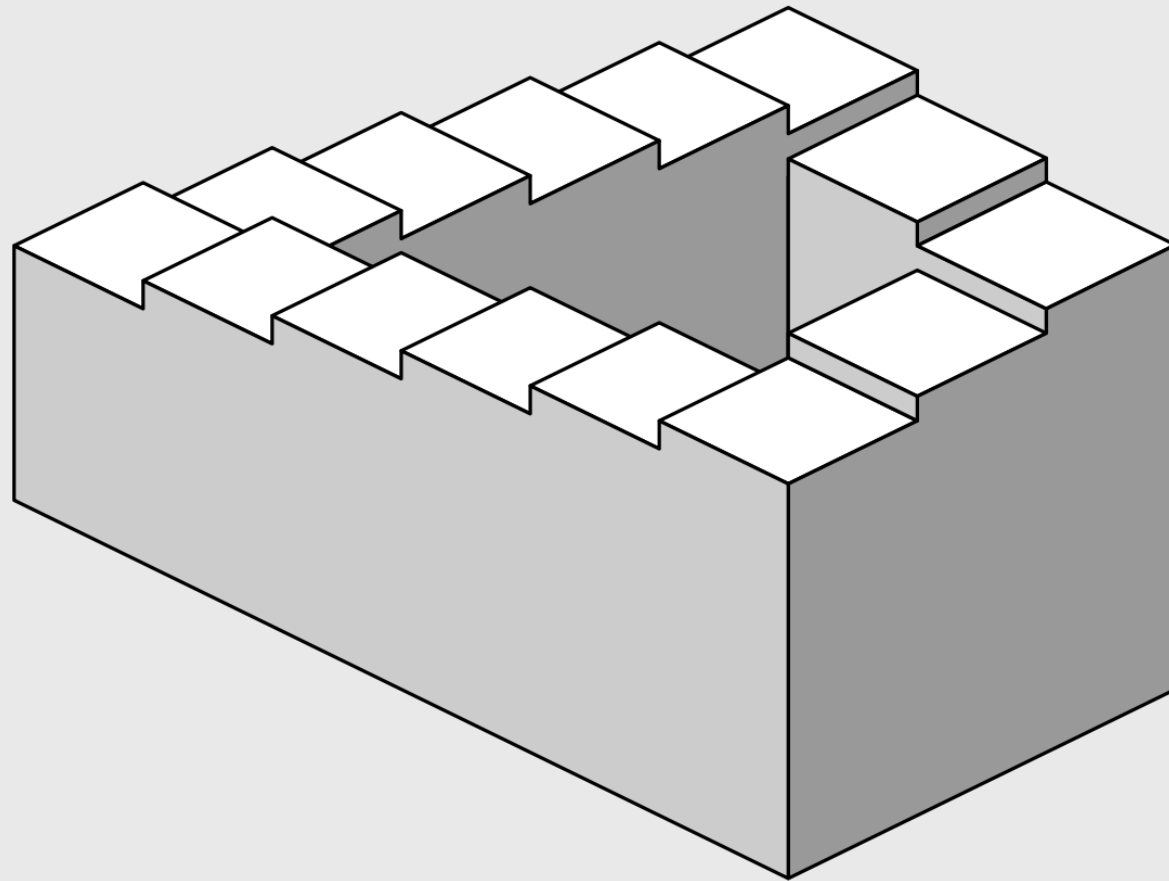


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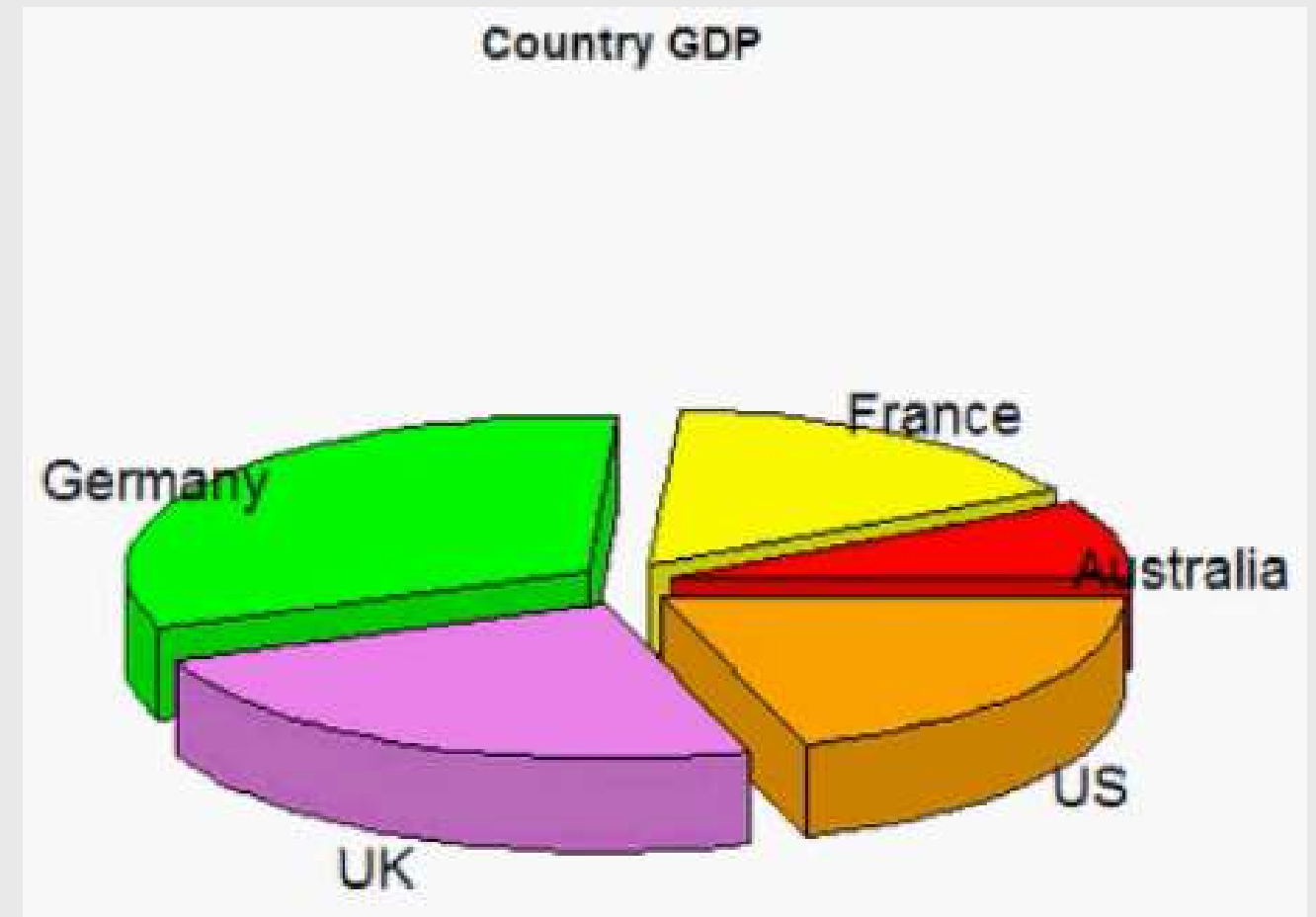
*most of the time

Humans aren't good at
distinguishing 3D space

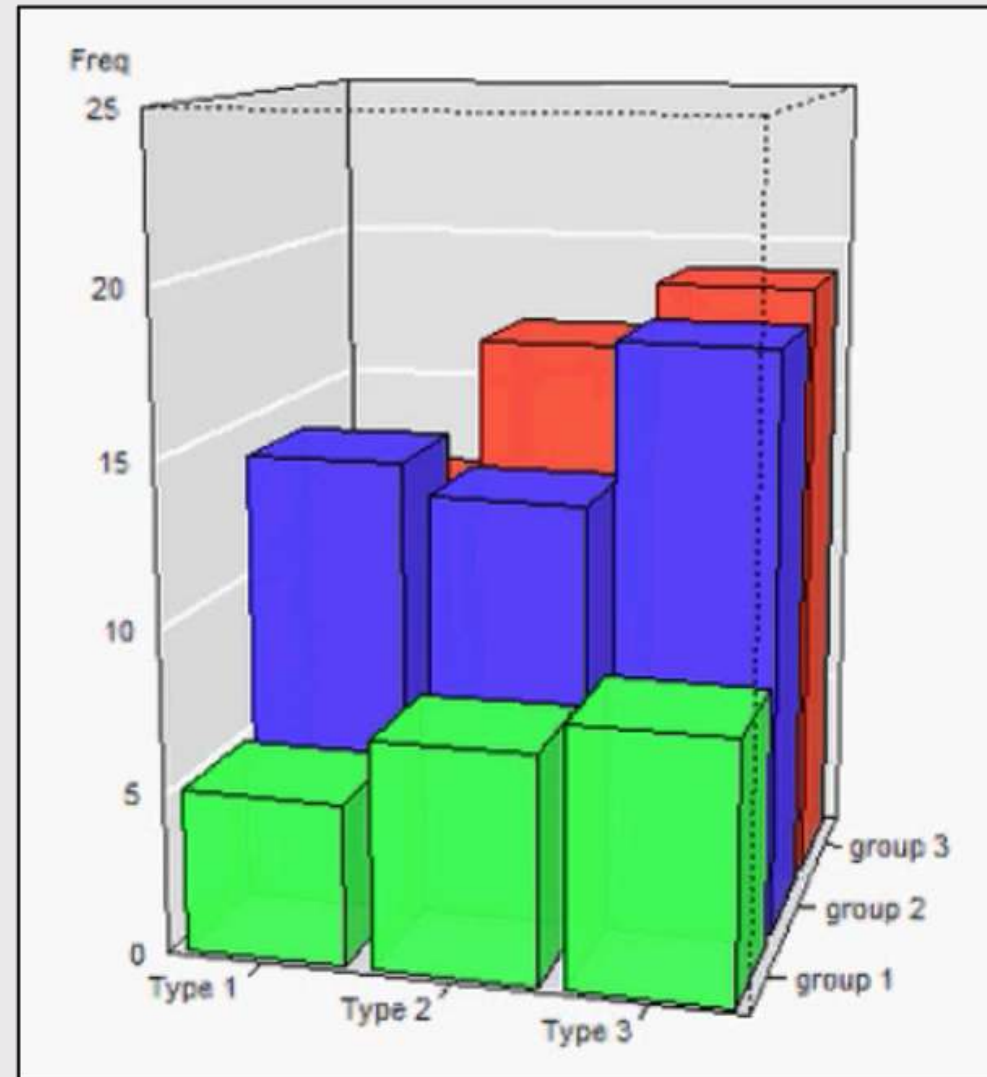


Penrose Stairs, made famous by
M.C. Escher (1898-1972)

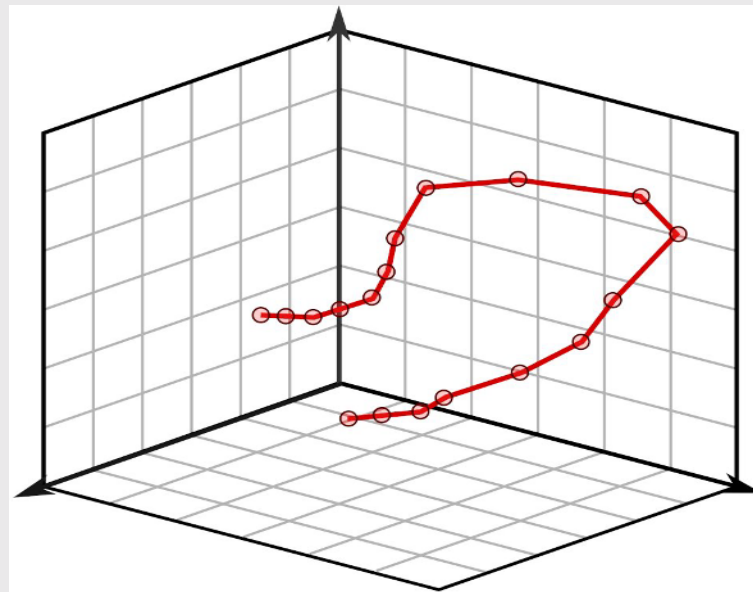
Ink proportions !=
true proportions



Occlusion: geoms are obscured



Multiple interpretations



Please never do this.

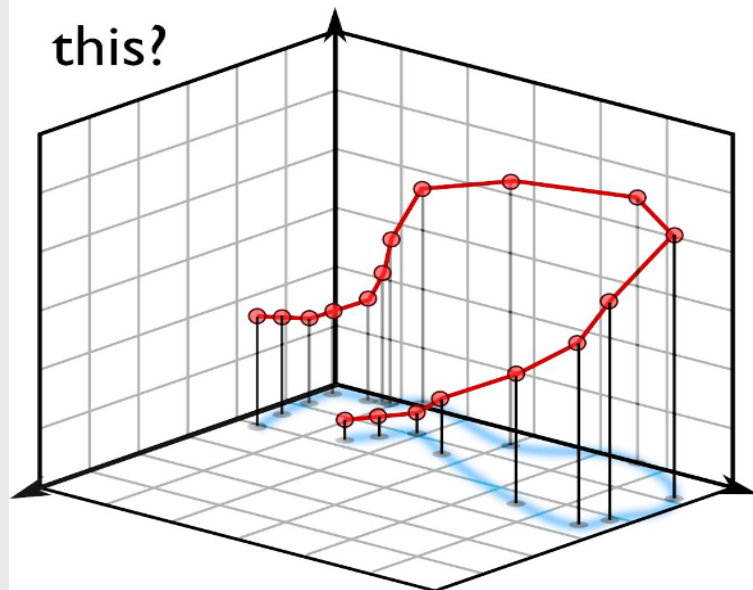


3D plots are ambiguous without a projection.

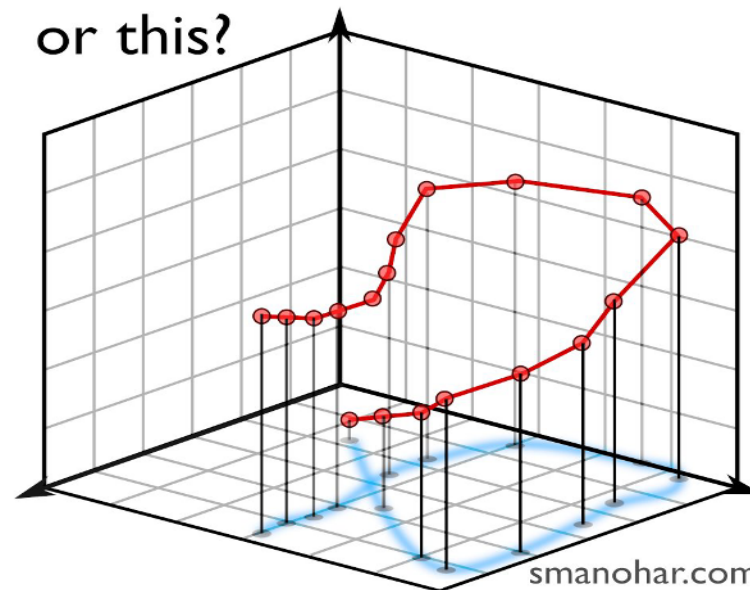
Each point has a whole line of possible 3D locations.

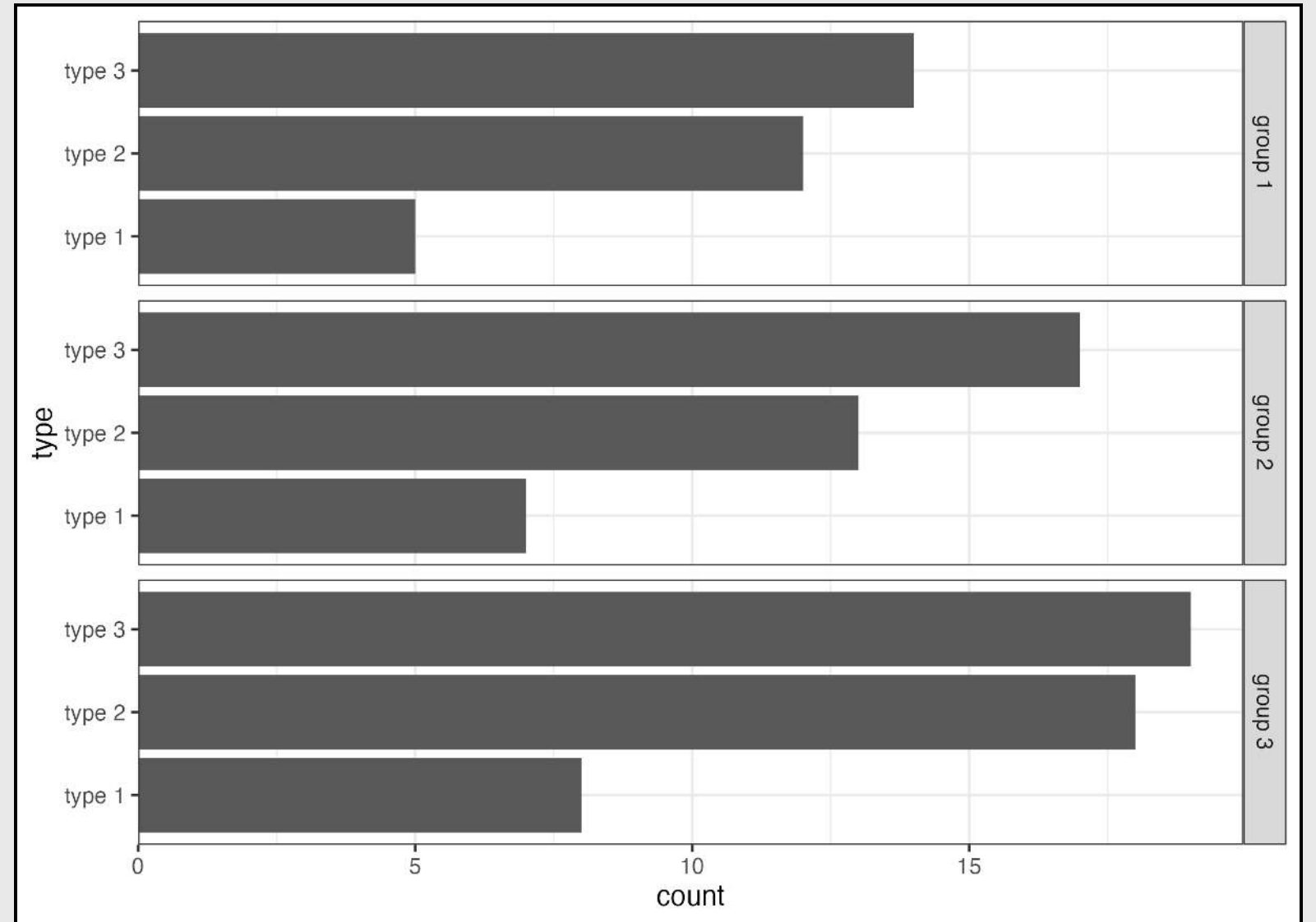
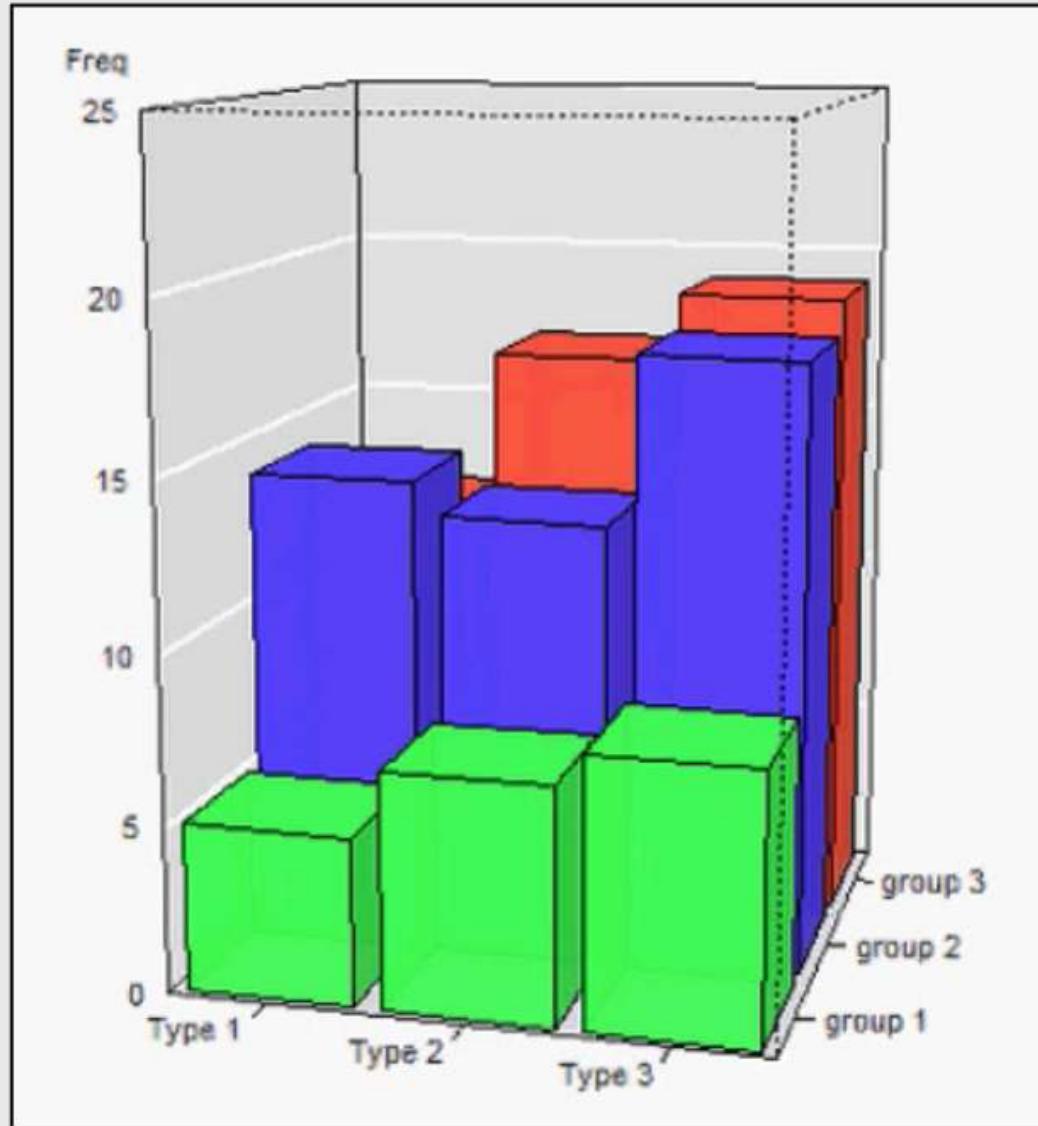
Do you mean...

this?



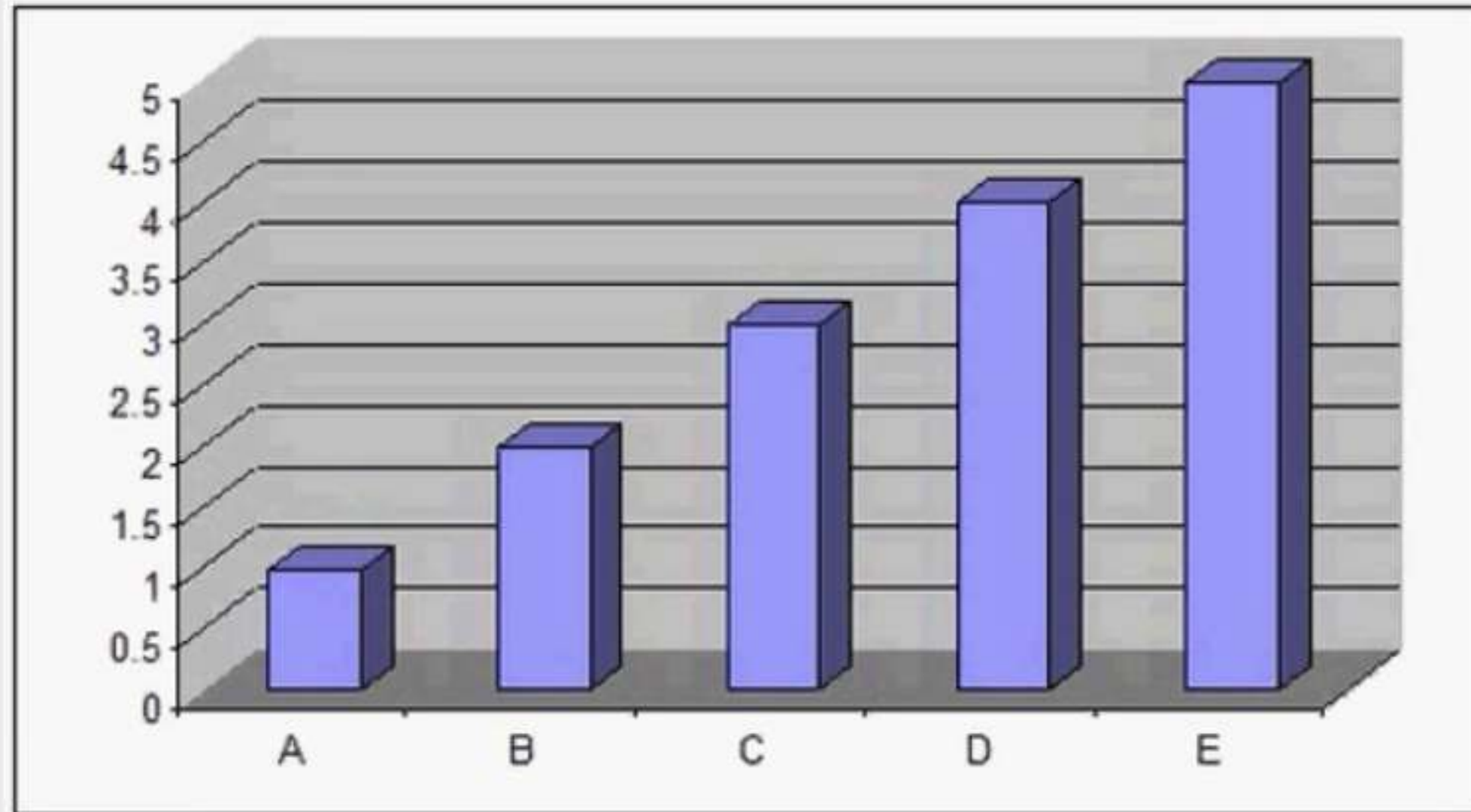
or this?

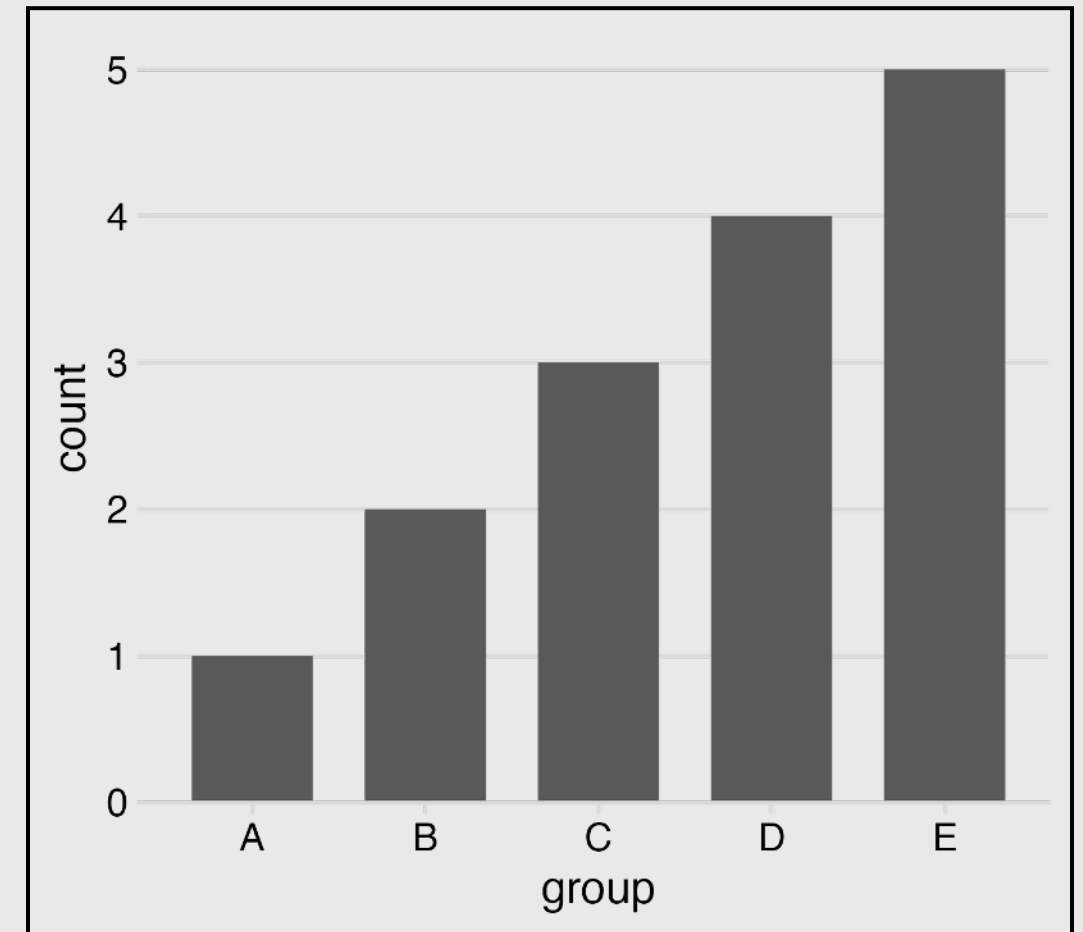
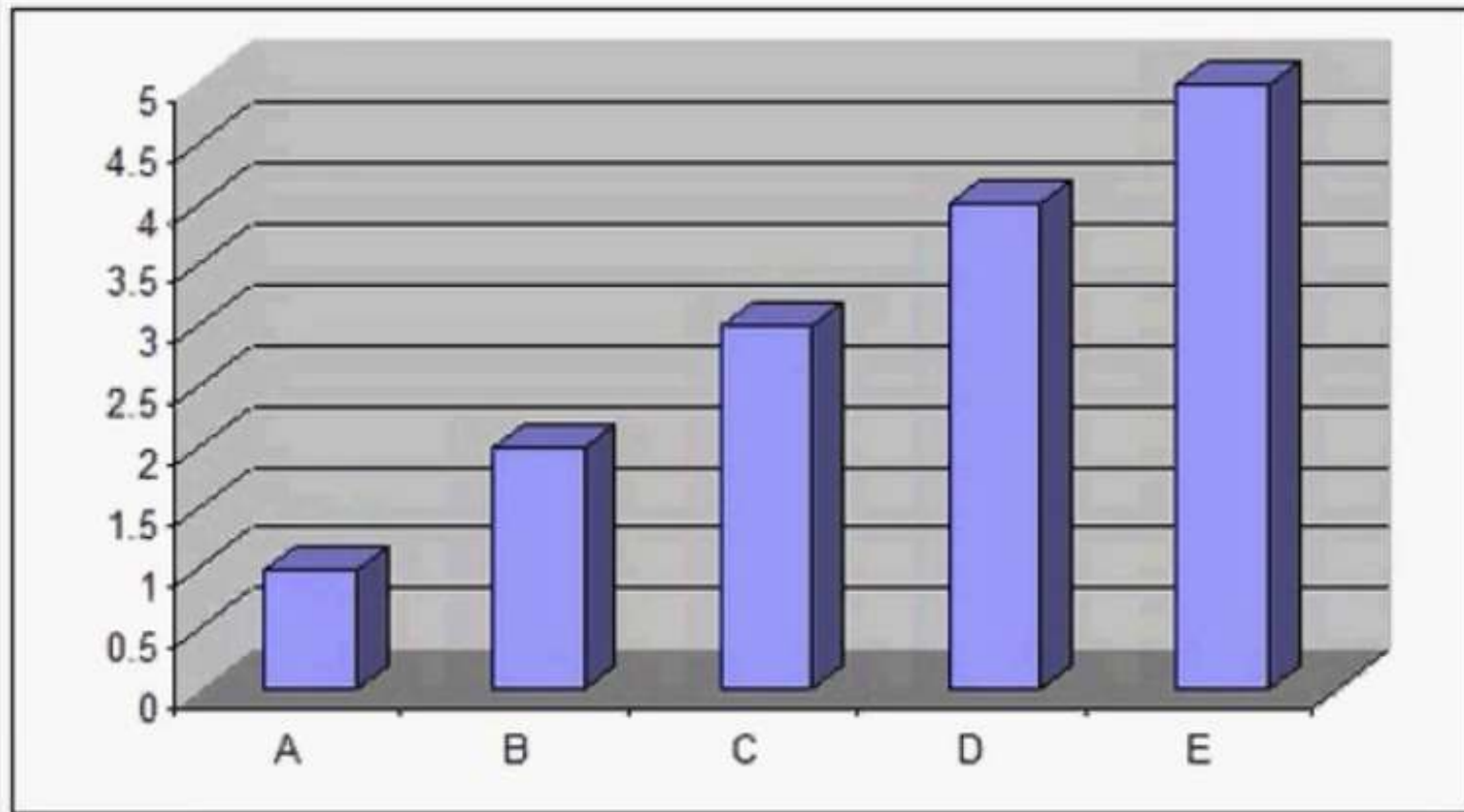




The third dimension distracts from the data

(this is what Tufte calls “chart junk”)

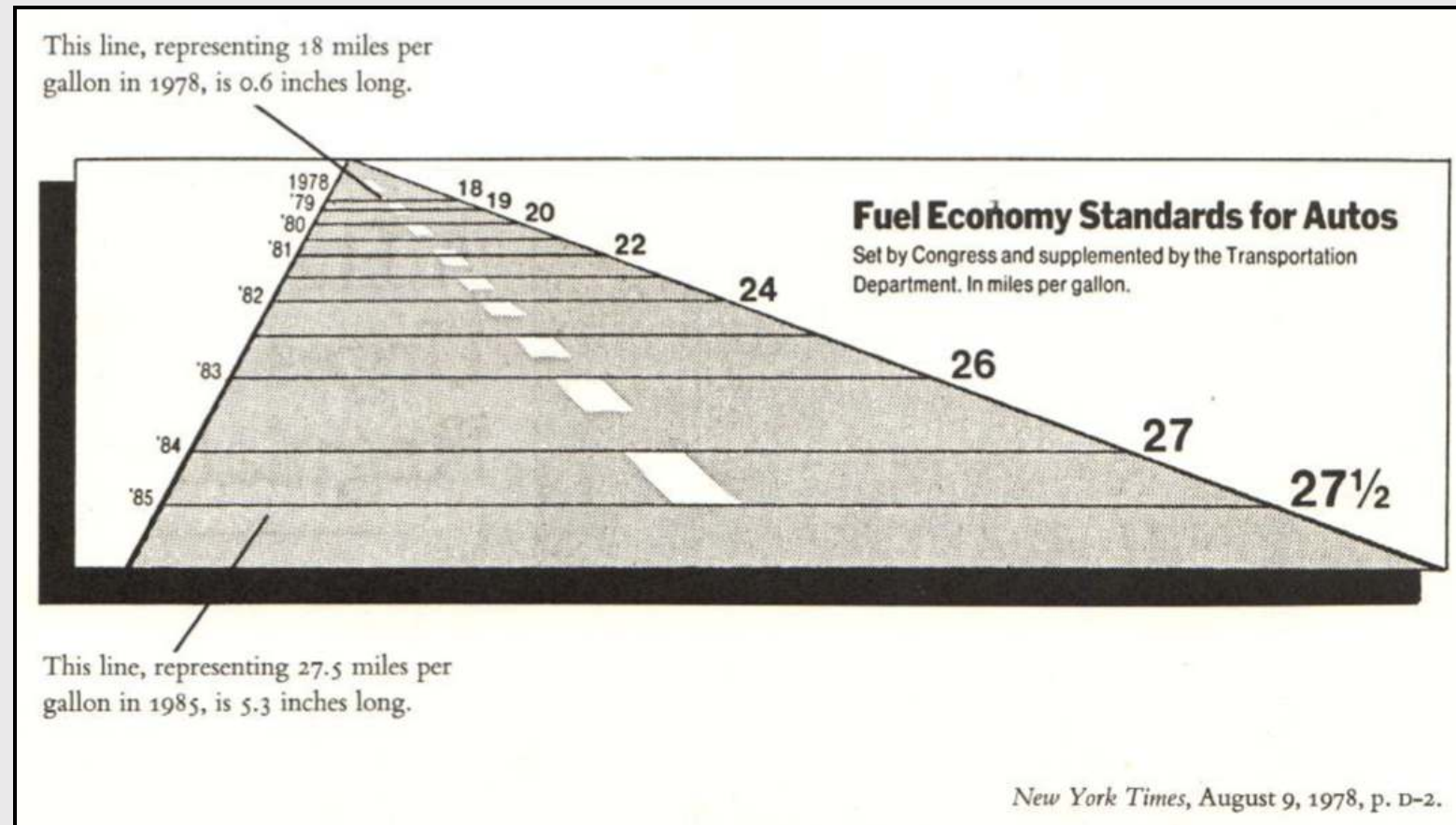




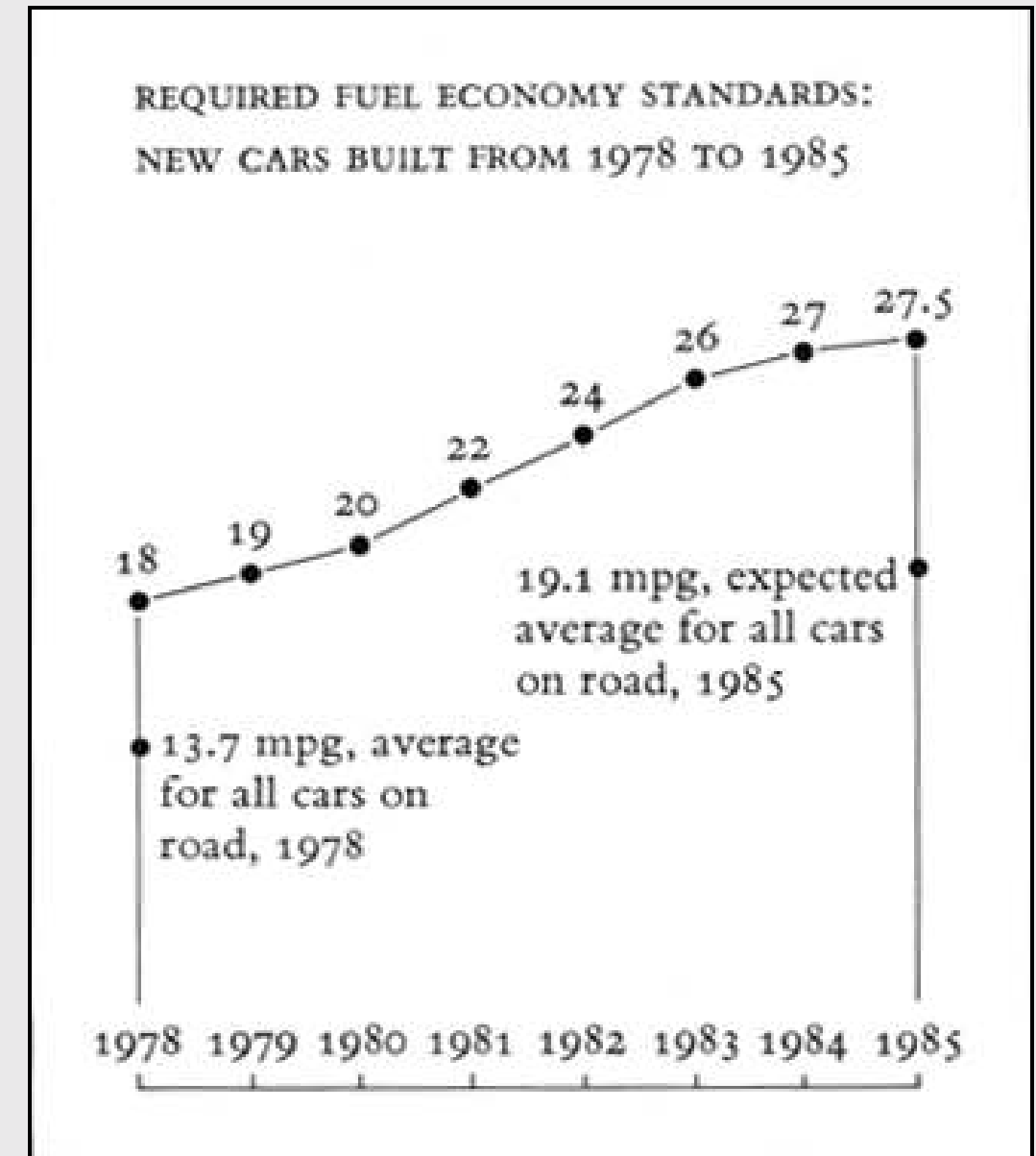
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*most of the time



Edward Tufte (2001) "The Visual Display of Quantitative Information", 2nd Edition, pg. 57-58.



Bar charts should always start at 0

Bar charts should always start at 0

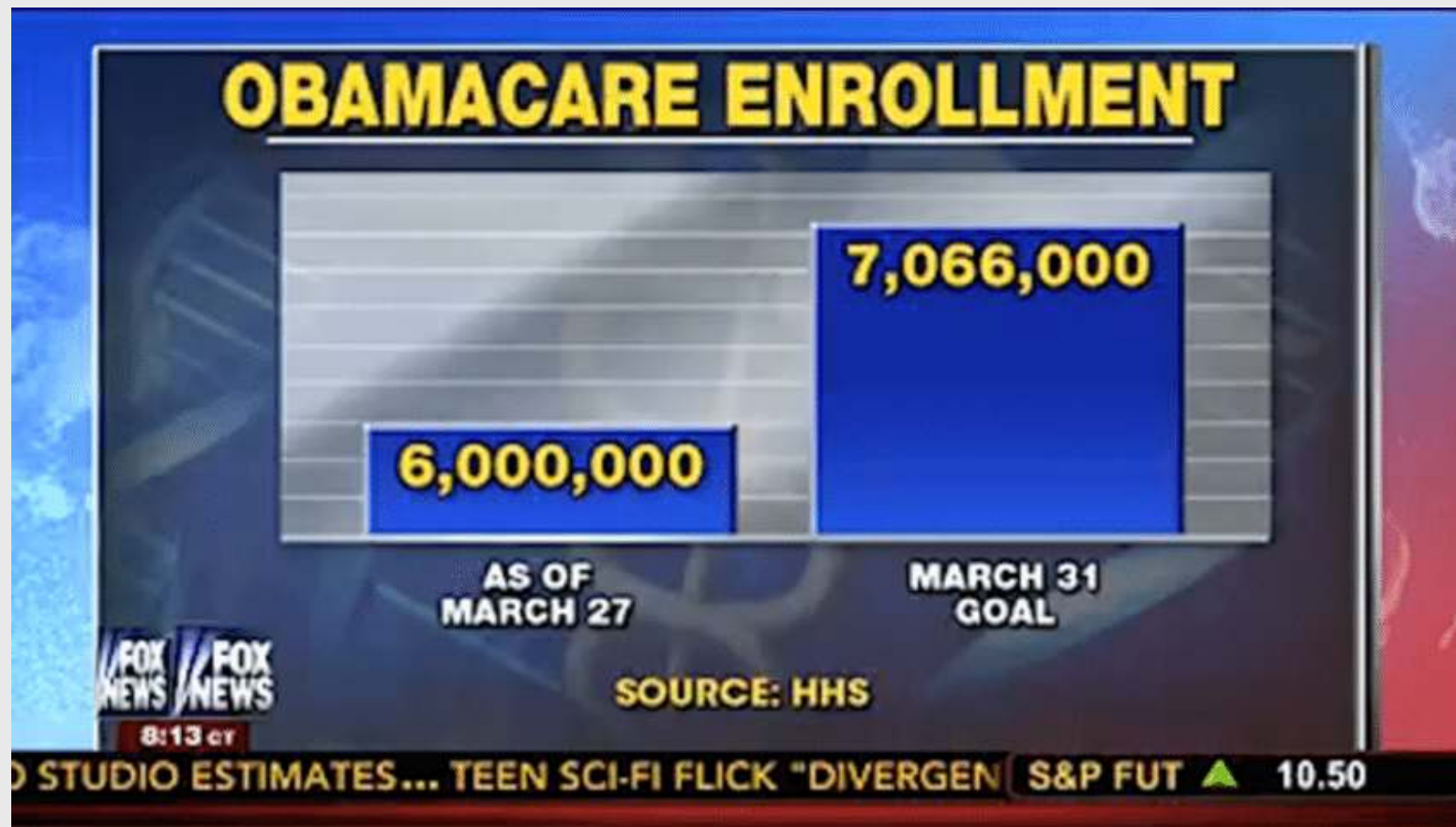
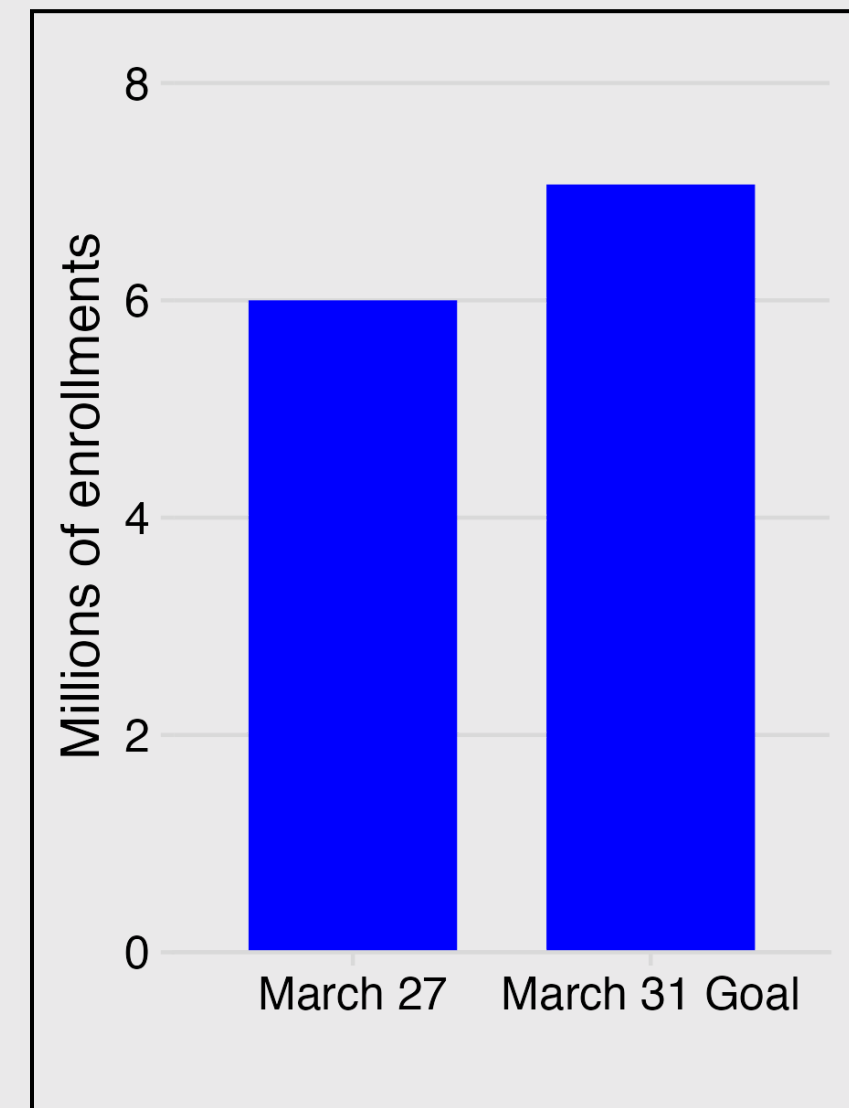


Image from <http://livingqlikview.com/the-9-worst-data-visualizations-ever-created/>



Don't cherry-pick your data

Don't cherry-pick your data

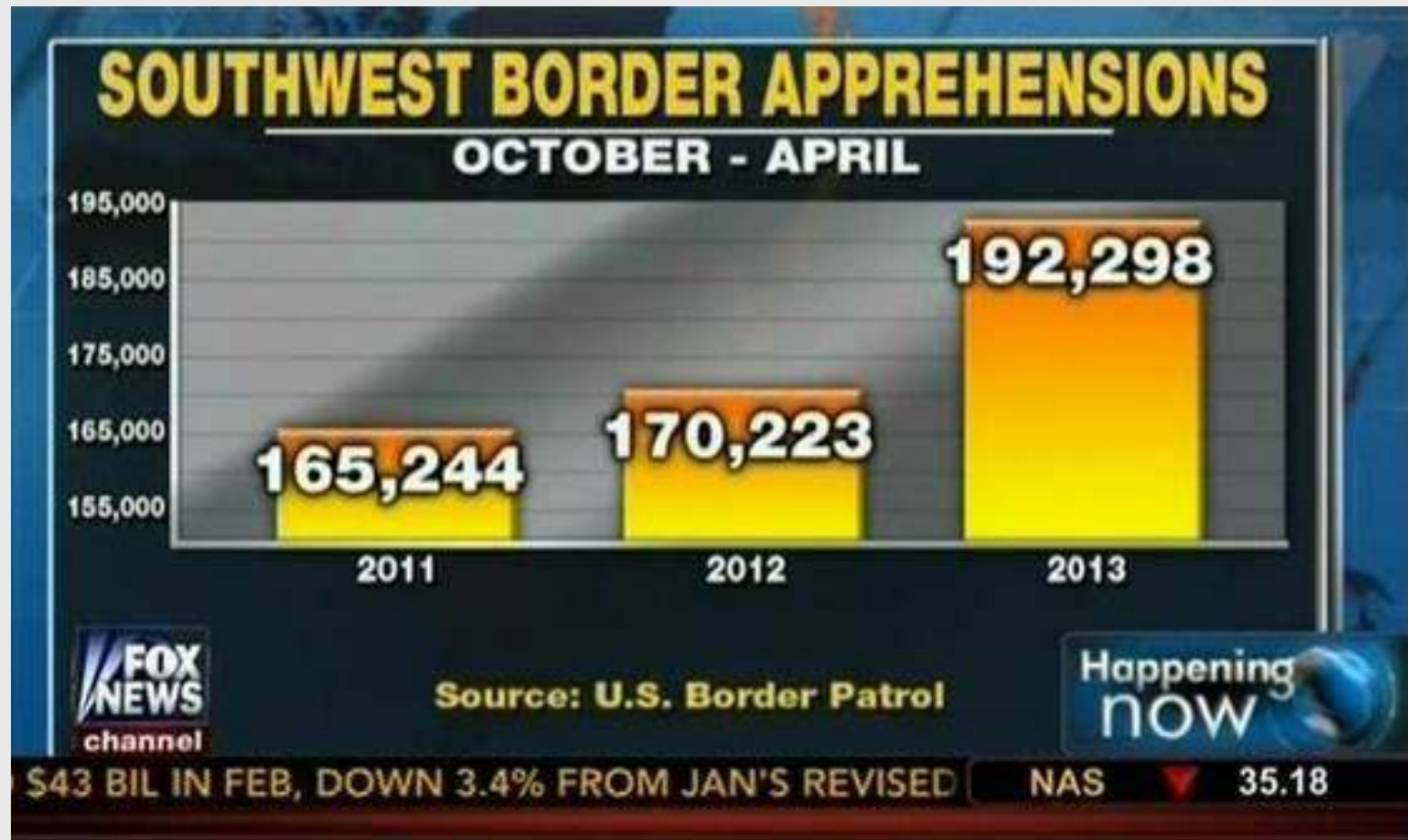
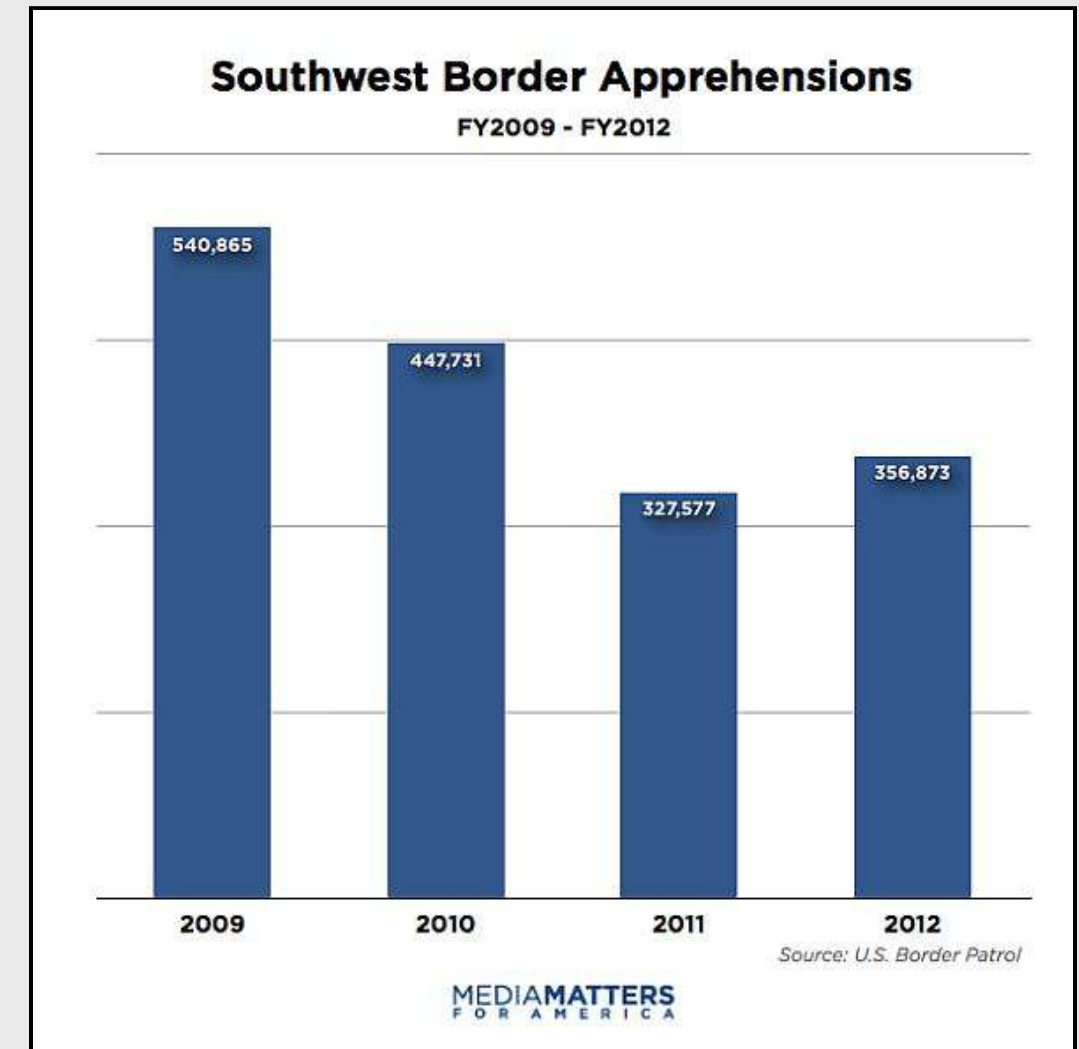


Image from <https://www.mediamatters.org/fox-news/fox-news-newest-dishonest-chart-immigration-enforcement>



Make sure your chart makes sense

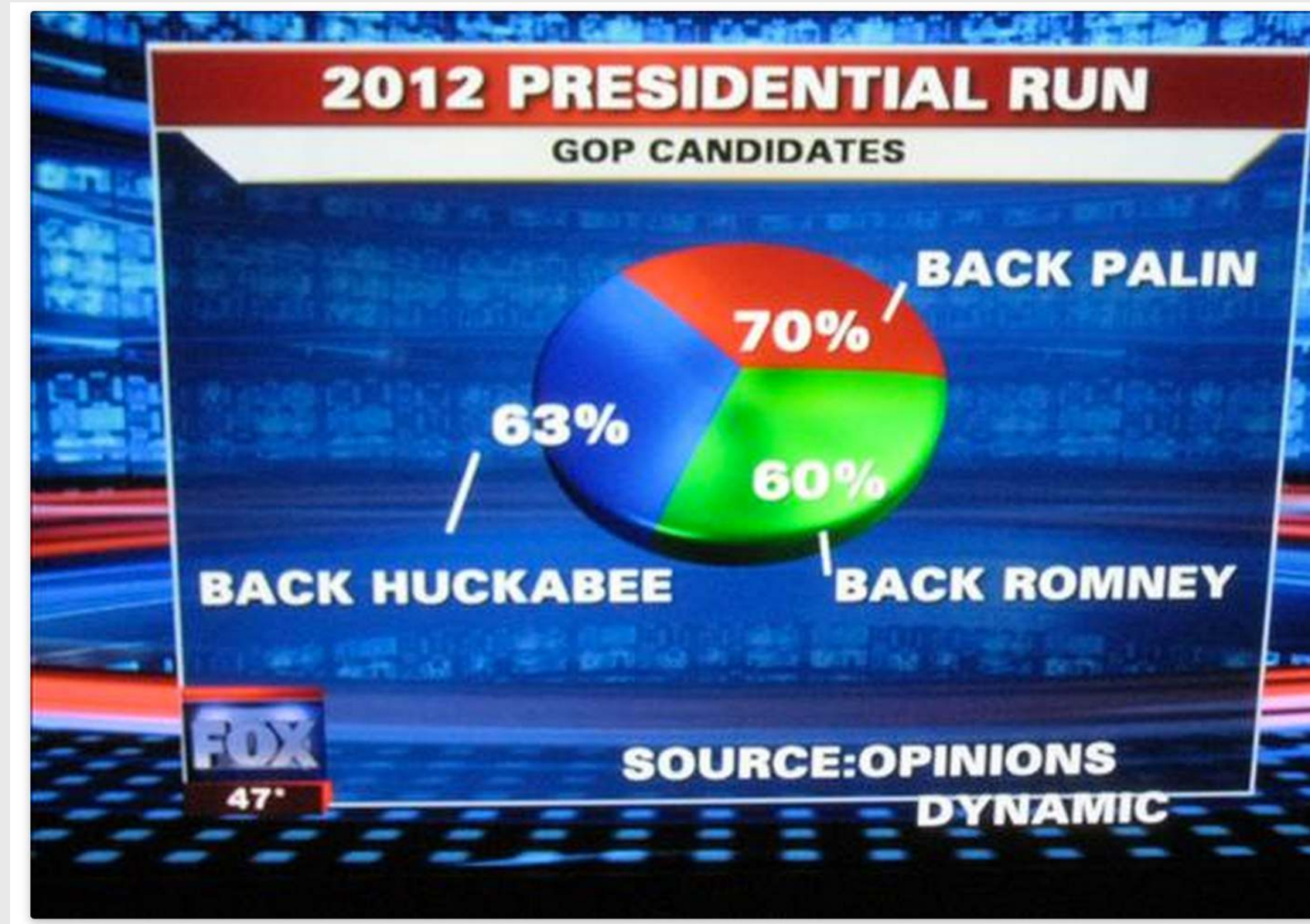
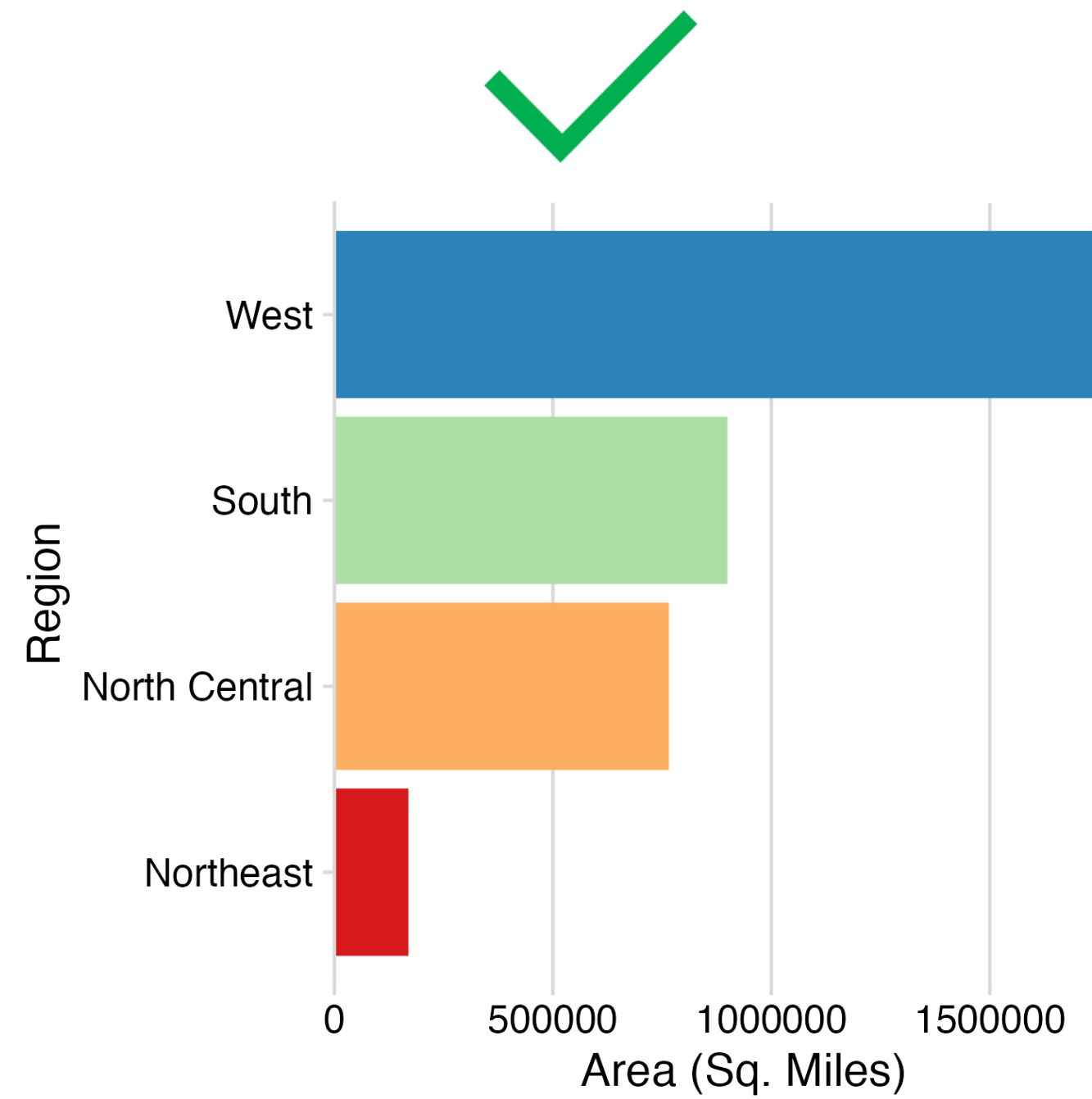
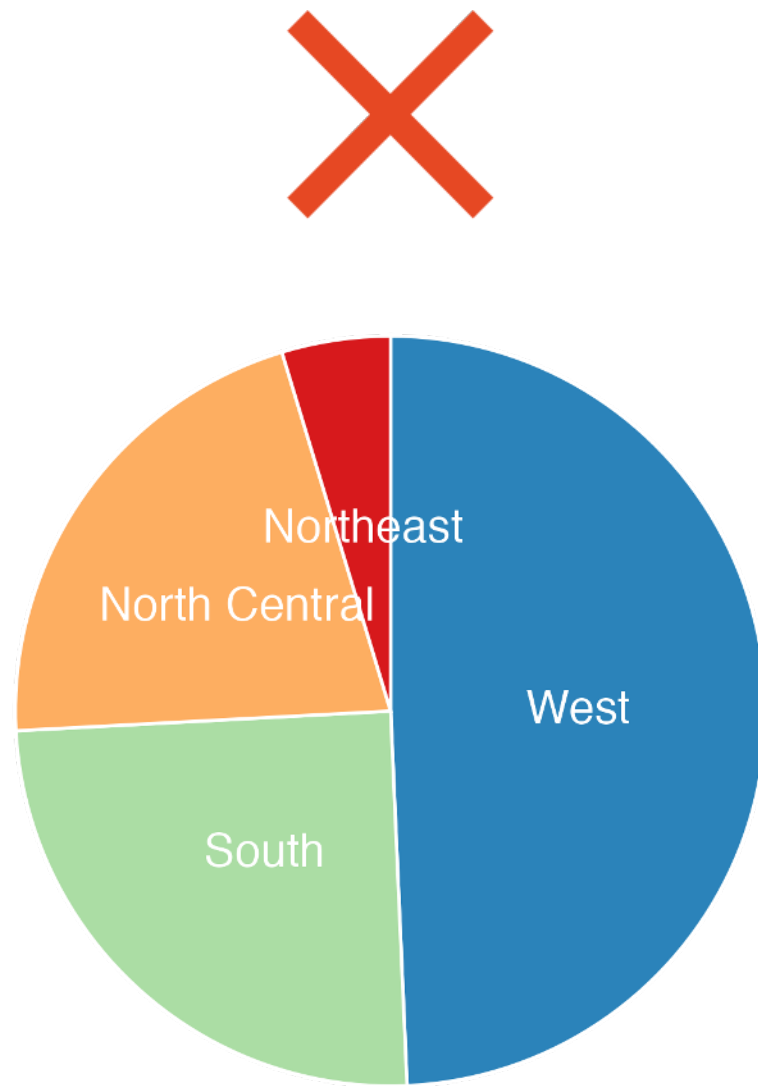


Image from <http://livingqlikview.com/the-9-worst-data-visualizations-ever-created/>

10 Data Viz Best Practices

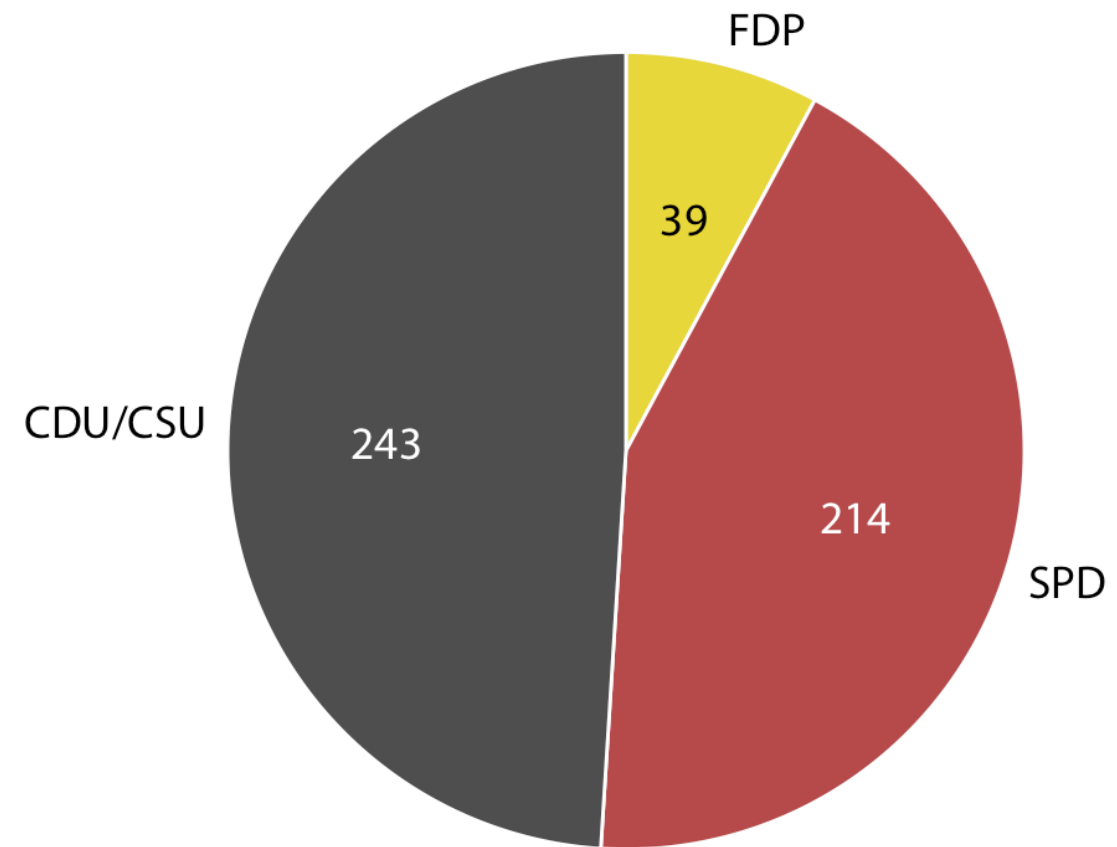
1. Remove chart chunk
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*most of the time

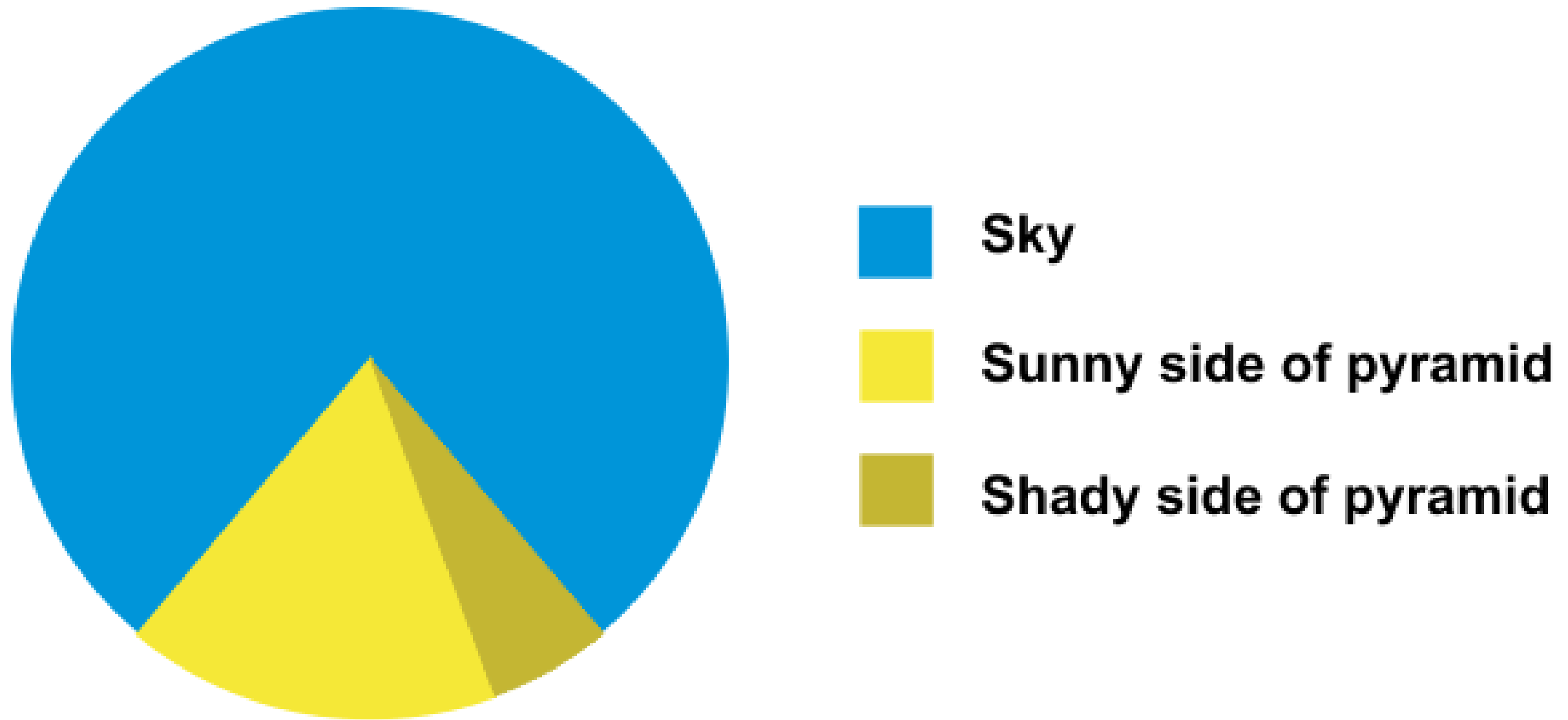


Exceptions:

- Small data
- Simple fractions
- If sum of parts matters



Best pie chart of all time

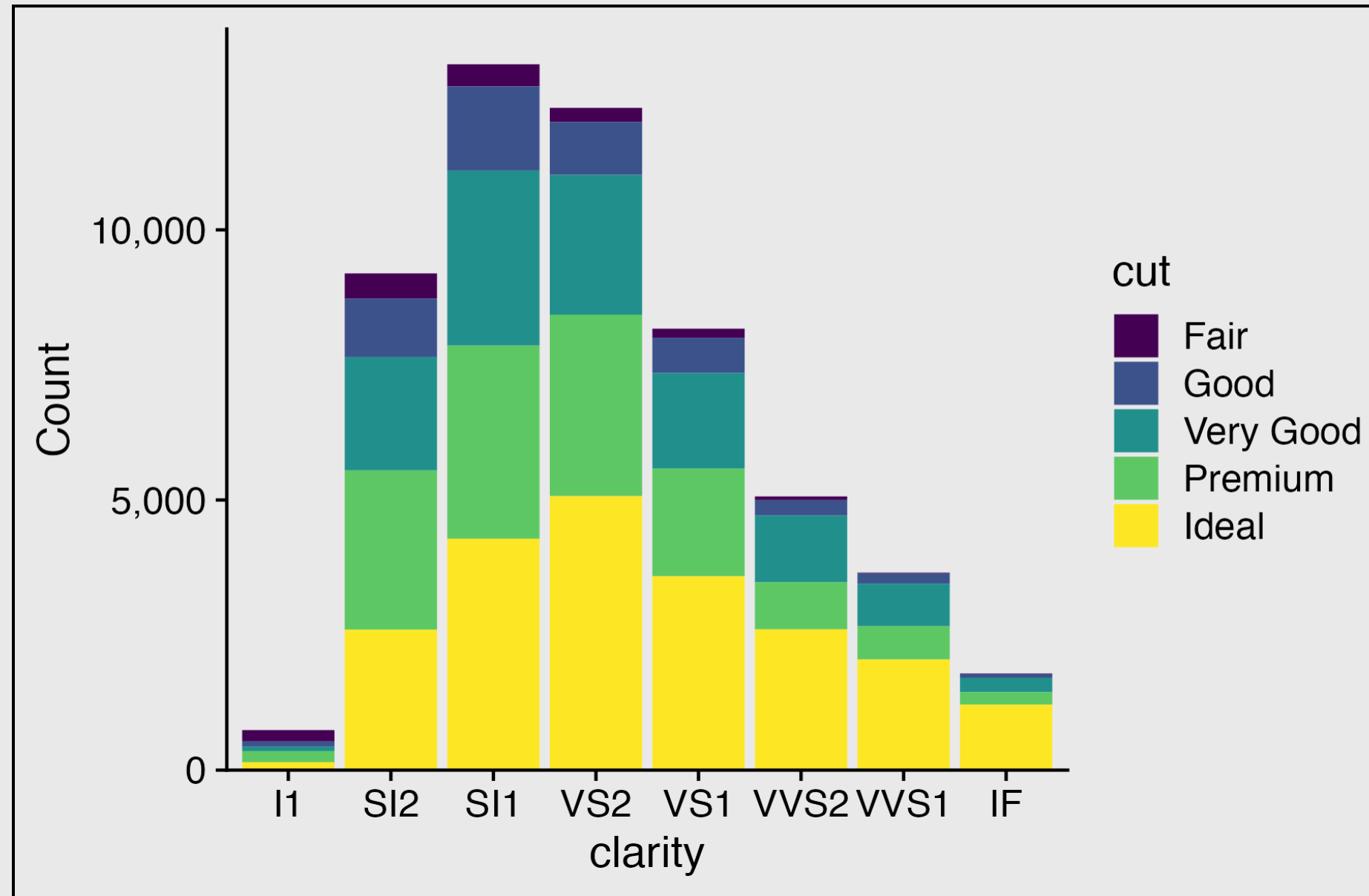


10 Data Viz Best Practices

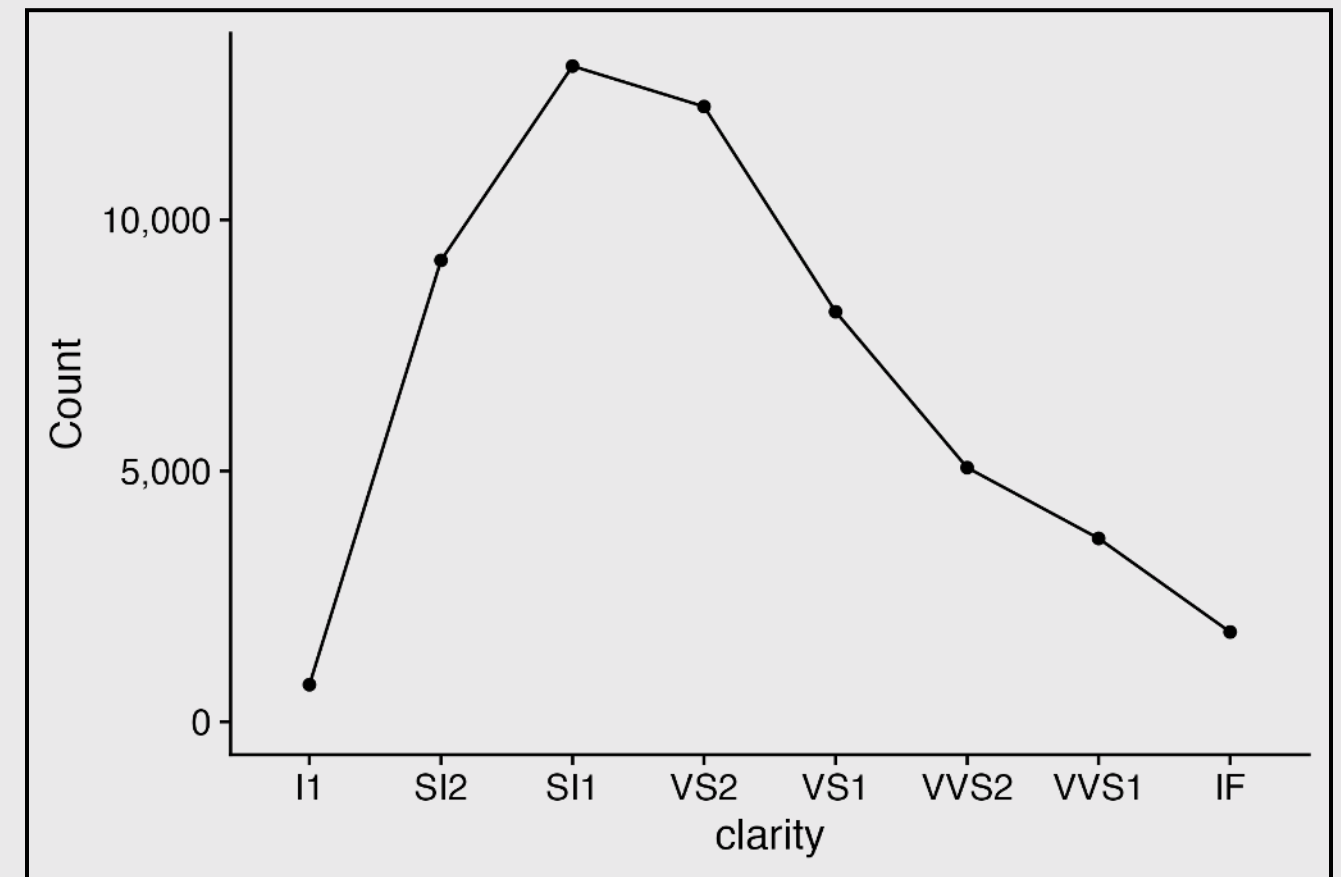
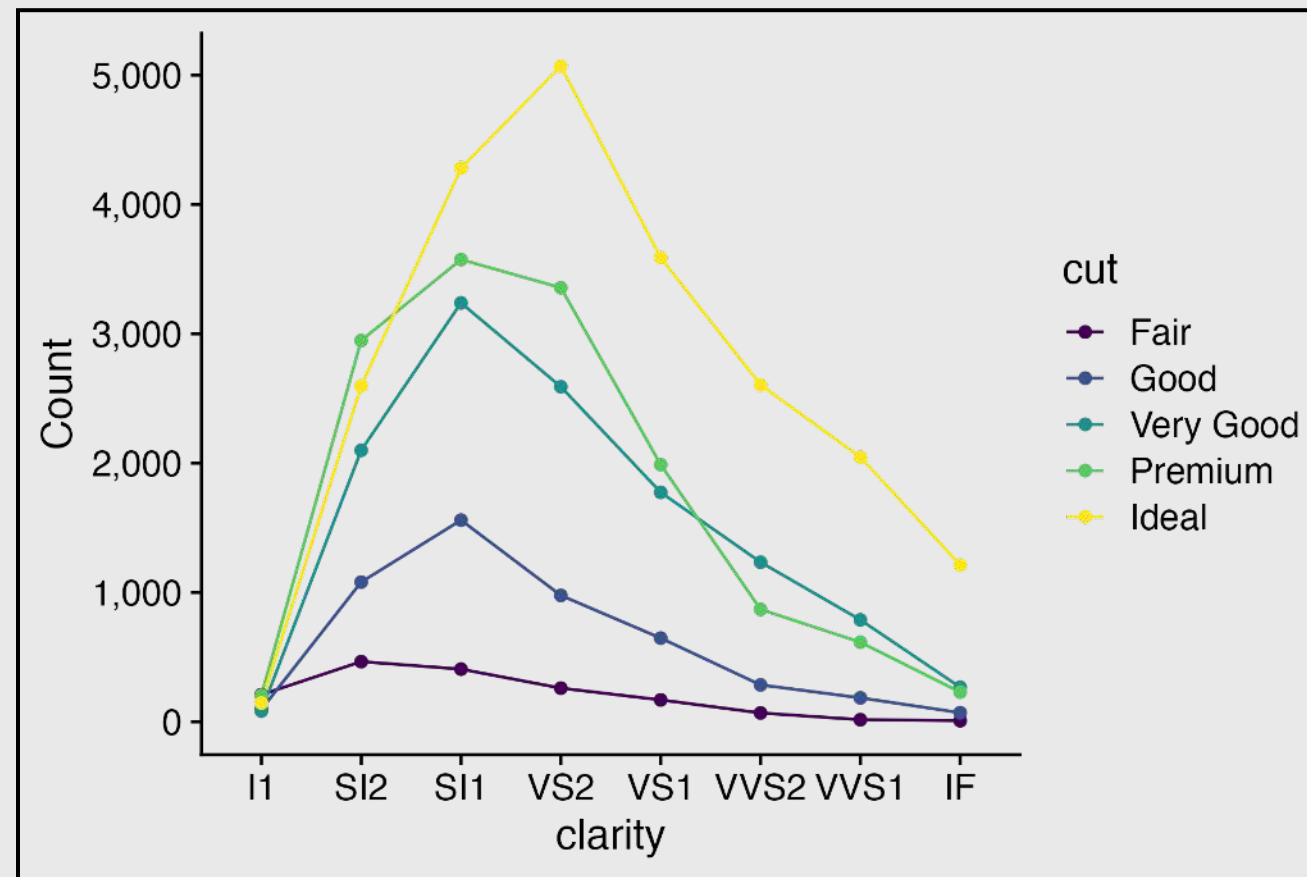
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Stacked bars are rarely a good idea

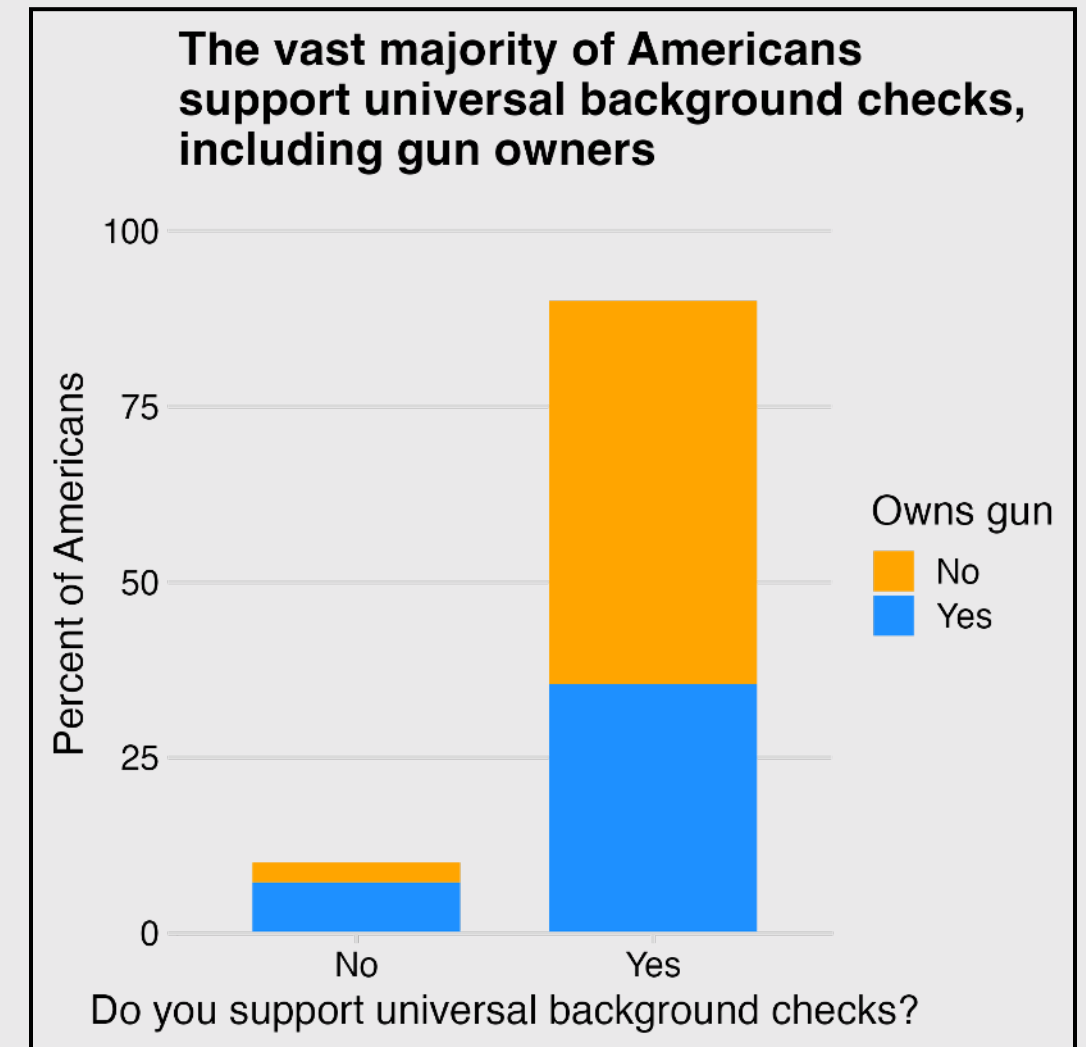
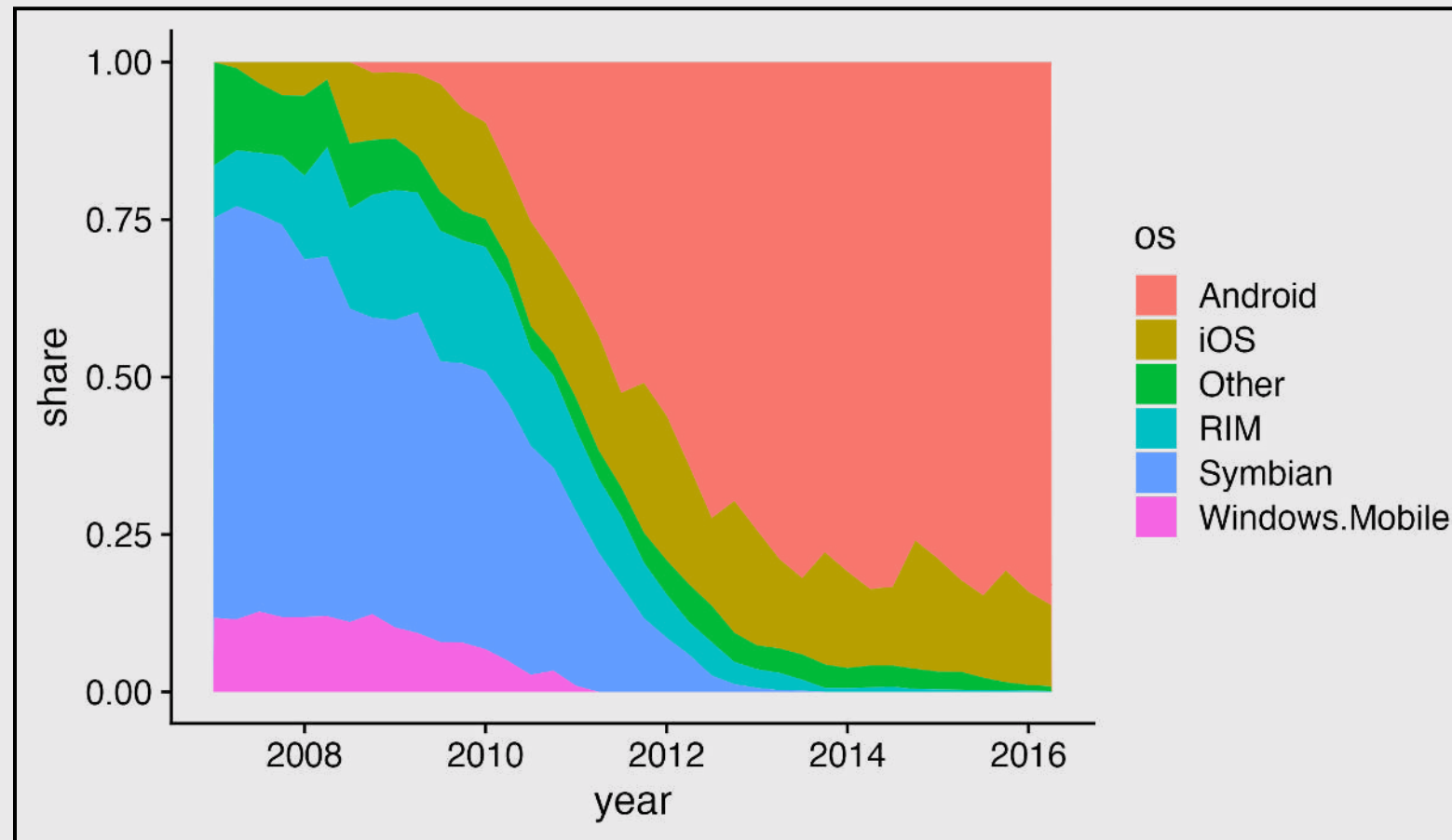


“Parallel coordinates” plot usually works better



Exception:

When you care about the *total* more than the categories

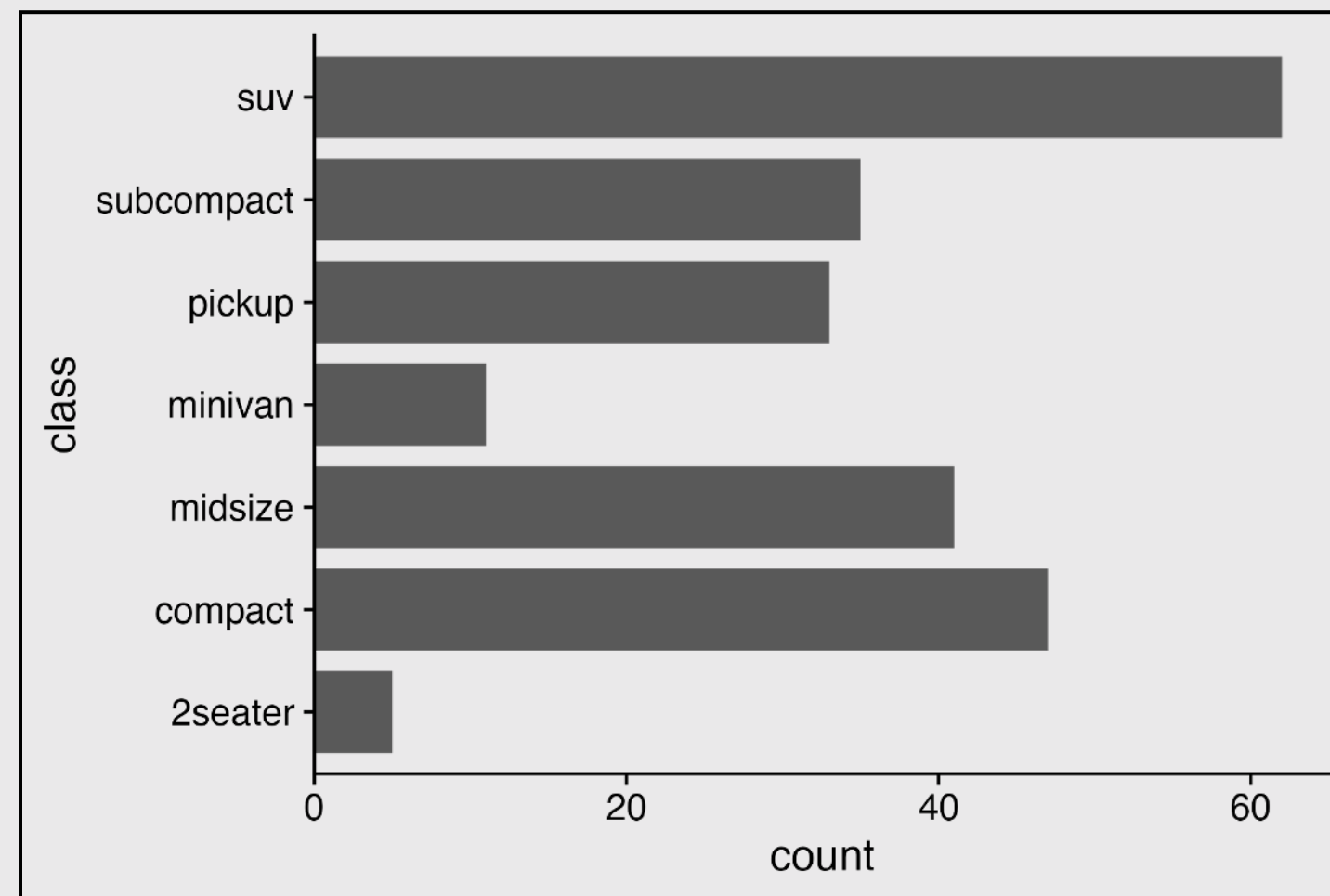
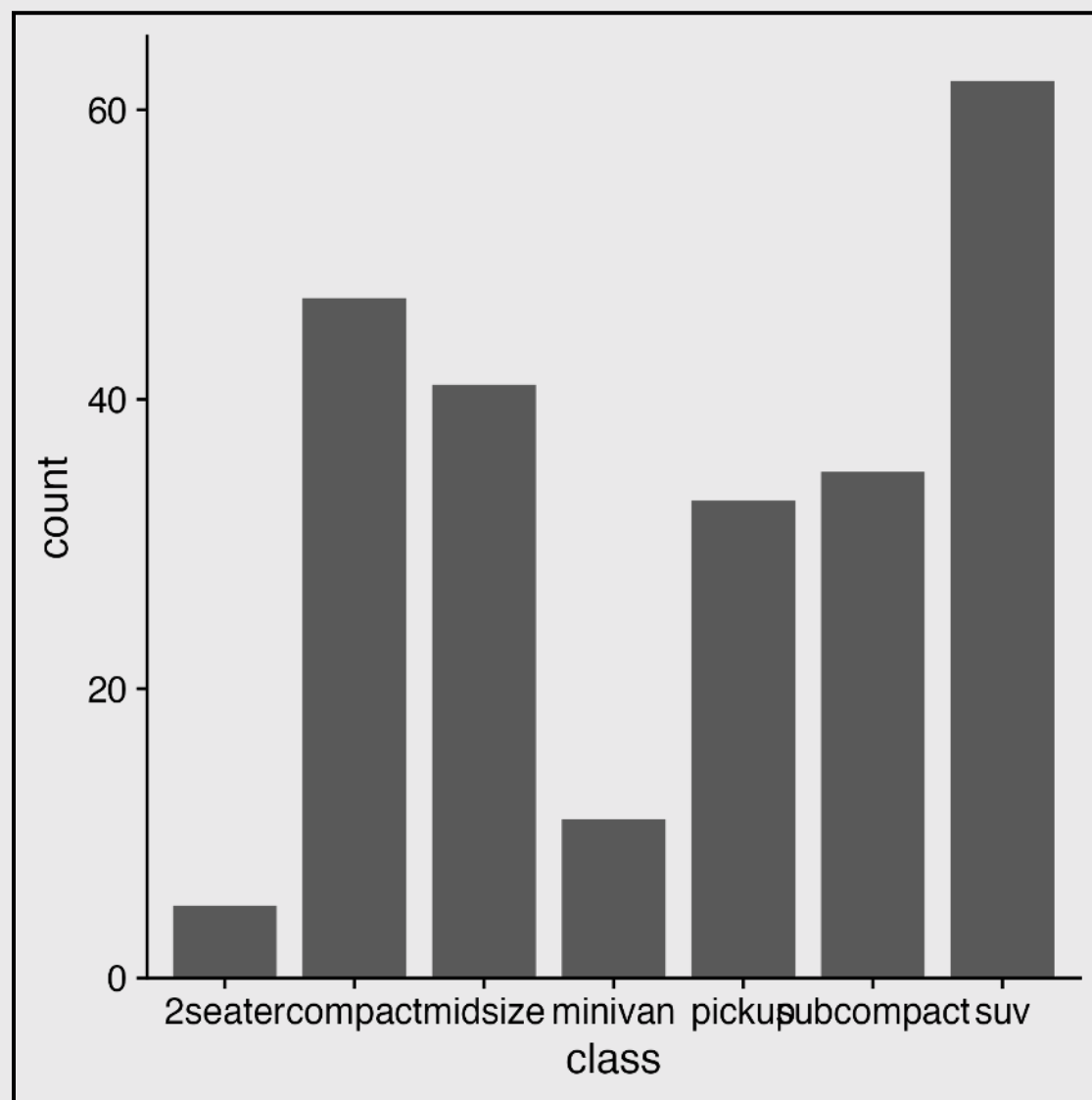


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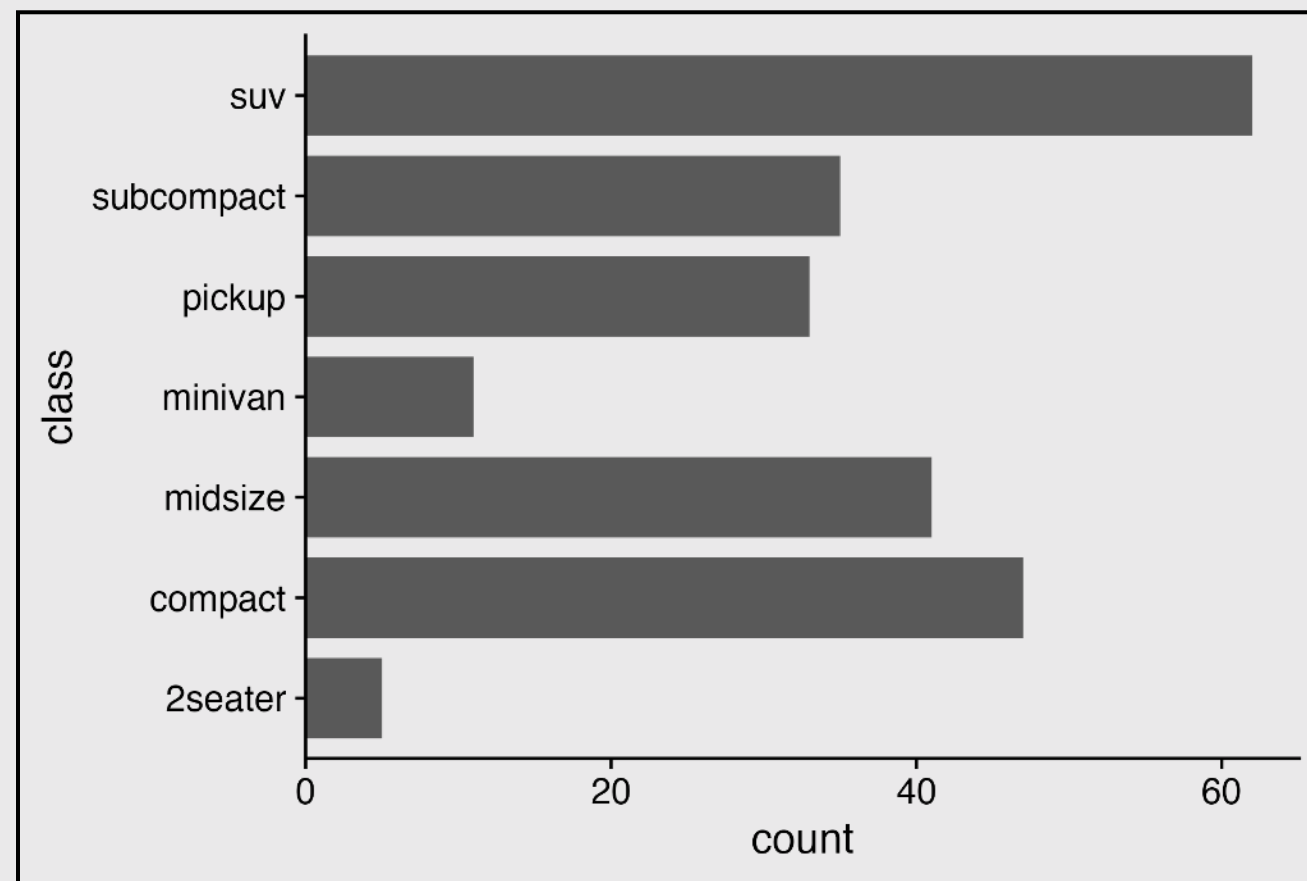
Rotate axes if you can't read them



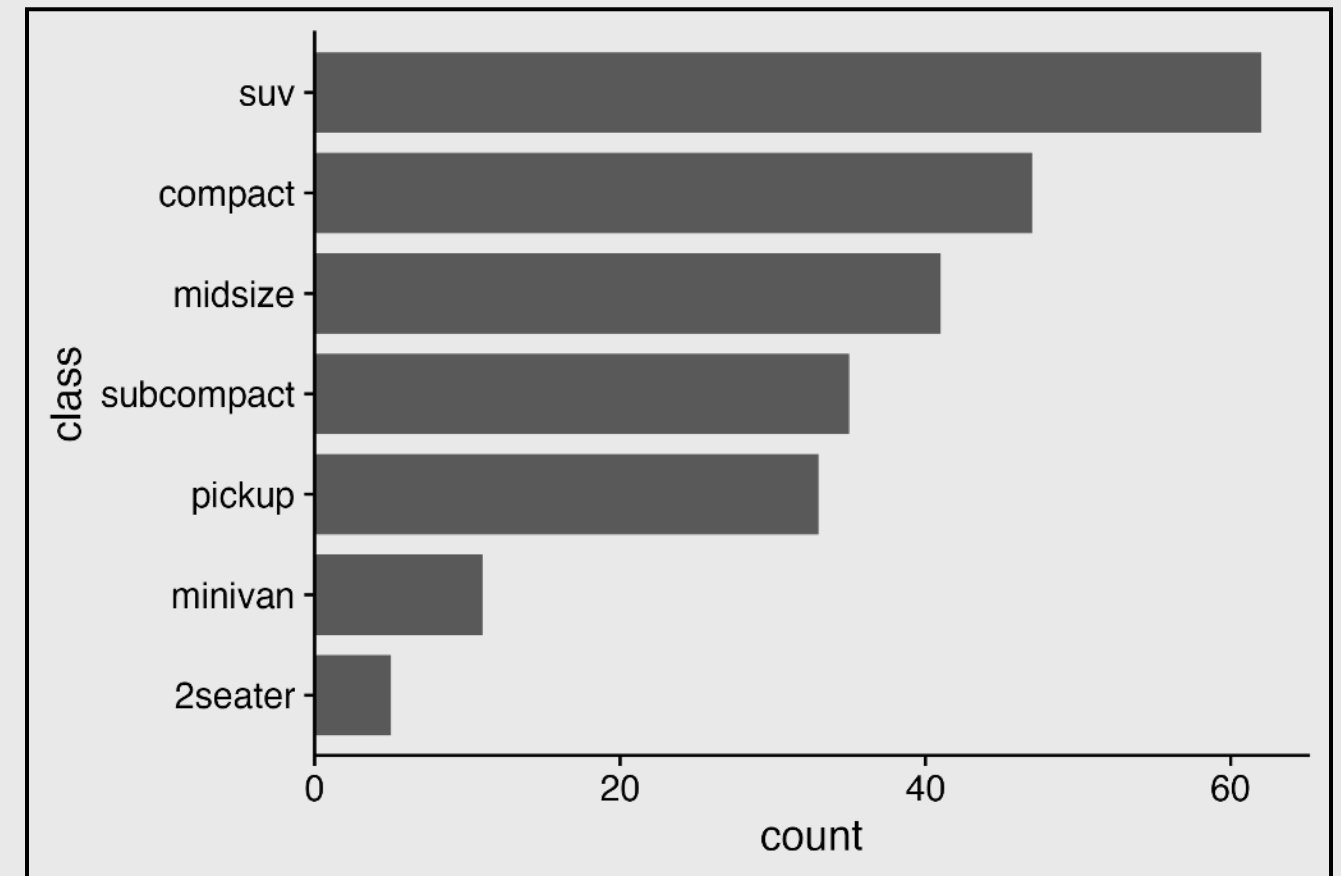
Default order is almost always wrong



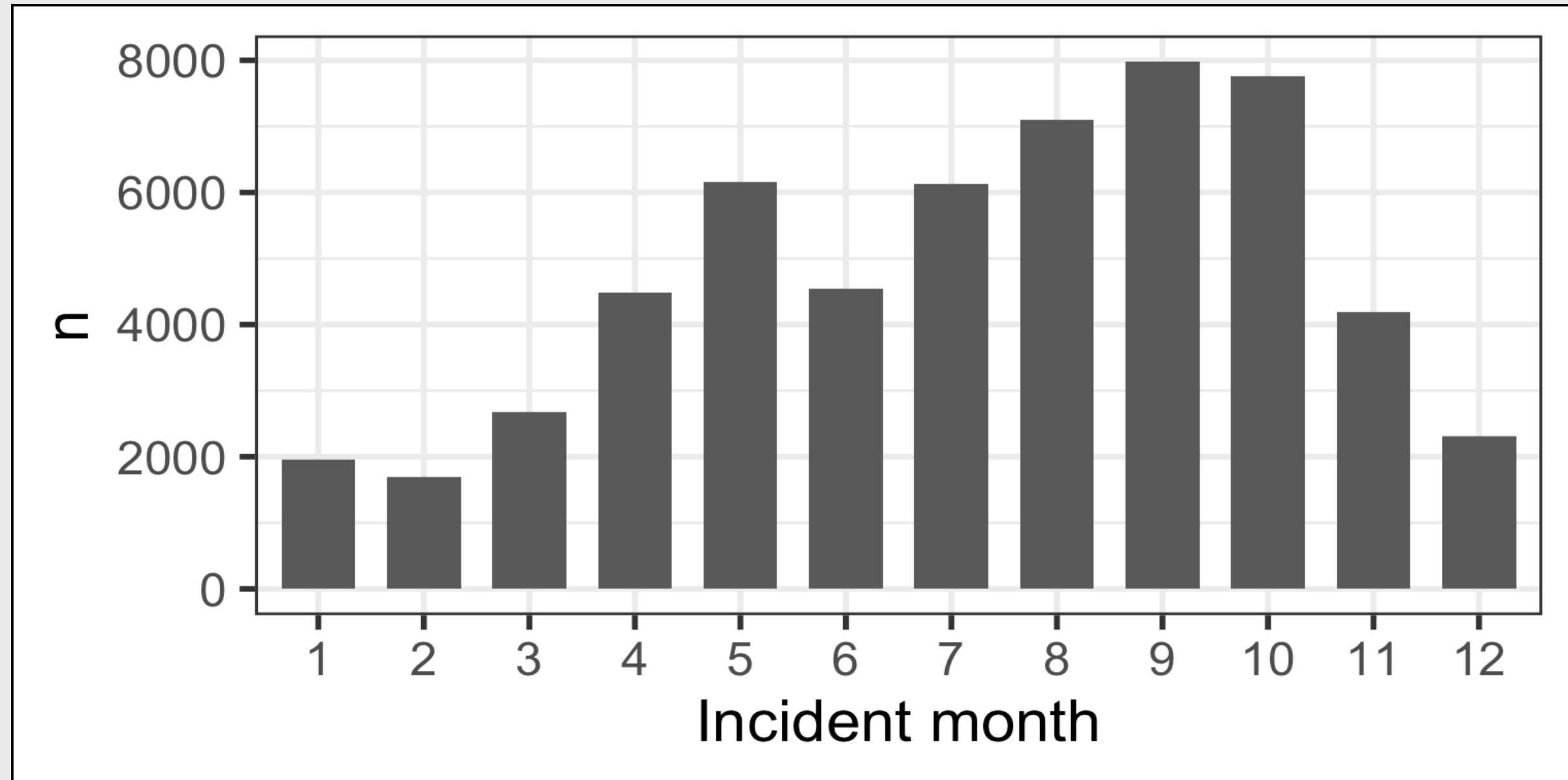
Ordered by alphabet (default)



Ordered by count



Exception: Ordinal variables

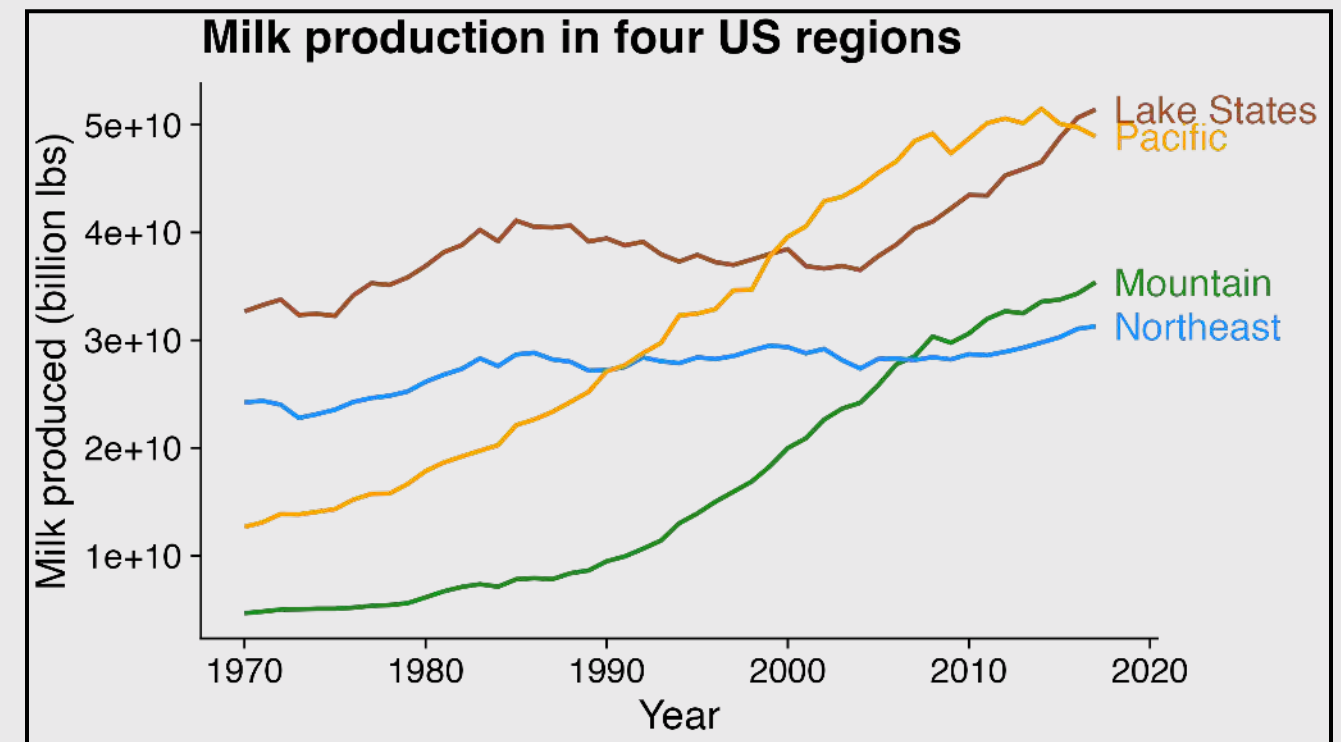
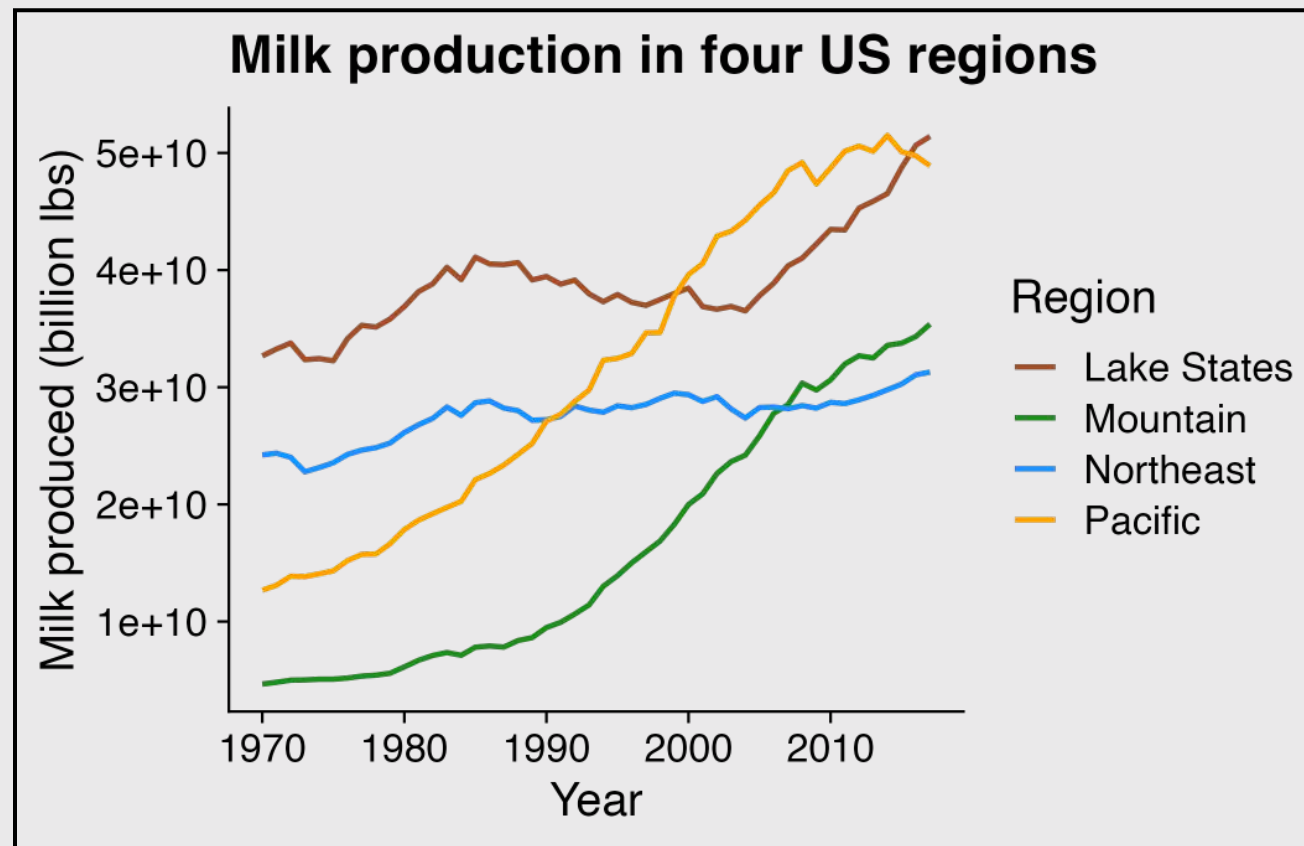


10 Data Viz Best Practices

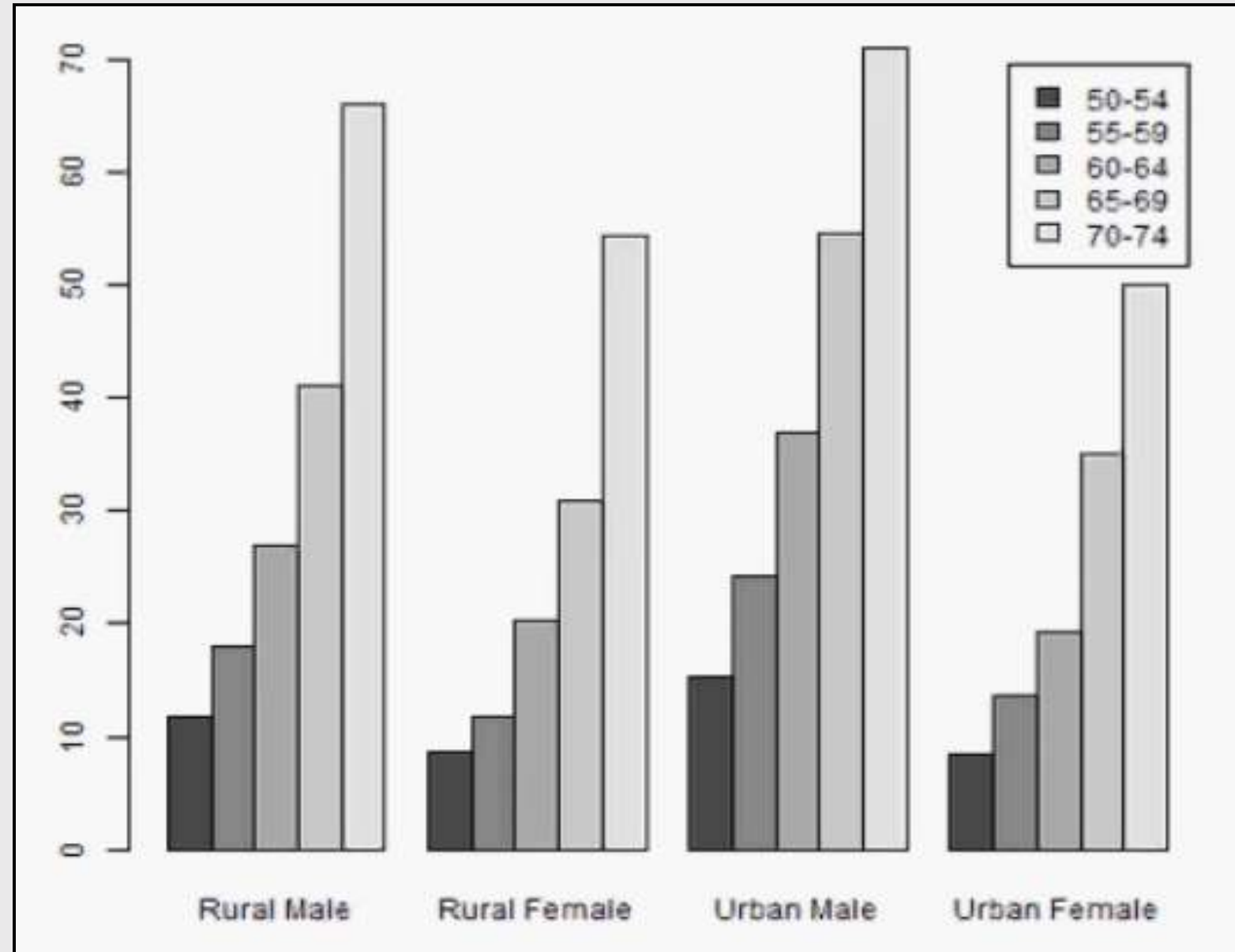
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Directly label geoms



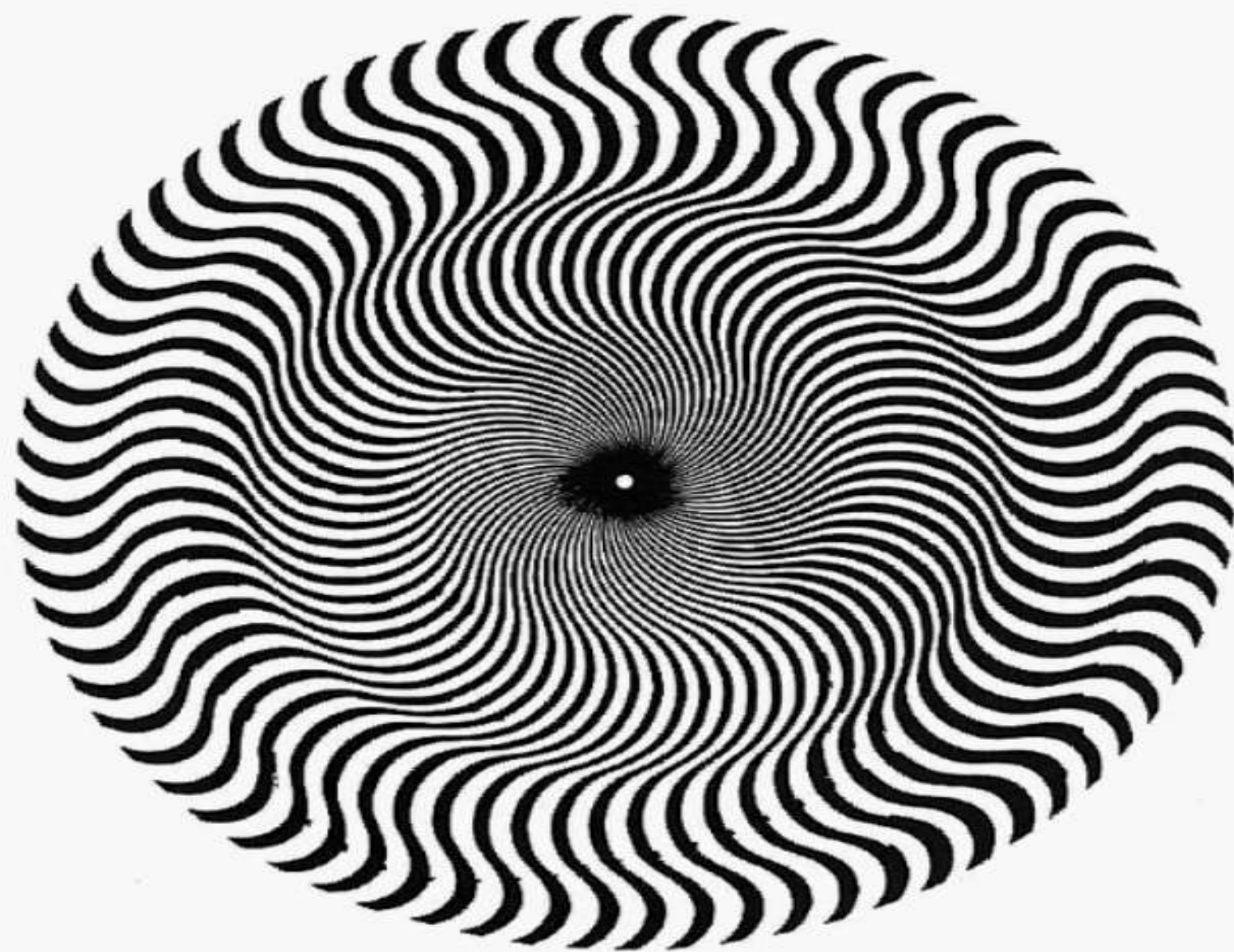
Exception: When you have repeated categories

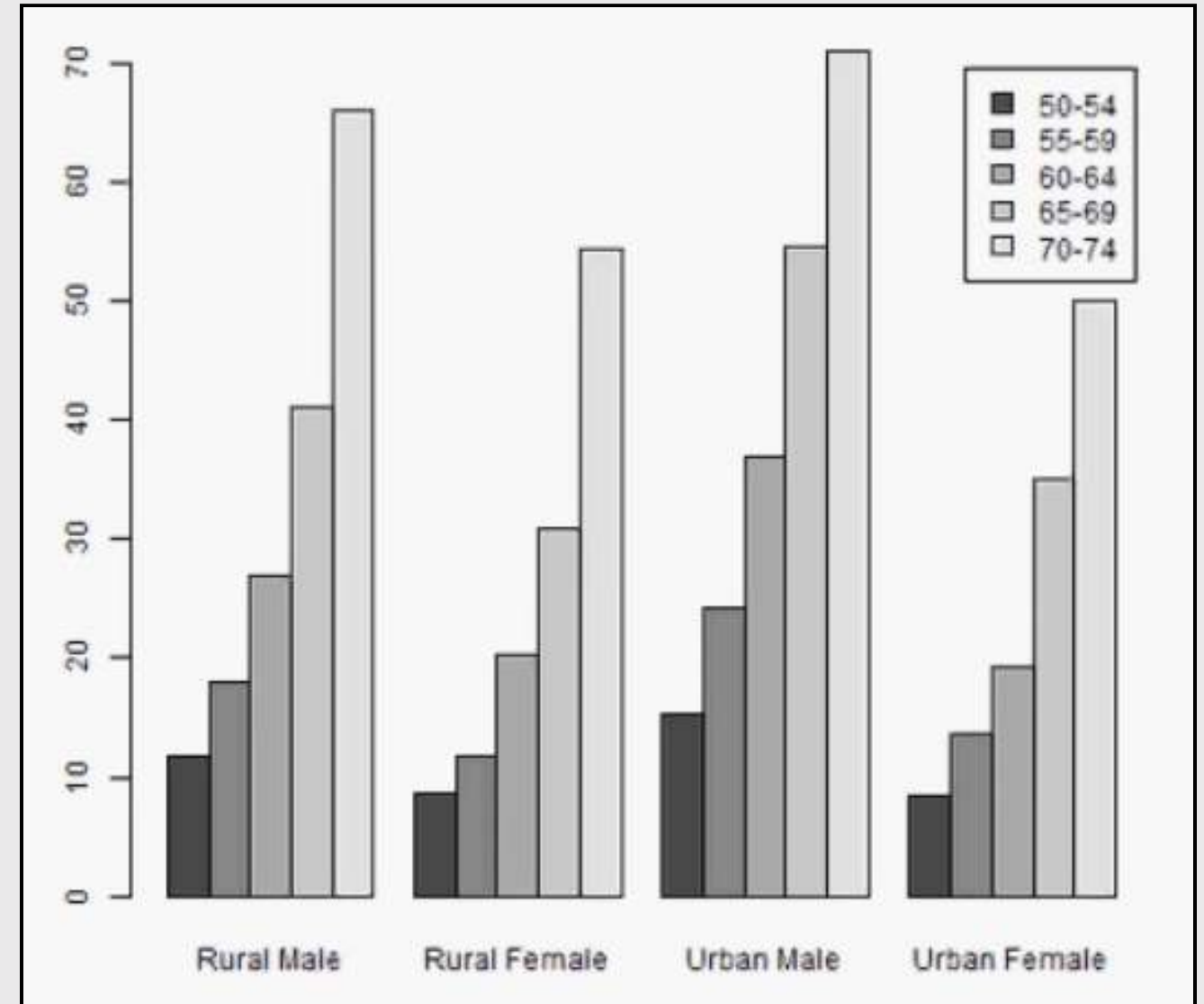
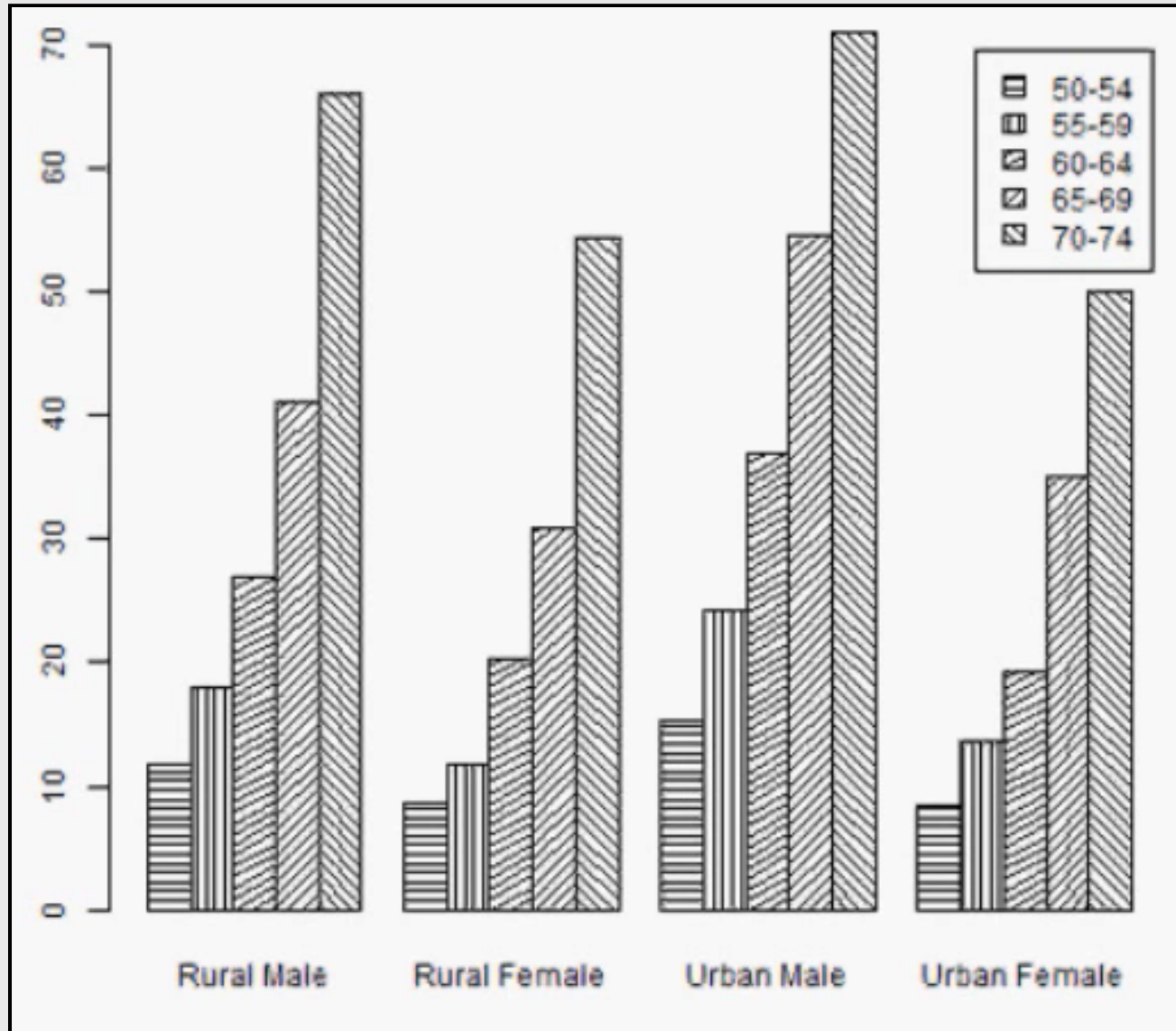


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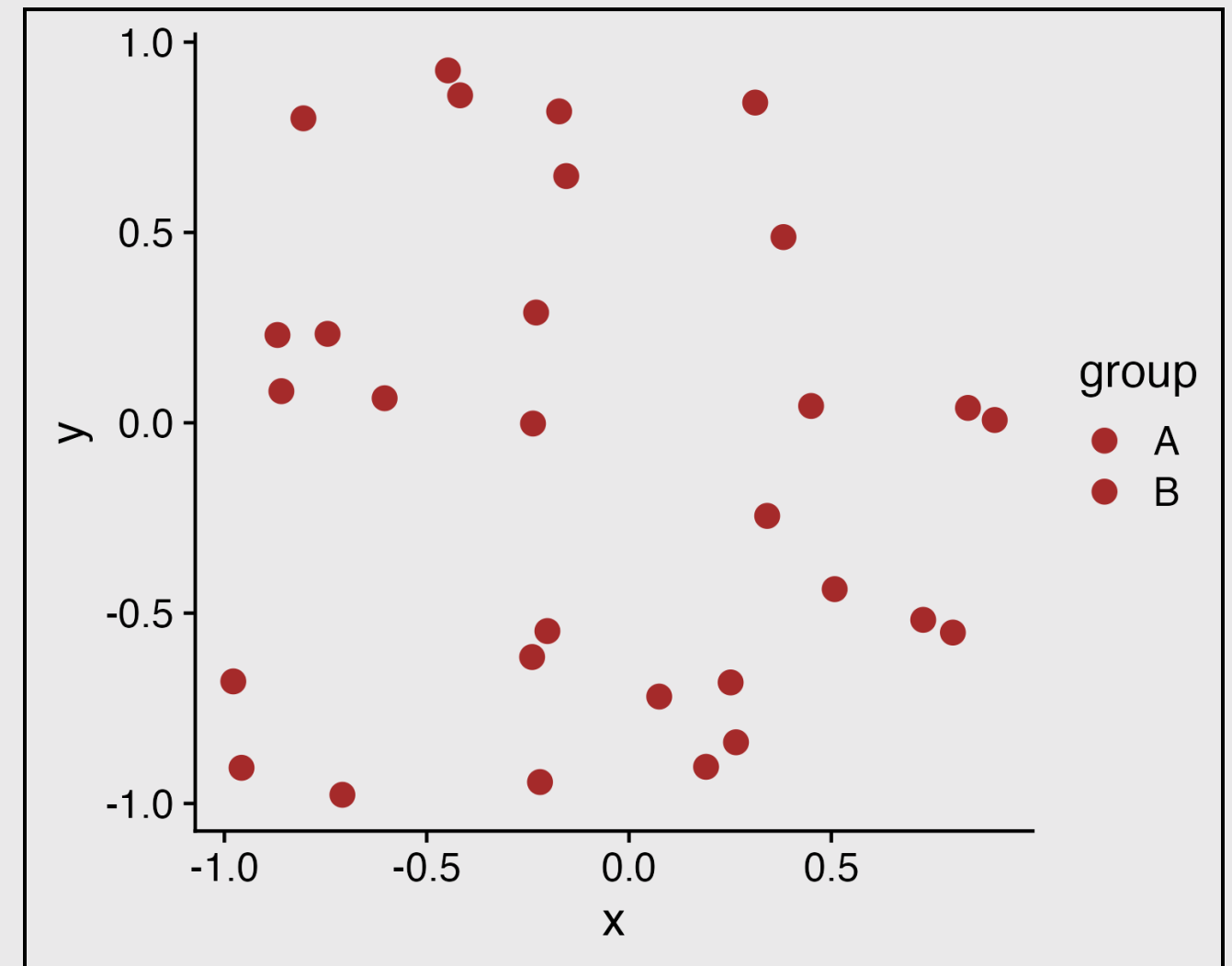
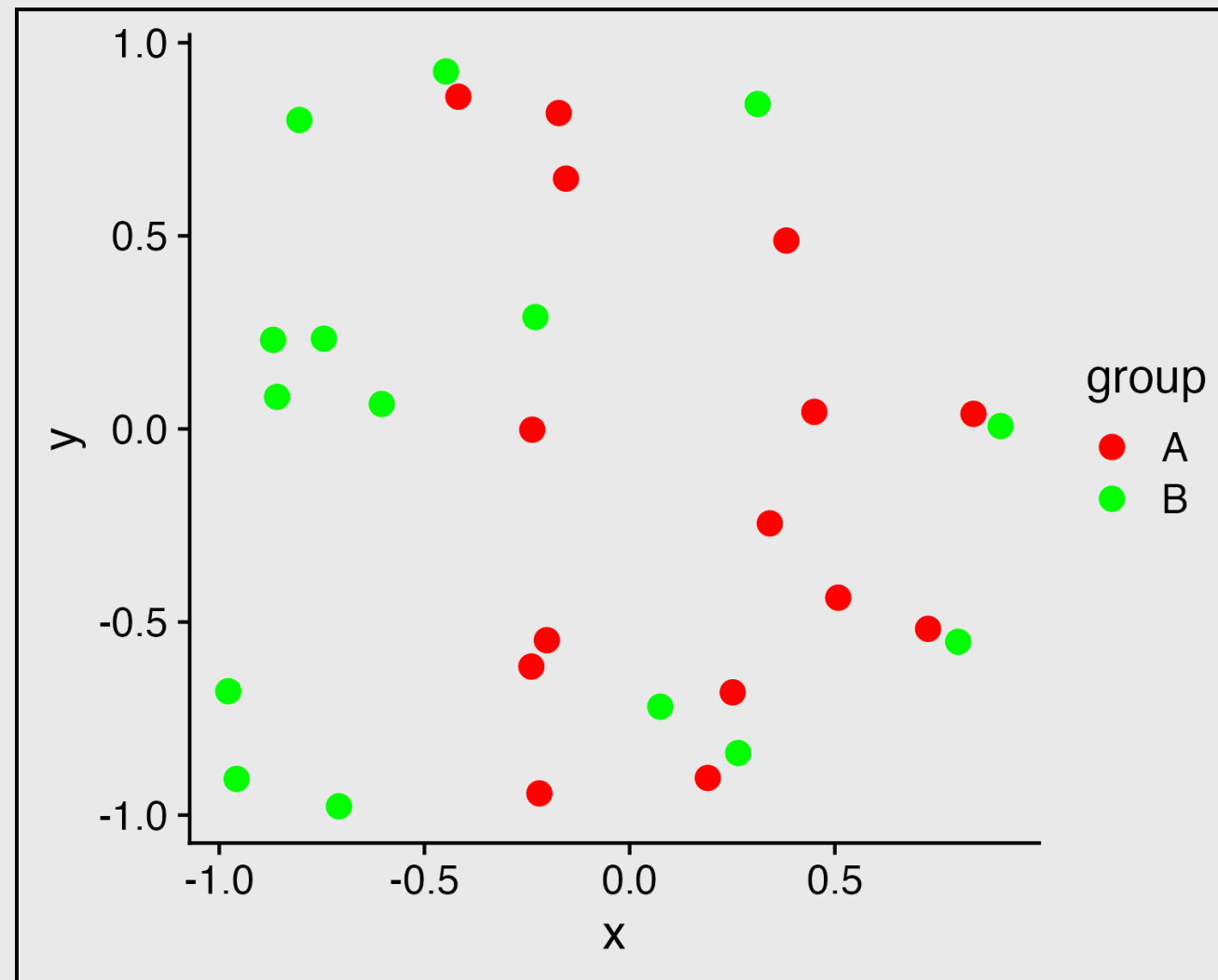


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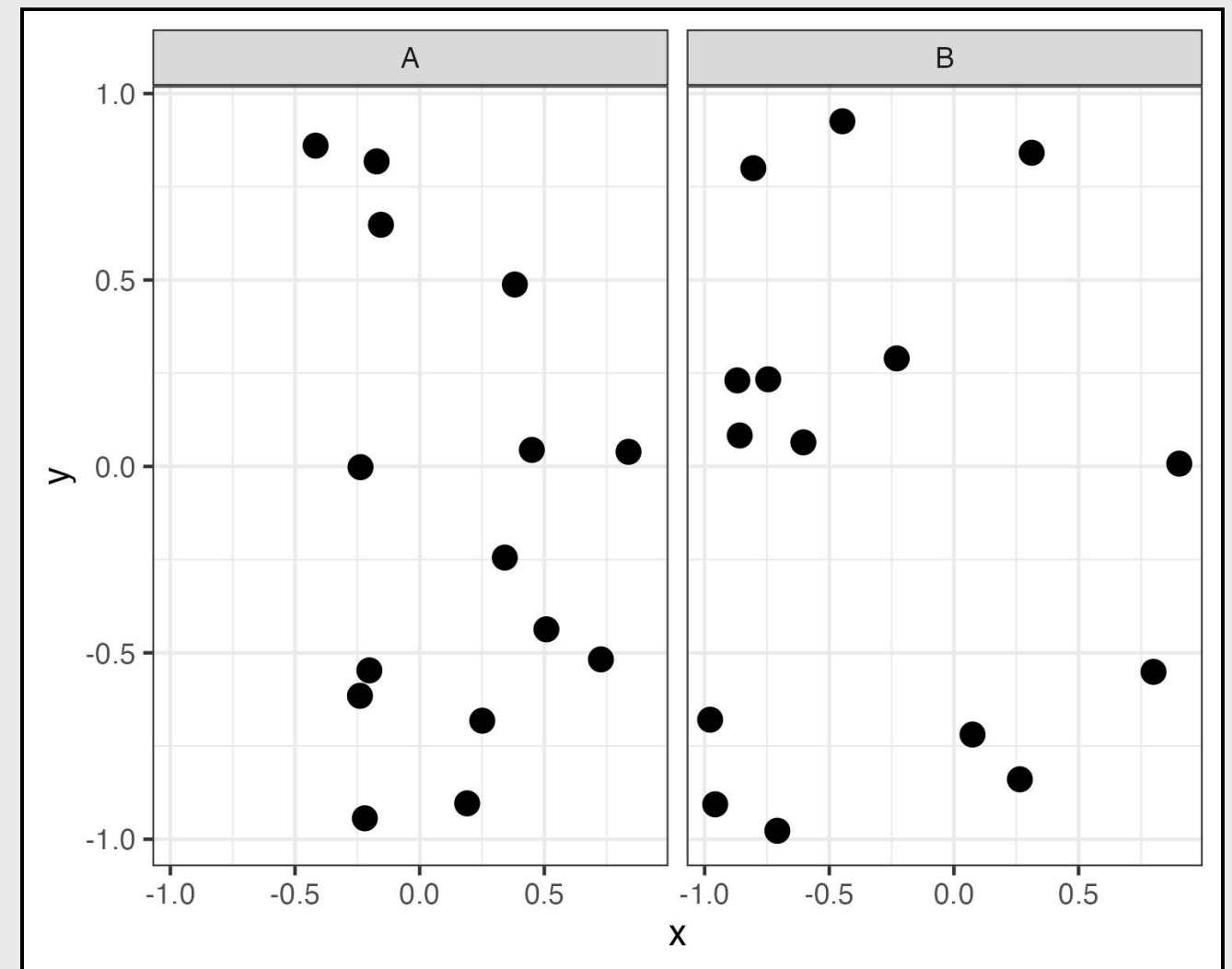
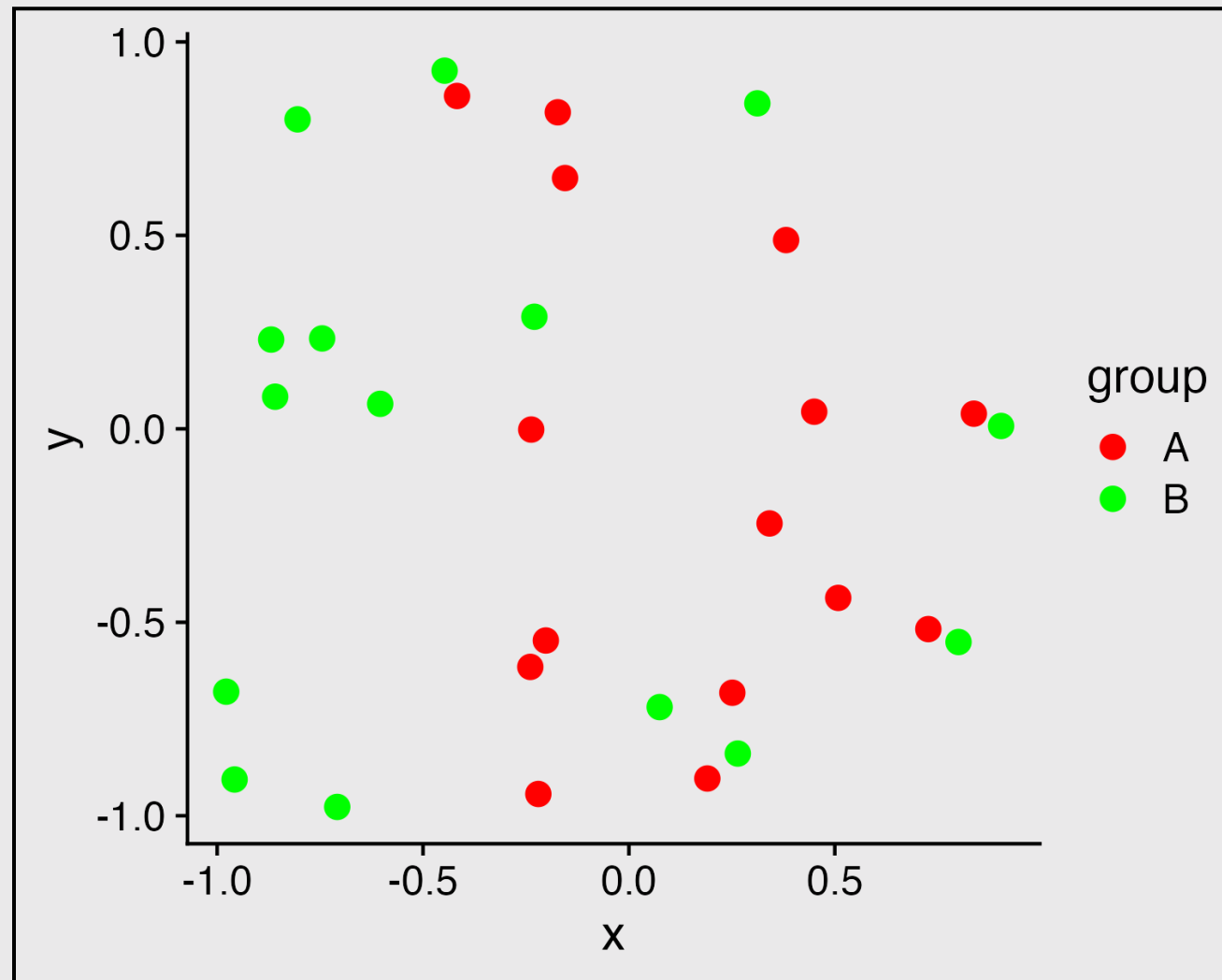
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10% of males and 1% of females are color blind



Facets can be used to avoid color altogether



10 Data Viz Best Practices

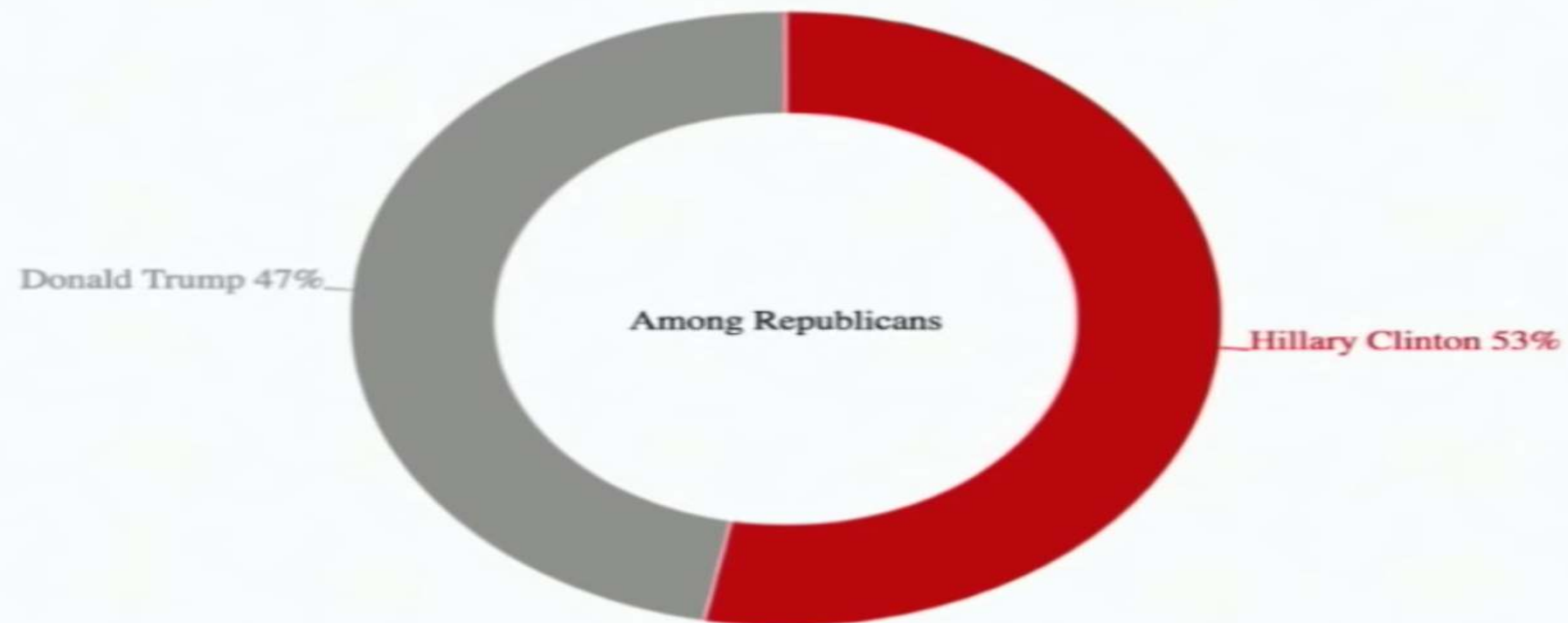
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Who do you think did a better job in tonight's debate?

Among Republicans

Among Democrats



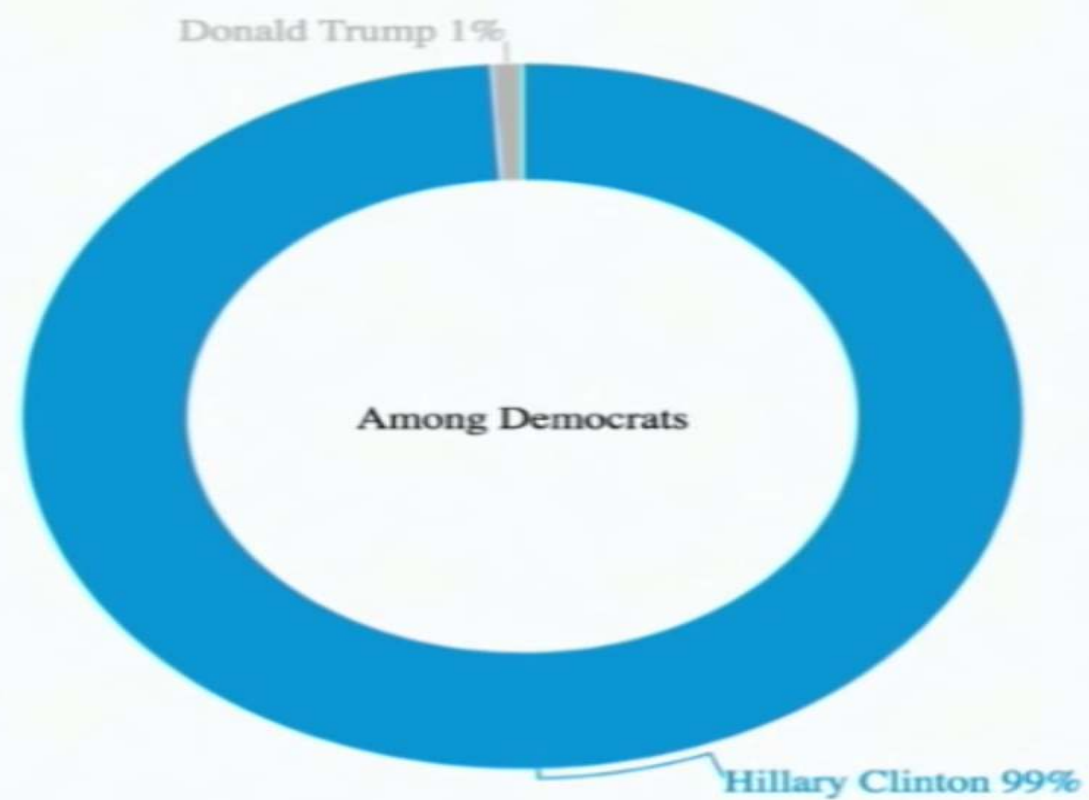
 Share

POLITICO

Who do you think did a better job in tonight's debate?

Among Republicans

Among Democrats



 Share

POLITICO

Who do you think did a better job in tonight's debate?

	Clinton	Trump
Among Democrats	99%	1%
Among Republicans	53%	47%

References:

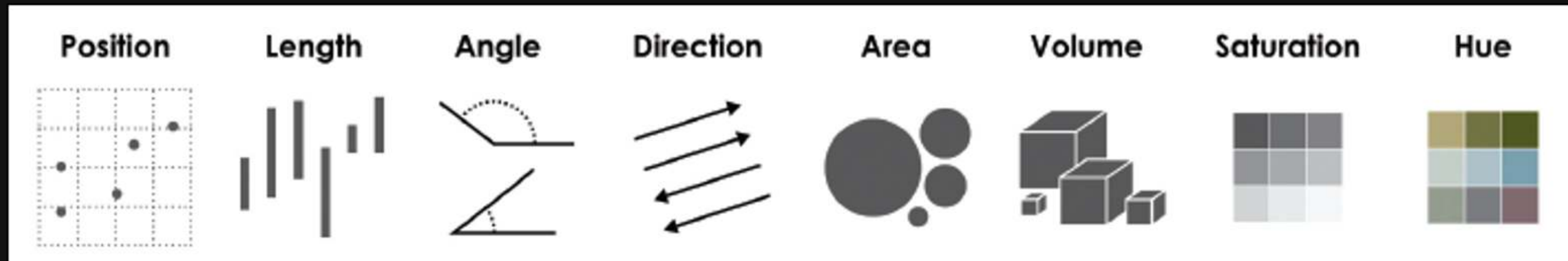
- Data Viz “Cheat Sheet”
- Data Viz Reference Page

Your turn - go [here](#)

10:00

For your “bad” visualization:

1. Identify where the graphic falls on Cleveland’s pattern recognition hierarchy

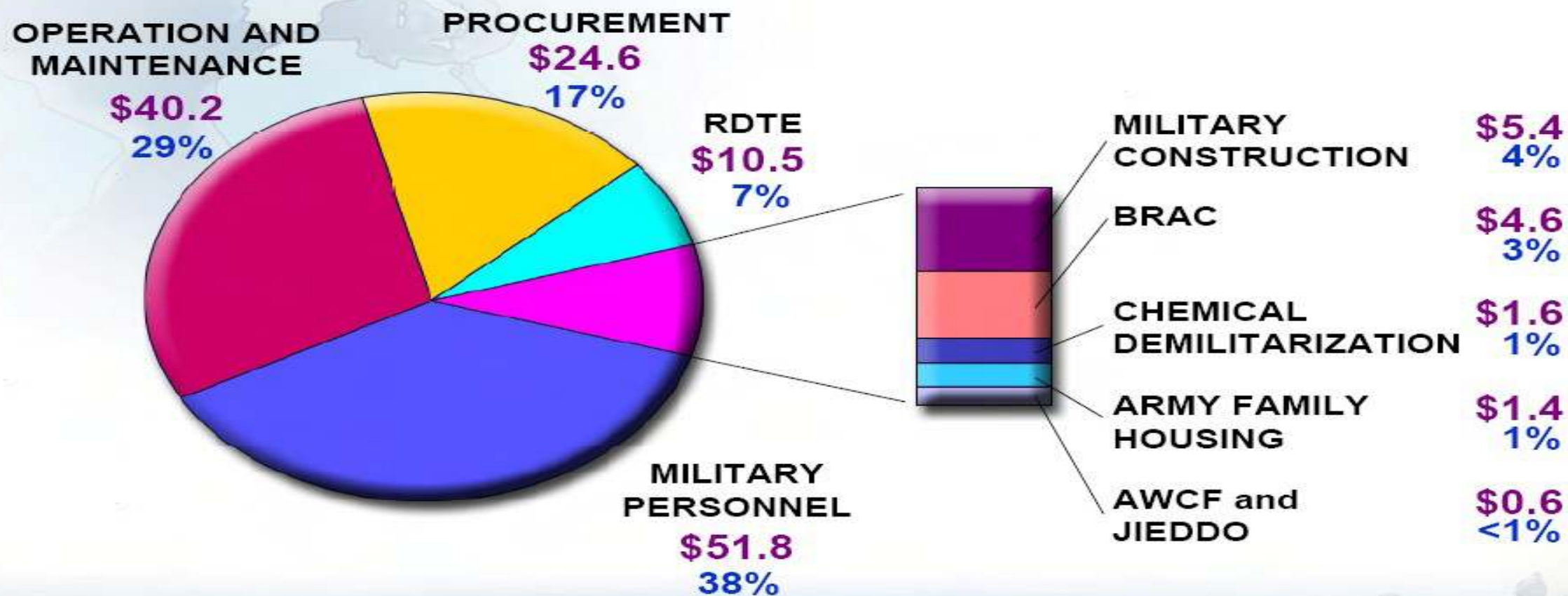


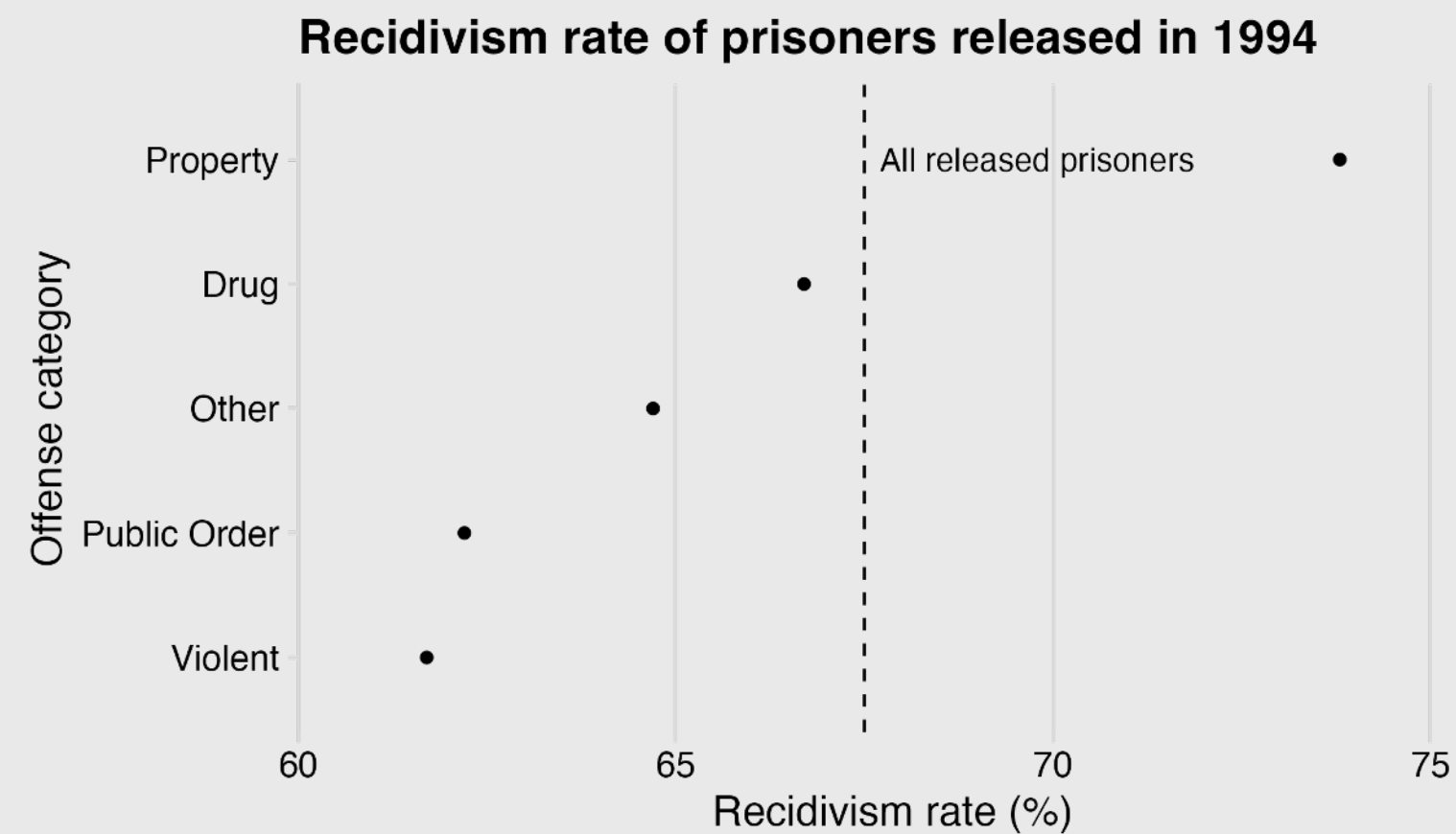
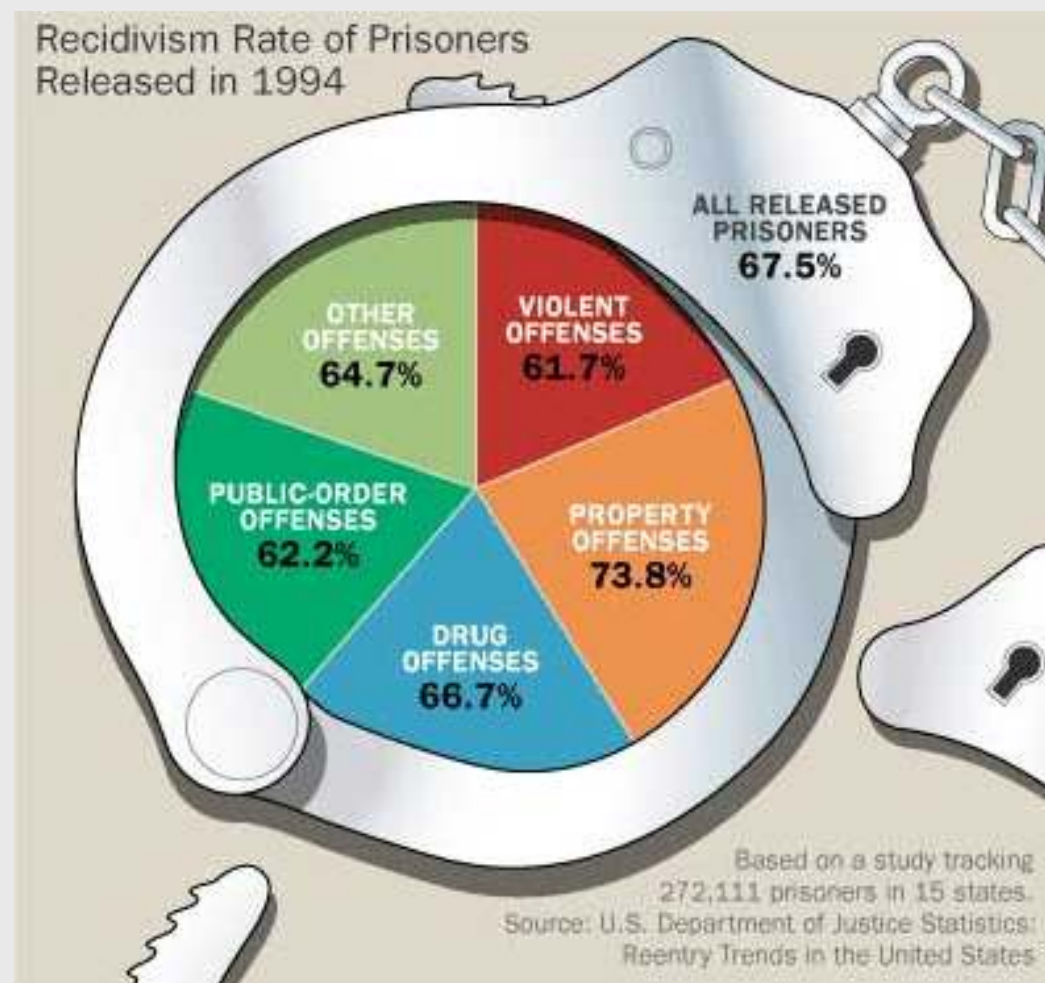
2. Any design rules that are broken

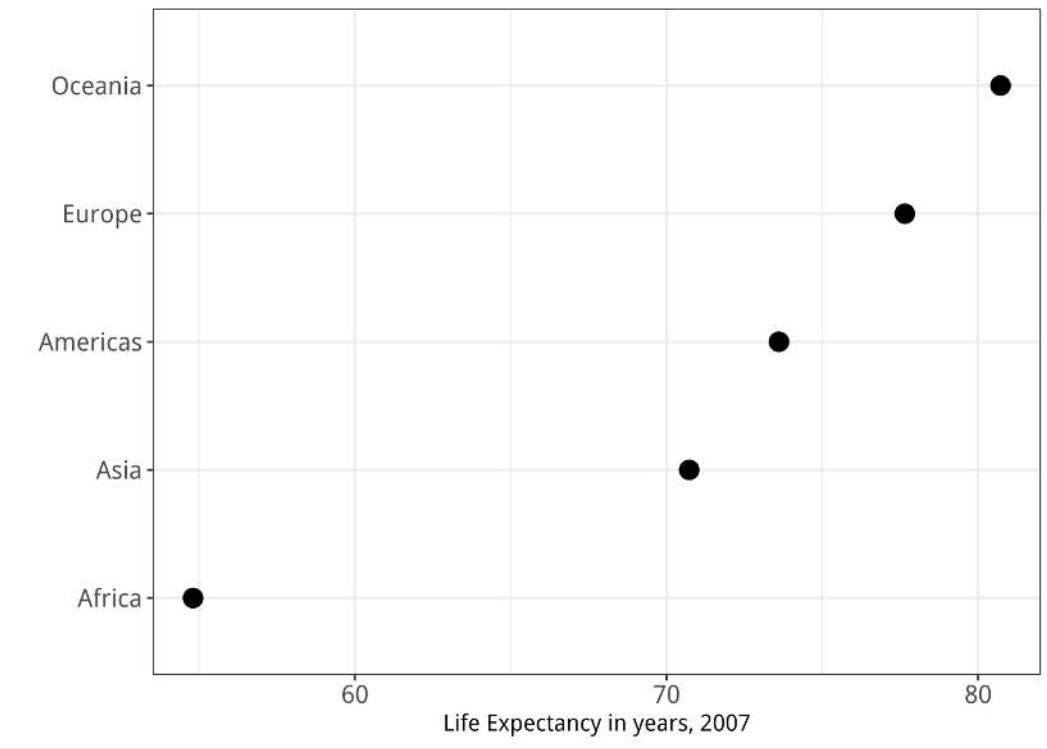
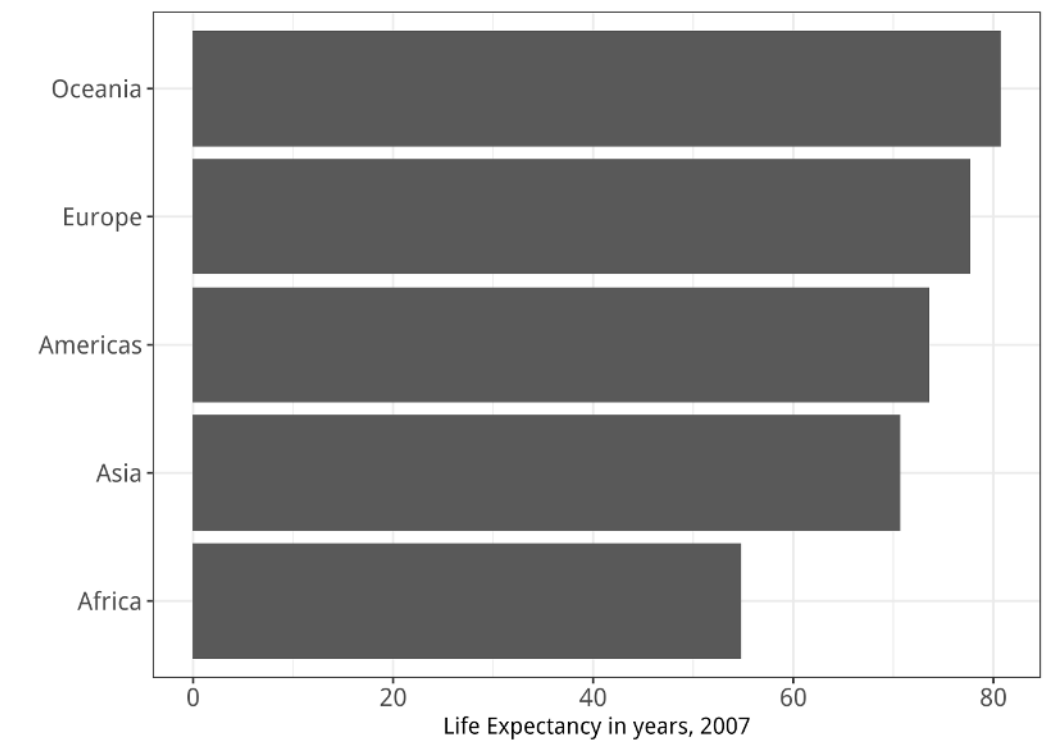
3. Suggest at least two improvements



FY09 Obligation Authority

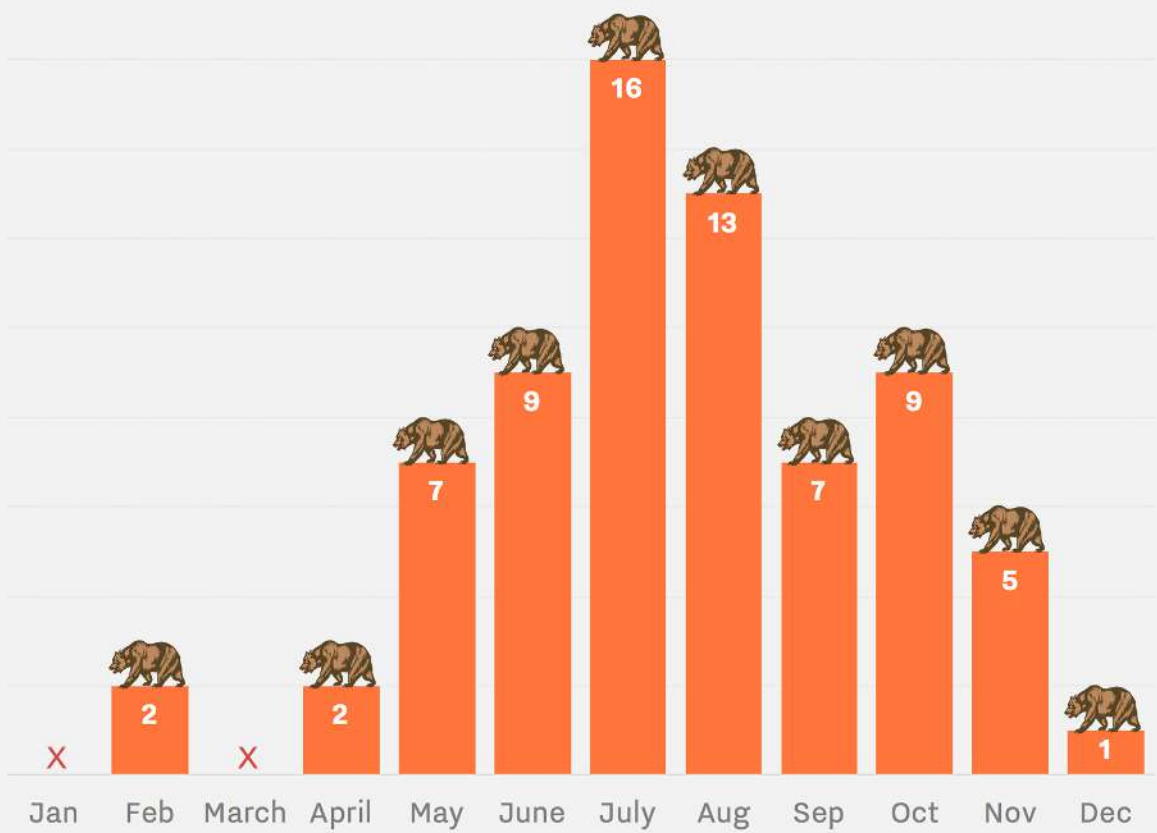






Most fatal bear attacks occur in July and August

Total fatal bear attacks (grizzly, black, and polar), 1900 to present



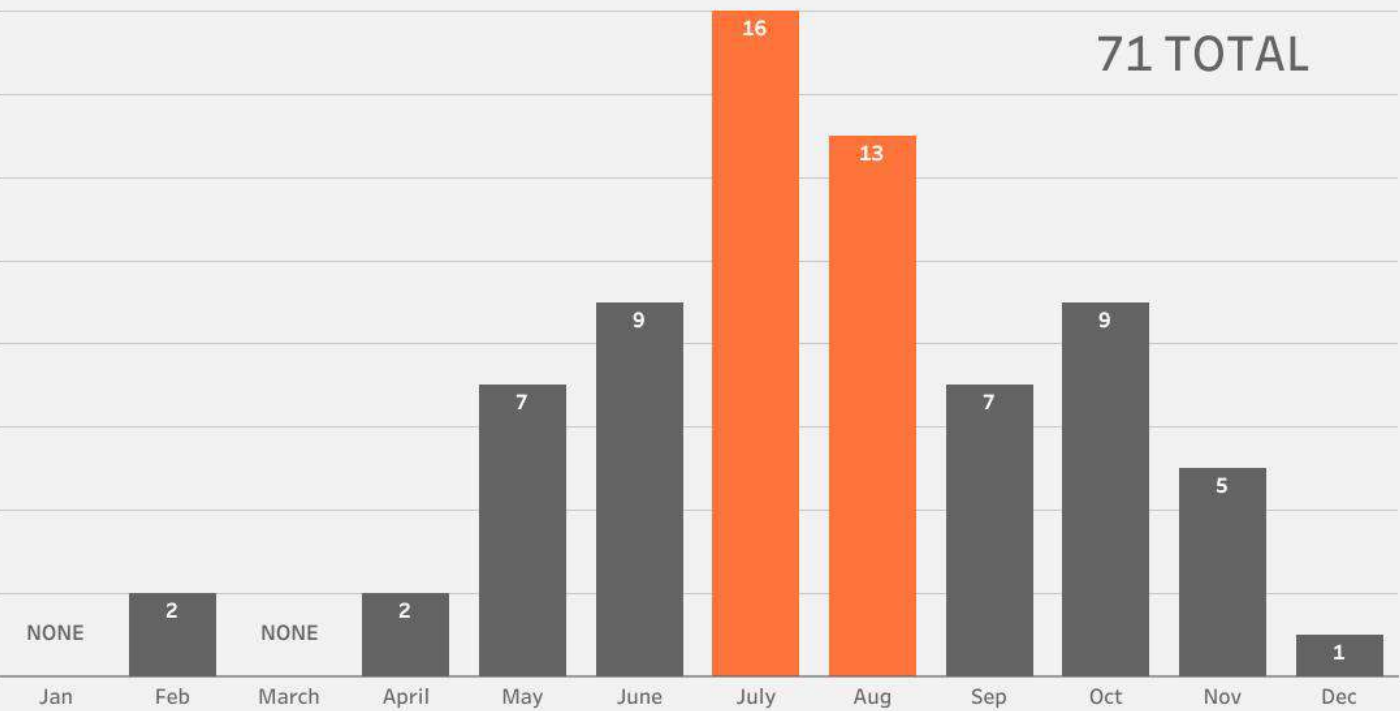
Source: News archives, Wikipedia



BEAR ATTACKS IN U.S. PARKS & WILDERNESS AREAS

Most fatal bear attacks occur in July and August

Total fatal bear attacks by grizzly, black and polar bears from 1900 to present



Source: News archives, Wikipedia (as of 10/2016)

Created by Jeffrey A. Shaffer | MakeoverMonday 2019WK21

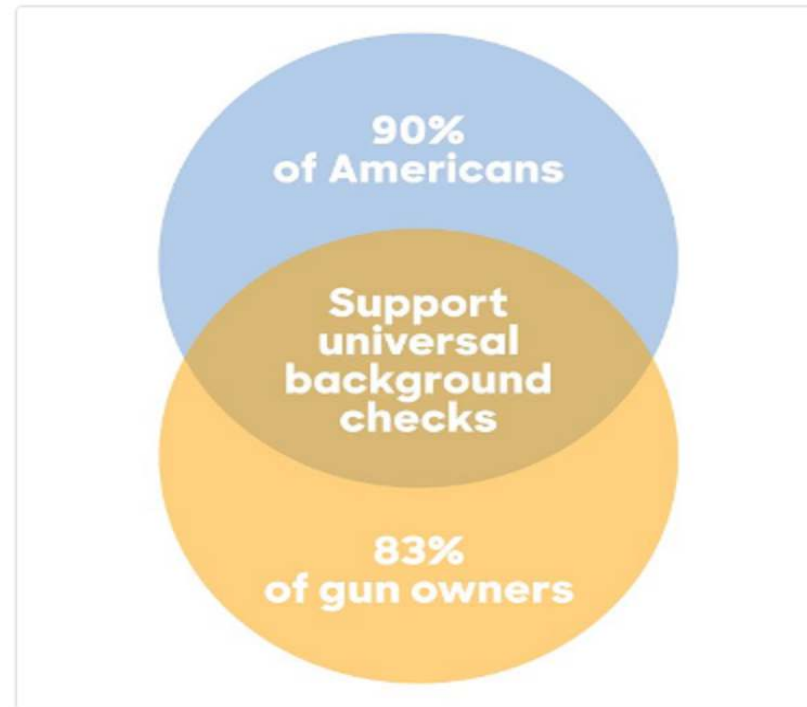


Hillary Clinton
@HillaryClinton

Follow

Dear Congress,
Let's get this done.
Thanks,

The vast majority of Americans



RETWEETS

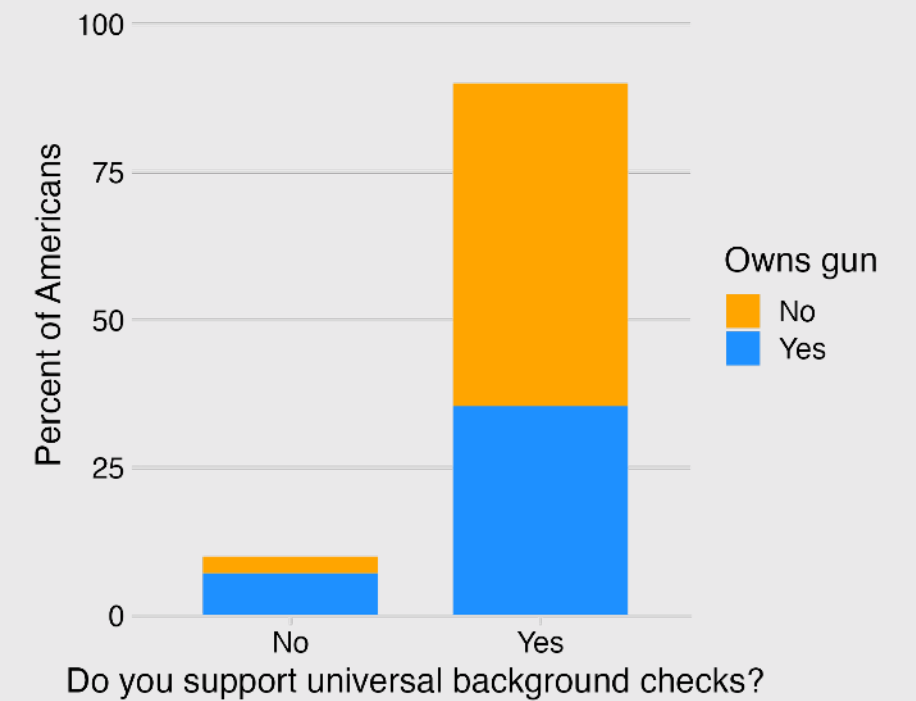
2,308

LIKES

5,333



The vast majority of Americans support universal background checks, including gun owners



People who know
how to make
Venn Diagrams

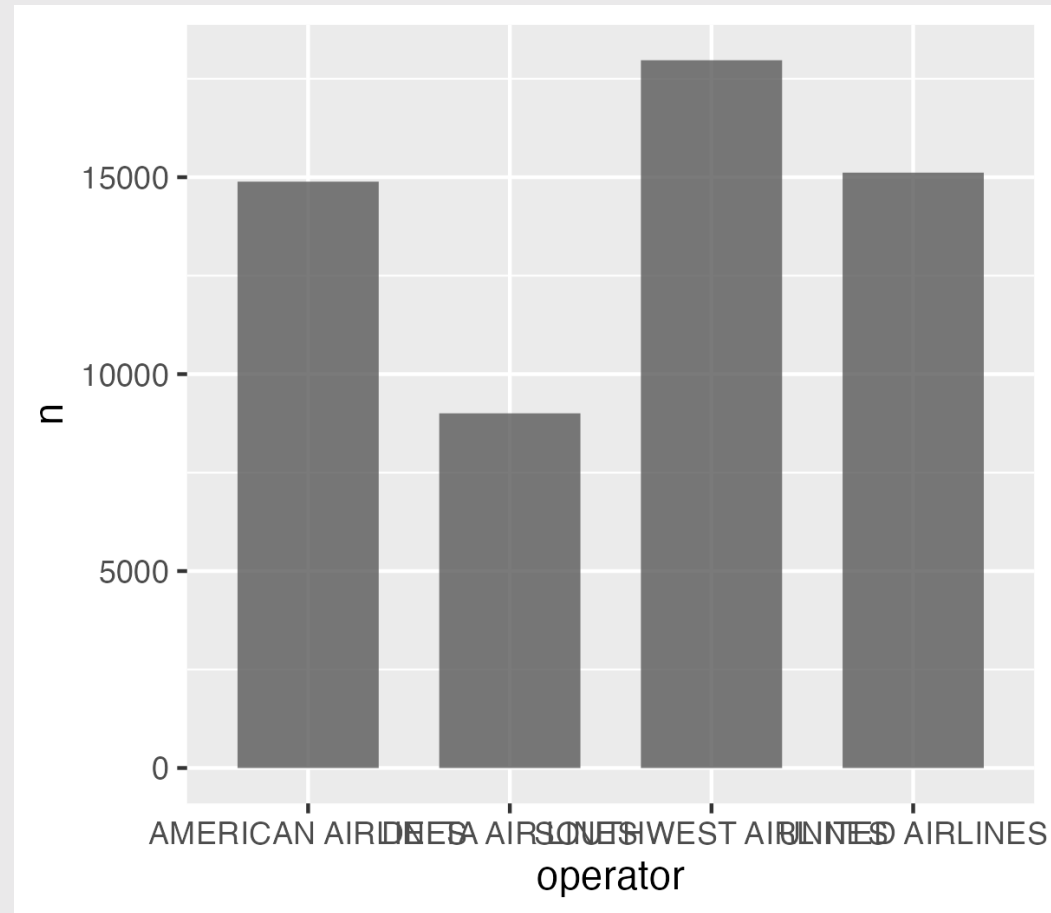
Hillary's graphic
design staff

Week 7: *Visualizing Information*

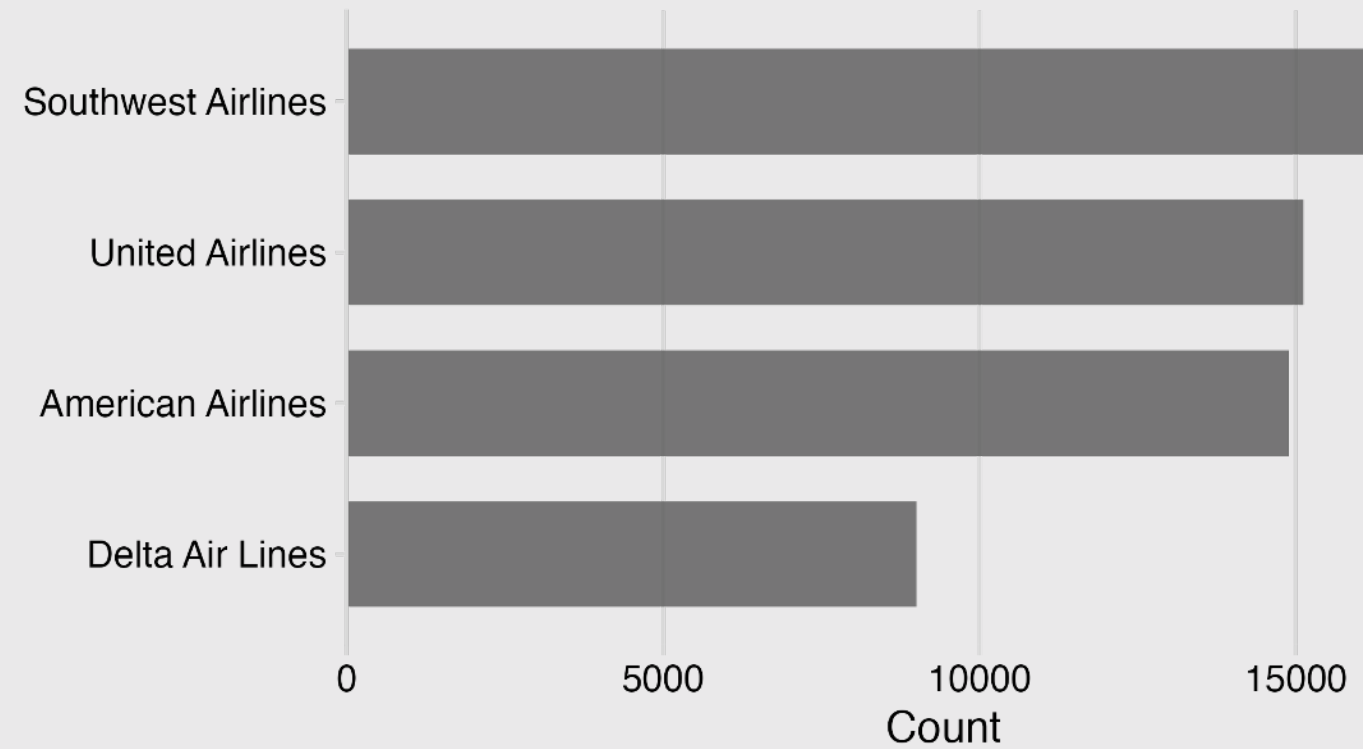
1. The Human Visual-Memory System
2. The Psychology of Data Viz
3. 10 Data Viz Best Practices
4. Making a (good) ggplot

Making a (good) ggplot

Before:



After:



Making a (good) ggplot

1. Format data frame
2. Add geoms
3. Flip coordinates?
4. Reorder factors?
5. Adjust scales
6. Adjust theme
7. Annotate

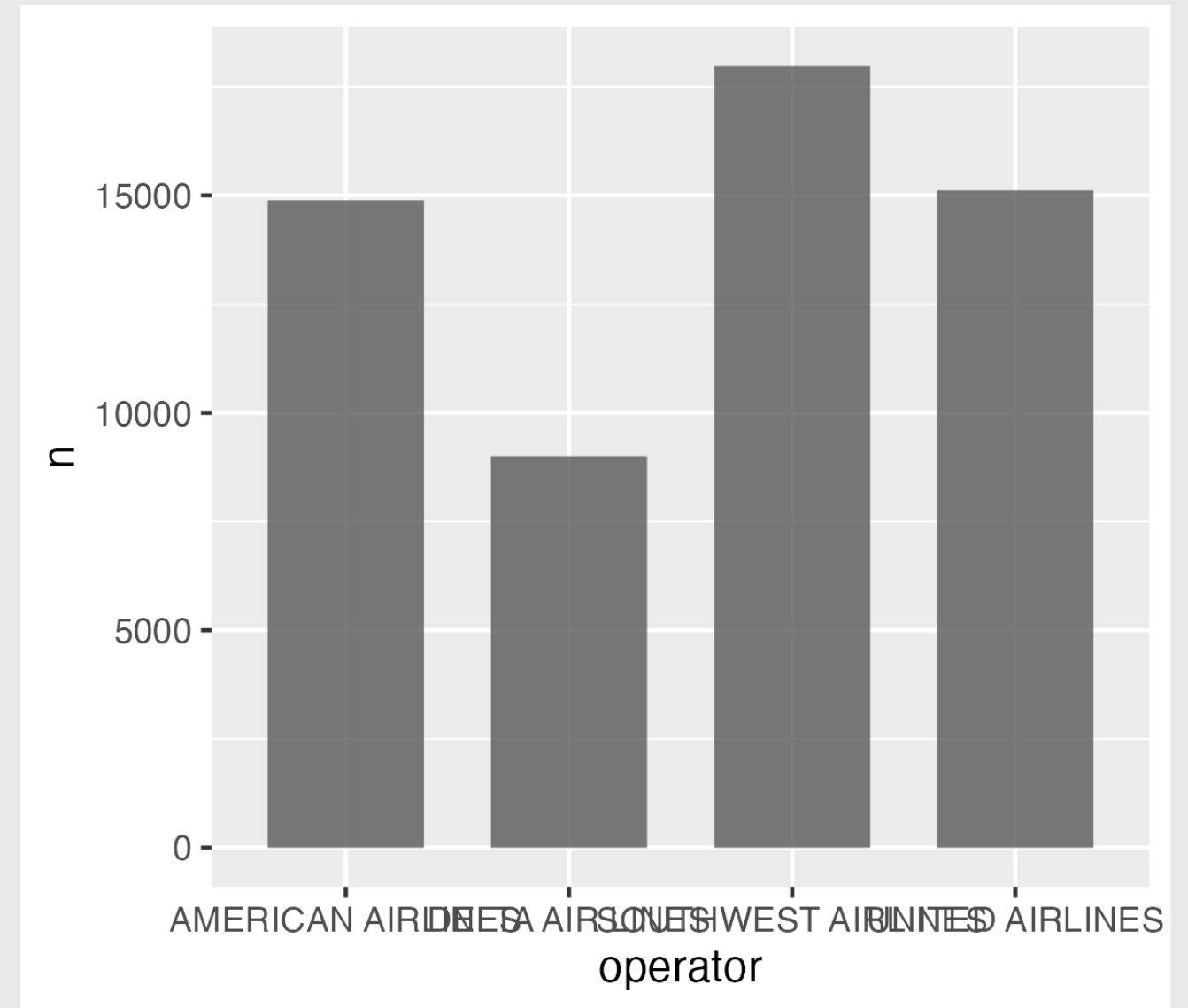
1) Format data frame

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator)
```

```
#> # A tibble: 4 × 2
#>   operator          n
#>   <chr>        <int>
#> 1 AMERICAN AIRLINES 14887
#> 2 DELTA AIR LINES   9005
#> 3 SOUTHWEST AIRLINES 17970
#> 4 UNITED AIRLINES   15116
```

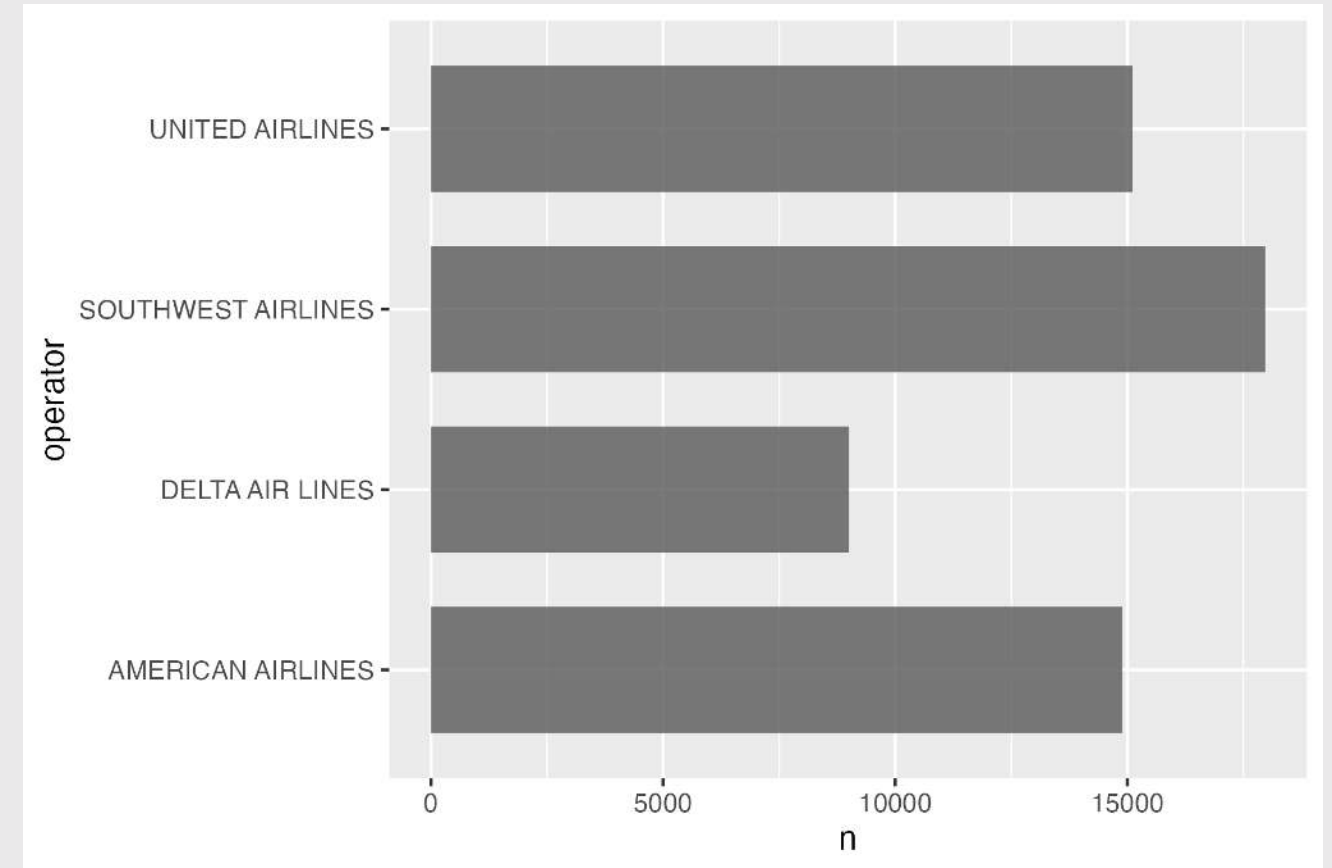
2) Add geoms

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = operator, y = n),
9     width = 0.7,
10    alpha = 0.8
11  )
```



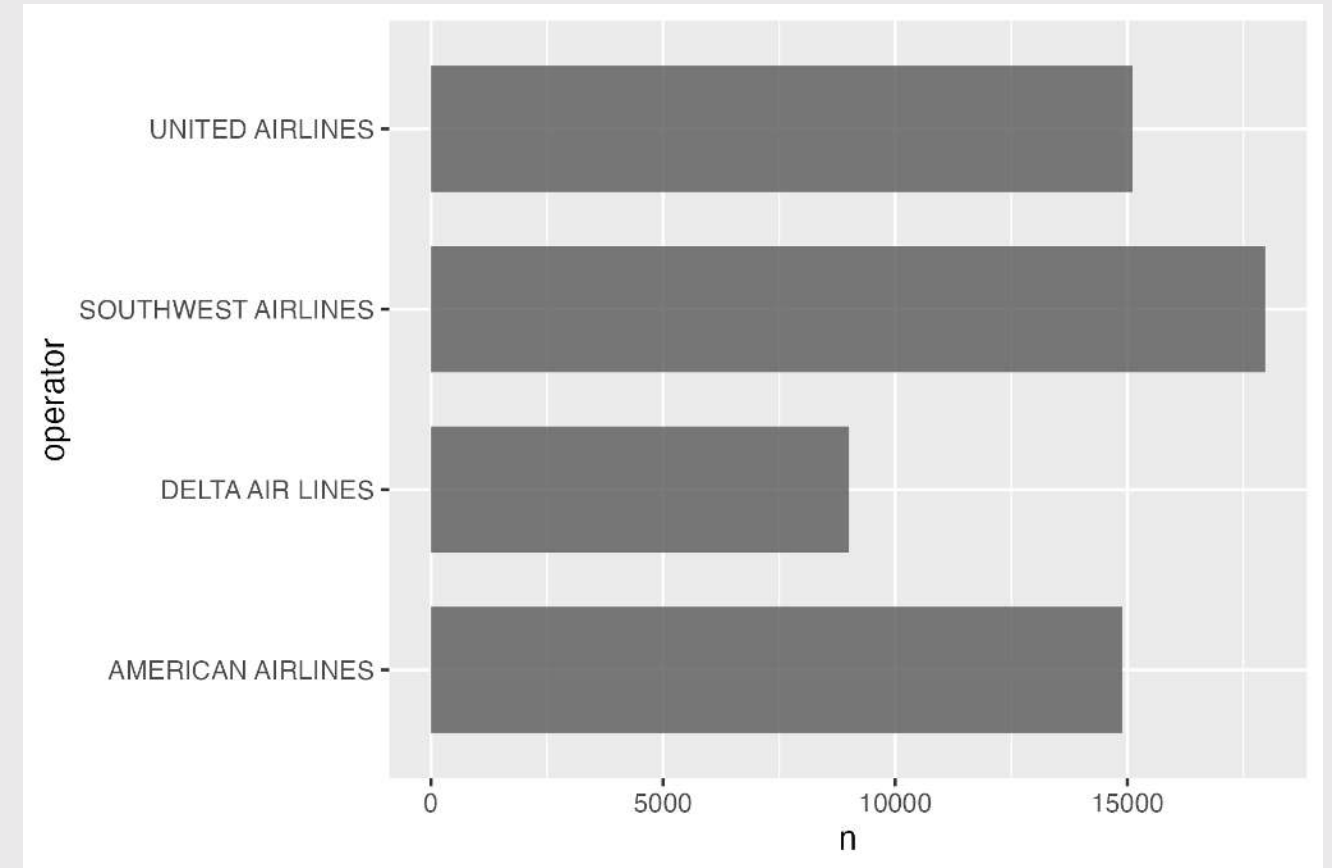
3) Flip coordinates - can you read the labels?

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = operator, y = n),
9     width = 0.7,
10    alpha = 0.8
11  ) +
12
13  # Flip coordinates
14  coord_flip()
```



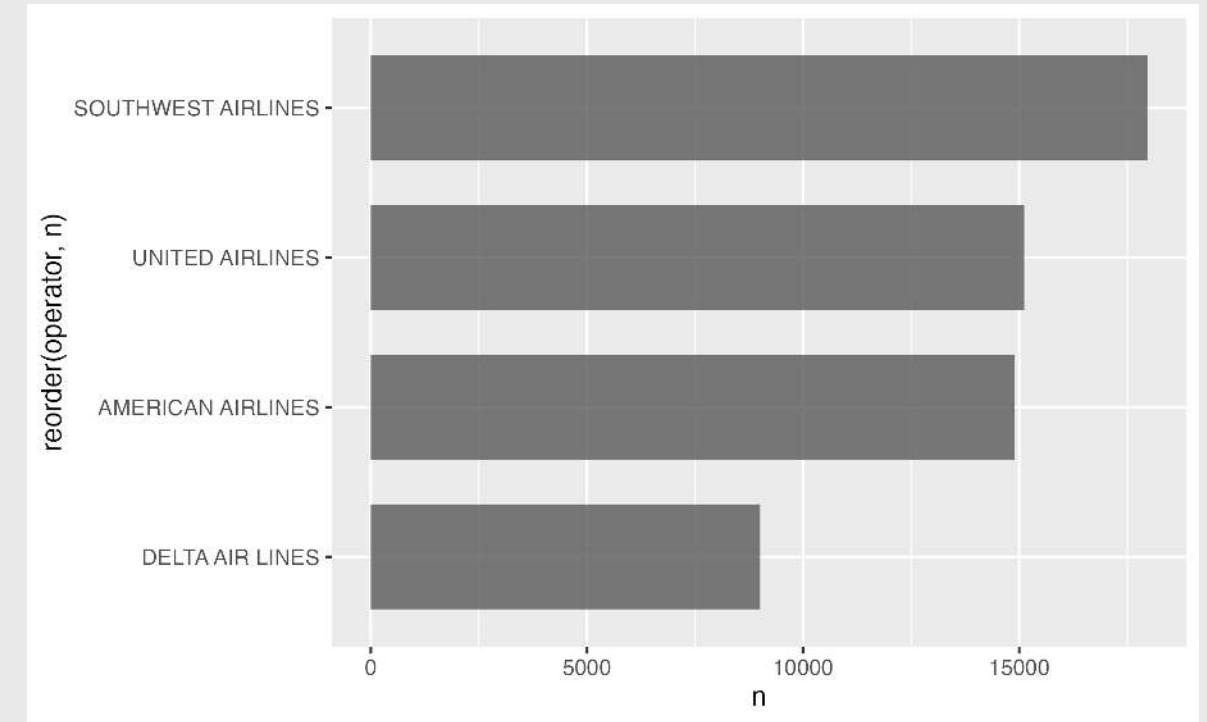
3) Flip coordinates - can you read the labels?

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = n, y = operator),
9     width = 0.7,
10    alpha = 0.8
11  )
```



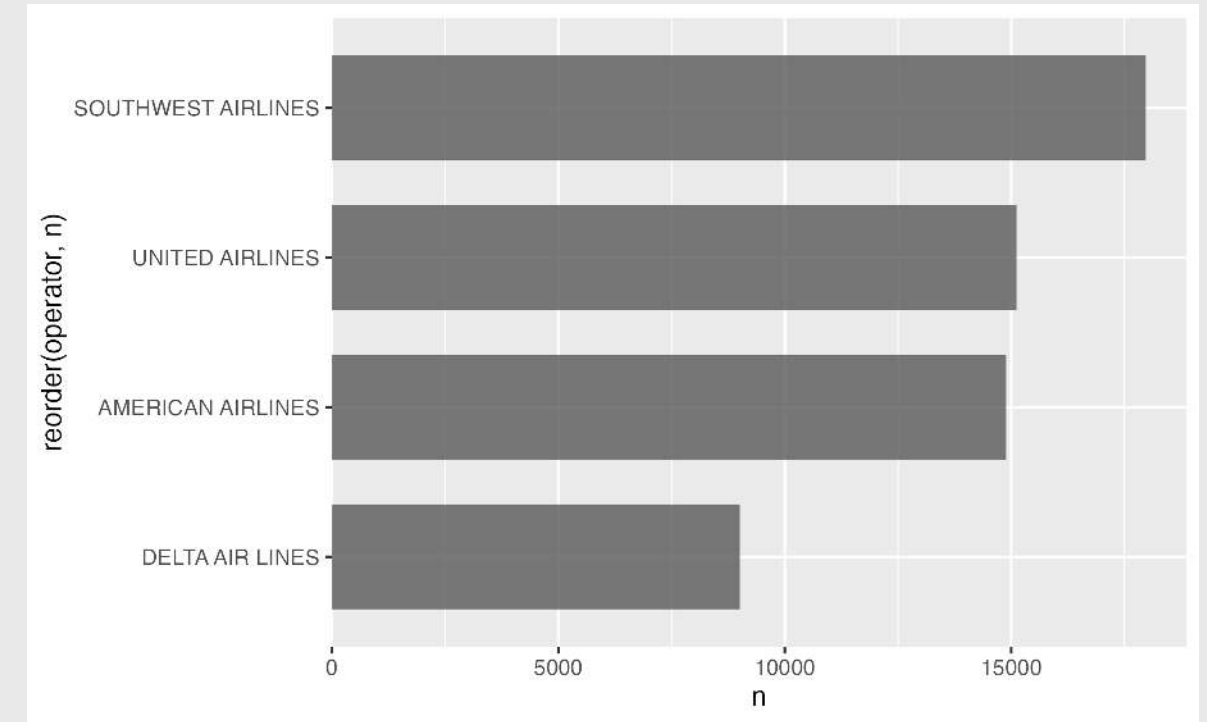
4) Reorder factors with `reorder()`

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = n, y = reorder(operator, n)),
9     width = 0.7,
10    alpha = 0.8
11  )
```



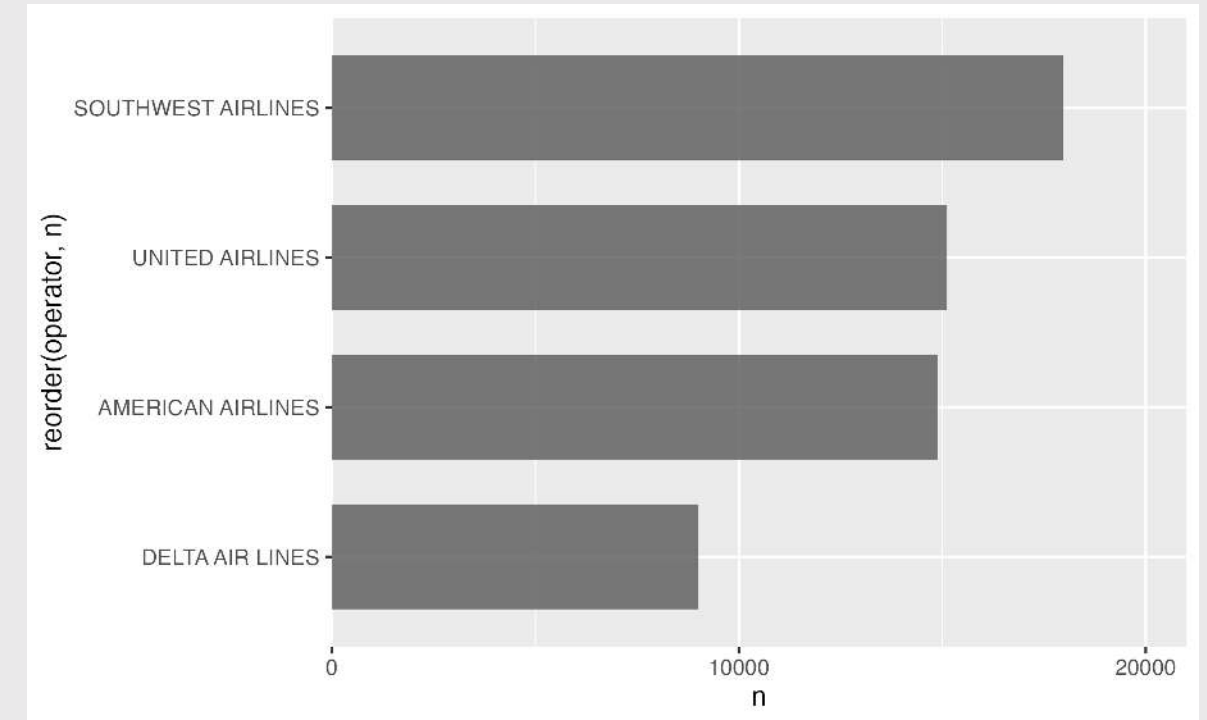
5) Adjust scales

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = n, y = reorder(operator, n)),
9     width = 0.7,
10    alpha = 0.8
11  ) +
12
13  # Adjust x axis scale
14  scale_x_continuous(
15    expand = expansion(mult = c(0, 0.05))
16  )
```



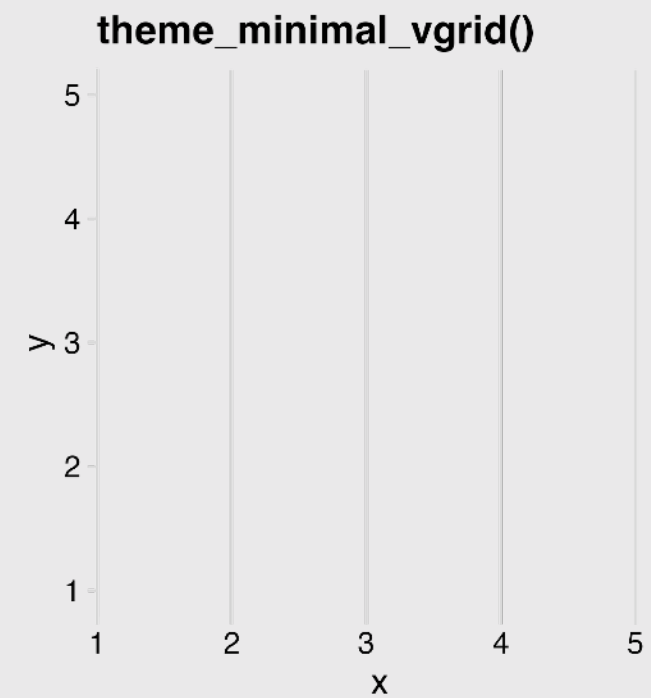
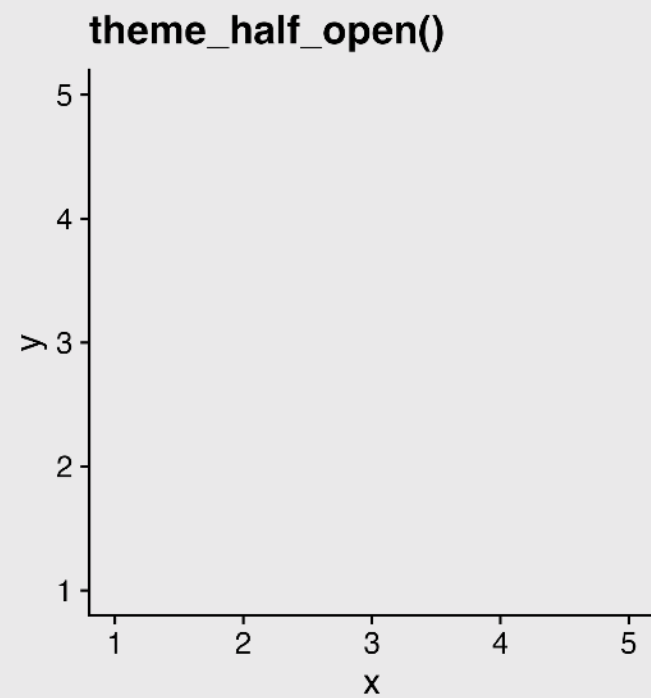
5) Adjust scales - customize break points (if you want)

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = n, y = reorder(operator, n)),
9     width = 0.7,
10    alpha = 0.8
11  ) +
12
13  # Adjust x axis scale
14  scale_x_continuous(
15    expand = expansion(mult = c(0, 0.05)),
16    breaks = c(0, 10000, 20000),
17    limits = c(0, 20000)
18  )
```



6) Adjust theme

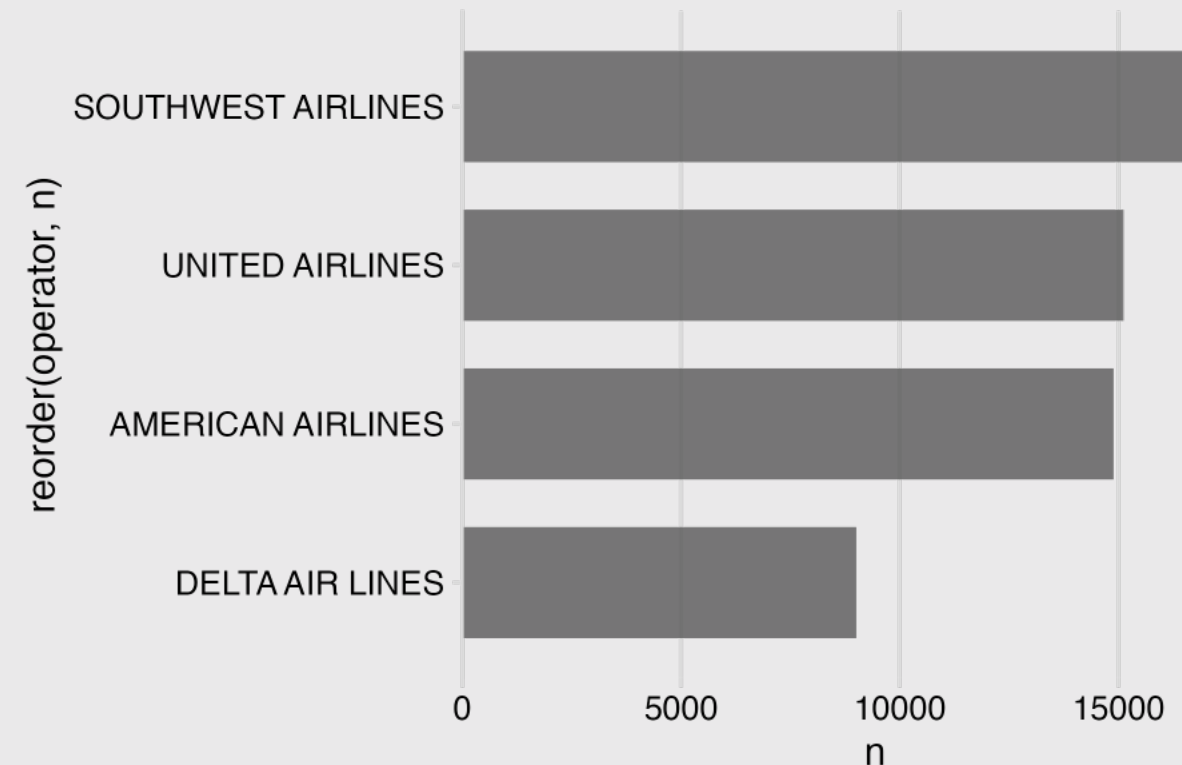
Four `cowplot` themes you should know



6) Adjust theme

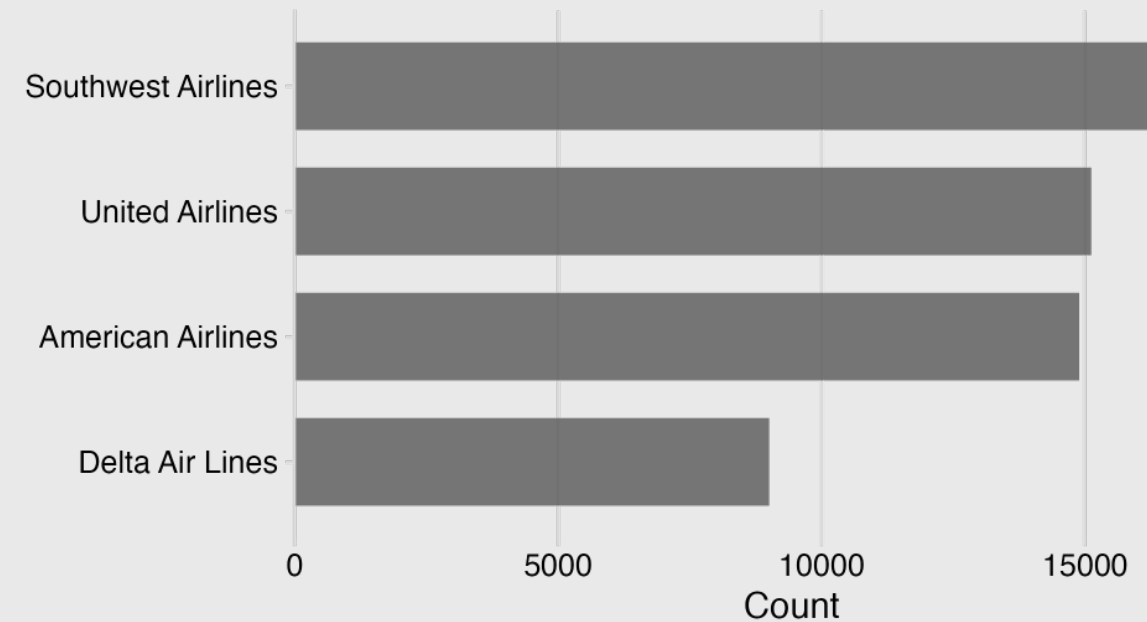
For horizontal bars, add only vertical grid

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4
5   # Add geoms
6   ggplot() +
7   geom_col(
8     aes(x = n, y = reorder(operator, n)),
9     width = 0.7,
10    alpha = 0.8
11  ) +
12
13  # Adjust x axis scale
14  scale_x_continuous(
15    expand = expansion(mult = c(0, 0.05))
16  ) +
17
18  # Adjust theme
19  theme_minimal_vgrid()
```



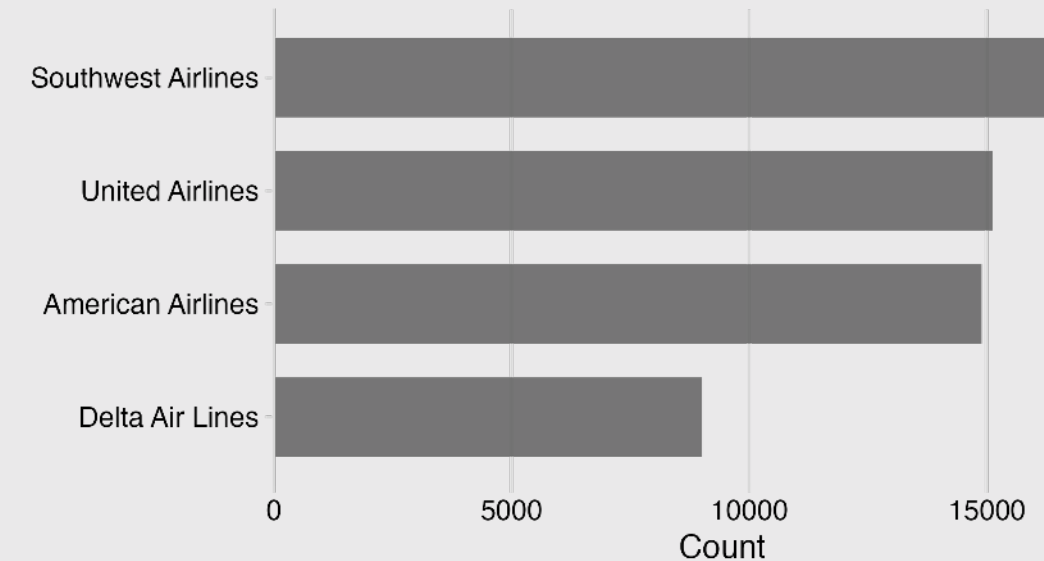
7) Annotate

```
1 # Format the data frame
2 wildlife_impacts %>%
3   count(operator) %>%
4   mutate(operator = str_to_title(operator)) %>%
5
6   # Add geoms
7   ggplot() +
8   geom_col(
9     aes(x = n, y = reorder(operator, n)),
10    width = 0.7,
11    alpha = 0.8
12  ) +
13
14  # Adjust x axis scale
15  scale_x_continuous(
16    expand = expansion(mult = c(0, 0.05))
17  ) +
18
19  # Adjust theme
20  theme_minimal_vgrid() +
21
22  # Annotate
23  labs(
24    x = 'Count',
```



Finished product

```
1 wildlife_impacts %>%
2   count(operator) %>%
3   mutate(operator = str_to_title(operator)) %>%
4   ggplot() +
5   geom_col(
6     aes(x = n, y = reorder(operator, n)),
7     width = 0.7,
8     alpha = 0.8
9   ) +
10  scale_x_continuous(
11    expand = expansion(mult = c(0, 0.05))
12  ) +
13  theme_minimal_vgrid() +
14  labs(
15    x = 'Count',
16    y = NULL
17  )
```



Your turn

15:00

Use the `gapminder.csv` data to create the following plot, following these steps:

1. Format data frame
2. Add geoms
3. Flip coordinates?
4. Reorder factors?
5. Adjust scales
6. Adjust theme
7. Annotate

